

An online retail company has more than 50 million active customers and receives more than 25,000 orders each day. The company collects purchase data for customers and stores this data in Amazon S3. Additional customer data is stored in Amazon RDS.

The company wants to make all the data available to various teams so that the teams can perform analytics. The solution must provide the ability to manage fine-grained permissions for the data and must minimize operational overhead.

Which solution will meet these requirements?

- A. Migrate the purchase data to write directly to Amazon RDS. Use RDS access controls to limit access.
- B. Schedule an AWS Lambda function to periodically copy data from Amazon RDS to Amazon S3. Create an AWS Glue crawler. Use Amazon Athena to query the data. Use S3 policies to limit access.
- C. Create a data lake by using AWS Lake Formation. Create an AWS Glue JDBC connection to Amazon RDS. Register the S3 bucket in Lake Formation. Use Lake Formation access controls to limit access. **Most Voted**
- D. Create an Amazon Redshift cluster. Schedule an AWS Lambda function to periodically copy data from Amazon S3 and Amazon RDS to Amazon Redshift. Use Amazon Redshift access controls to limit access.

**Correct Answer: C**

Community vote distribution

C (100%)

## Comments

**anhike** **Highly Voted** 3 years, 6 months ago

Answer : C keyword "manage-fine-grained"

<https://aws.amazon.com/blogs/big-data/manage-fine-grained-access-control-using-aws-lake-formation/>

upvoted 27 times

**markw92** 2 years, 11 months ago

You can manage fine grained using redshift as well - <https://aws.amazon.com/blogs/big-data/achieve-fine-grained-data-security-with-row-level-access-control-in-amazon-redshift/>

But, I believe the keyword to look for is "minimize operational overhead", which lakeformation does without duplicating much of the data. Redshift is operational overhead and duplication of data. not sure why the answer is D. i vote C as well.

upvoted 9 times

**Olaunfazed** 2 years, 11 months ago

yeah, most of examtopics answers are wrong

upvoted 9 times

**Buruguduystunstugudunstuy** **Highly Voted** 3 years, 5 months ago

**Selected Answer: C**

Option C. Create a data lake by using AWS Lake Formation. Create an AWS Glue JDBC connection to Amazon RDS. Register the S3 bucket in Lake Formation. Use Lake Formation access controls to limit access.

To make all the data available to various teams and minimize operational overhead, the company can create a data lake by using AWS Lake Formation. This will allow the company to centralize all the data in one place and use fine-grained access controls to manage access to the data.

To meet the requirements of the company, the solutions architect can create a data lake by using AWS Lake Formation, create an AWS Glue JDBC connection to Amazon RDS, and register the S3 bucket in Lake Formation. The solutions architect can then use Lake Formation access controls to limit access to the data. This solution will provide the ability to manage fine-grained permissions for the data and minimize operational overhead.

upvoted 6 times

**ike001** **Most Recent** 1 year, 11 months ago

Answer is C

upvoted 3 times

**MehulKapadia** 2 years, 1 month ago

**Selected Answer: C**

Keyword: "manage fine-grained permissions for data"

Data Lake Using Lake Formation: manage fine-grained permissions for the data with ease.

Fine grained permissions for data = Lake Formation

Answer: C

upvoted 4 times

**LoXoL** 2 years, 4 months ago

**Selected Answer: C**

C represents the easiest way to ingest data from S3 and control accesses.

upvoted 1 times

**karloscetina007** 2 years, 8 months ago

**Selected Answer: C**

a fine grained permissions is one of the conditions to accomplish with the requirement.

With the use of AWS Glue you can get accomplish with this requirement.

My answer is: C

upvoted 4 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: C**

With Lake formation you can scale permissions more easily with fine-grained security capabilities, including row- and cell-level permissions and tag-based access control.

upvoted 5 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: C**

Lake Formation enables the creation of a secure and scalable data lake on AWS, allowing centralized access controls for both S3 and RDS data. By using Lake Formation, the company can manage permissions effectively and integrate RDS data through the AWS Glue JDBC connection. Registering the S3 in Lake Formation ensures unified access control. This solution reduces operational overhead while providing fine-grained permissions management.

upvoted 4 times

**cookieMr** 2 years, 11 months ago

**Selected Answer: C**

Lake Formation enables the creation of a secure and scalable data lake on AWS, allowing centralized access controls for both S3 and RDS data. By using Lake Formation, the company can manage permissions effectively and integrate RDS data through the AWS Glue JDBC connection. Registering the S3 in Lake Formation ensures unified access control. This solution reduces operational overhead while providing fine-grained permissions management.

A. Directly writing purchase data to Amazon RDS with RDS access controls lacks comprehensive permissions management for both S3 and RDS data.

B. Periodically copying data from RDS to S3 using Lambda and using AWS Glue and Athena for querying does not offer fine-grained permissions management and introduces data synchronization complexities.

D. Creating an Redshift cluster and copying data from S3 and RDS to Redshift adds complexity and operational overhead without the flexibility of Lake Formation's permissions management capabilities.

upvoted 4 times

**pisica134** 2 years, 11 months ago

Answer is C AWS Lake Formation provides a comprehensive solution for building and managing a data lake. It simplifies data ingestion, organization, and access control. By creating a data lake using AWS Lake Formation, you can centralize and govern access to your data across multiple sources.

upvoted 2 times

**Bmarodi** 2 years, 12 months ago

**Selected Answer: C**

Option C is right answer: <https://docs.aws.amazon.com/lake-formation/latest/dg/what-is-lake-formation.html>

upvoted 2 times

**Abrar2022** 3 years ago

Lake Formation helps you manage fine-grained access for internal and external customers from a centralized location and in a scalable way.

upvoted 2 times

**doorahmie** 3 years, 4 months ago

<https://docs.aws.amazon.com/lake-formation/latest/dg/access-control-overview.html>

upvoted 3 times

**LuckyAro** 3 years, 4 months ago

**Selected Answer: C**

To me, the give-away was: "The company wants to make all the data available to various teams" - Data-Lake - All data in one place.  
upvoted 5 times

**master1004** 3 years, 5 months ago

The correct answer is D.

The company uses all the data from various teams so that the teams can do their analysis.

Therefore, it is the best way to separately configure redshift for data warehousing and for all employees to connect to the redshift DB and perform analysis tasks without burdening the operating DB (must minimize operational overhead).

upvoted 4 times

**ruqui** 3 years ago

I don't think that "periodically copy data from Amazon S3 and RDS to Redshift" minimize the operational overhead. The correct answer for me is C  
upvoted 2 times

**aba2s** 3 years, 5 months ago

**Selected Answer: C**

Manage fine-grained access control using AWS Lake Formation

<https://aws.amazon.com/blogs/big-data/manage-fine-grained-access-control-using-aws-lake-formation/>

upvoted 2 times

**kvenikoduru** 3 years, 5 months ago

**Selected Answer: C**

a combination of the following 2 URLs I believe it is C

<https://aws.amazon.com/lake-formation/>

<https://aws.amazon.com/blogs/big-data/manage-fine-grained-access-control-using-aws-lake-formation/>

upvoted 3 times

A company hosts a marketing website in an on-premises data center. The website consists of static documents and runs on a single server. An administrator updates the website content infrequently and uses an SFTP client to upload new documents.

The company decides to host its website on AWS and to use Amazon CloudFront. The company's solutions architect creates a CloudFront distribution. The solutions architect must design the most cost-effective and resilient architecture for website hosting to serve as the CloudFront origin.

Which solution will meet these requirements?

- A. Create a virtual server by using Amazon Lightsail. Configure the web server in the Lightsail instance. Upload website content by using an SFTP client.
- B. Create an AWS Auto Scaling group for Amazon EC2 instances. Use an Application Load Balancer. Upload website content by using an SFTP client.
- C. Create a private Amazon S3 bucket. Use an S3 bucket policy to allow access from a CloudFront origin access identity (OAI). Upload website content by using the AWS CLI. **Most Voted**
- D. Create a public Amazon S3 bucket. Configure AWS Transfer for SFTP. Configure the S3 bucket for website hosting. Upload website content by using the SFTP client.

**Correct Answer: C**

Community vote distribution

C (82%)

D (18%)

## Comments

**bjexamprep** **Highly Voted** 2 years, 10 months ago

**Selected Answer: C**

The question here is whether the solution architect can change the requirement. The requirement says very clear about SFTP which cannot be addressed by option C. But the question also gives very clear hint about OAI which cannot be addressed by option D. Option D also doesn't mention anything about CloudFront which is part of the requirement of the question.

So, if the requirement cannot be changed, D is the answer; if the requirement can be changed, C is the answer. But if the requirement can be changed, what's the limitation? That will be a Chaos.

I'm voting C, and curse the question designer.

upvoted 25 times

**Iconique** 2 years, 8 months ago

"The solutions architect must design the most cost-effective and resilient architecture for website hosting to serve as the CloudFront origin." The solution architect is looking for a solution that can fit with CloudFront as origin! So it doesn't matter that option D does not mention CF, CF is part of the solution!

Having a marketing website on-premise clearly indicates having S3 as static content.

AWS Transfer Family is the way to upload files via FTP to S3!

So the answer is D.

Why not C?

User is already uploading content via FTP, option C is eliminating this option for him and forces using the CLI. The solution from C does not meet the requirements of having FTP.

upvoted 8 times

**Tsige** 1 year, 7 months ago

The main reason Option D is not the best answer is because it involves additional complexity and cost compared to Option C. Here's a breakdown

Option D requires setting up AWS Transfer for SFTP, which adds an extra service to manage and incurs additional costs. While it allows the administrator to continue using the SFTP client, it introduces more operational overhead.

Option C leverages Amazon S3 and CloudFront directly, which is simpler and more cost-effective. By using an S3 bucket policy and CloudFront

origin access identity (OAI), you ensure secure access without the need for additional services.

In summary, Option C provides a more streamlined and cost-effective solution for hosting the website and serving as the CloudFront origin.

upvoted 5 times

**cookieMr** Highly Voted 2 years, 11 months ago

**Selected Answer: C**

Hosting the website in a private S3 provides cost-effective and highly available storage for the static website content. By configuring a bucket policy to allow access from a CloudFront OAI, the S3 can be securely accessed only through CloudFront. This ensures that the website content is served through CloudFront while keeping the S3 private. Uploading website content using the AWS CLI allows for easy and efficient content management.

A. Hosting the website on an Lightsail virtual server would introduce additional management overhead and costs compared to using S3 directly for static content hosting.

B. Using an AWS ASG with EC2 instances and an ALB is not necessary for serving static website content. It would add unnecessary complexity and cost.

D. While using AWS Transfer for SFTP allows for SFTP uploads, it introduces additional costs and complexity compared to directly uploading content to an S3 using the AWS CLI. Additionally, hosting the website content in a public S3 may not be desirable from a security standpoint.

upvoted 9 times

**jayessh** Most Recent 1 year, 7 months ago

**Selected Answer: C**

Options A and B involve more complexity and higher costs due to the need for managing instances and load balancers. Option D, while feasible, involves making the S3 bucket public, which is less secure compared to using a private bucket with an OAI.

correct Answer : C

upvoted 2 times

**JohnYu** 1 year, 8 months ago

the option D creating a public S3 bucket could expose the website content to the public, which is a security risk. Using an OAI with a private S3 bucket (as in Option C) is more secure. AWS Transfer for SFTP adds unnecessary complexity unless SFTP specifically is required.

upvoted 2 times

**MatAlves** 1 year, 9 months ago

**Selected Answer: C**

"The company... use Amazon CloudFront" = C is the only option that mentions CloudFront.

upvoted 2 times

**Pr1est** 2 years, 1 month ago

**Selected Answer: C**

This is another great example of how AWS creates crappy tests. Even internal

Tests for employees have so many flaws that people is always creating tickets challenging Questions poorly worded.

upvoted 5 times

**f761d0e** 2 years, 1 month ago

Another reason why A is better than C:

" OAC helps you secure your origins, such as for Amazon S3. We recommend using OAC"

" If your origin is an Amazon S3 bucket configured as a website endpoint, you must set it up with CloudFront as a custom origin. That means you can't use OAC (or OAI)"

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/private-content-restricting-access-to-s3.html>

upvoted 1 times

**djgodzilla** 2 years, 5 months ago

**Selected Answer: D**

you can see in this figure that transfer family framework allows for the data to be available for a broad variety of use cases including content distribution (CF) <https://d1.awsstatic.com/HIW%20SFTP%20Connectors%20v3.920176622d281d0bd087518827314169b496a055.png>

upvoted 2 times

**awsgeek75** 2 years, 4 months ago

Two main problems with D:

It's public S3 behind CloudFront which is not well-architected

Infrequent site updates using SFTP so with S3 it will be cli changes. They don't need fancy transfer for this. Right?

upvoted 2 times

**MiniYang** 2 years, 6 months ago

**Selected Answer: C**

I think the this is a big misleading " SFTP" ( doesn't usally upload) , and it said clearly need Cloudfront and want a cheep solution. So I chose "C"

upvoted 3 times

**rlamberti** 2 years, 7 months ago

**Selected Answer: C**

Transferring via AWS CLI is cheaper than via Transfer Family.  
It is not the best option, but will do the job of uploading the data to S3.

upvoted 2 times

**juanrasus2** 2 years, 7 months ago

I'd go with D. In C there is no mention to S3 bucket being configured for web hosting. Simply adding the Cloudfront distribution and pointing that to the S3 won't work out of the box.

upvoted 2 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: D**

D - SFTP client to upload new documents.

upvoted 2 times

**Guru4Cloud** 2 years, 9 months ago

I changed C. is better than D

upvoted 4 times

**baku98** 2 years, 5 months ago

D is the only one possible.

C cannot be because: In Amazon CloudFront: For Restricting access to an Amazon S3 origin: If your origin is an Amazon S3 bucket configured as a website endpoint, you must set it up with CloudFront as a custom origin. That means you can't use OAC (or OAI).

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/private-content-restricting-access-to-s3.html>

upvoted 2 times

**eugene\_stalker** 3 years ago

**Selected Answer: D**

D - SFTP client to upload new documents.

upvoted 2 times

**bdp123** 3 years, 3 months ago

**Selected Answer: C**

AWS transfer is a cost and doesn't mention using CloudFront

<https://aws.amazon.com/aws-transfer-family/pricing/>

upvoted 4 times

**Yelizaveta** 3 years, 3 months ago

**Selected Answer: C**

If you don't want to disable block public access settings for your bucket but you still want your website to be public, you can create a Amazon CloudFront distribution to serve your static website. For more information, see Use an Amazon CloudFront distribution to serve a static website in the Amazon Route 53 Developer Guide.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/WebsiteAccessPermissionsReqd.html>

upvoted 3 times

**PDR** 3 years, 4 months ago

**Selected Answer: C**

I at first thought D but it is in fact C because

"D: Create a public Amazon S3 bucket. Configure AWS Transfer for SFTP. Configure the S3 bucket for website hosting. Upload website content by using the SFTP client." questions says that the company has decided to use Amazon Cloudfront and this answer does not reference using CF and setting S3 as the Origin

"C. Create a private Amazon S3 bucket. Use an S3 bucket policy to allow access from a CloudFront origin access identity (OAI). Upload website content by using the AWS CLI." - mentions CF and the origin and the AWS CLI does infact support transfer by SFTP (which was the part I originally doubted but this link evidences that it does:

<https://docs.aws.amazon.com/cli/latest/reference/transfer/describe-server.html>

upvoted 5 times

**bullrem** 3 years, 4 months ago

**Selected Answer: D**

Option C, creating a private Amazon S3 bucket and using an S3 bucket policy to allow access from a CloudFront origin access identity (OAI), would not be the most cost-effective solution. While it would allow the company to use Amazon S3 for storage, it would also require additional setup and maintenance of the OAI, which would add additional cost. Additionally, this solution would not allow the use of SFTP client for uploading content which is the current method used by the company.

upvoted 1 times



A company wants to manage Amazon Machine Images (AMIs). The company currently copies AMIs to the same AWS Region where the AMIs were created. The company needs to design an application that captures AWS API calls and sends alerts whenever the Amazon EC2 CreateImage API operation is called within the company's account.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Create an AWS Lambda function to query AWS CloudTrail logs and to send an alert when a CreateImage API call is detected.
- B. Configure AWS CloudTrail with an Amazon Simple Notification Service (Amazon SNS) notification that occurs when updated logs are sent to Amazon S3. Use Amazon Athena to create a new table and to query on CreateImage when an API call is detected.
- C. Create an Amazon EventBridge (Amazon CloudWatch Events) rule for the CreateImage API call. Configure the target as an Amazon Simple Notification Service (Amazon SNS) topic to send an alert when a CreateImage API call is detected. **Most Voted**
- D. Configure an Amazon Simple Queue Service (Amazon SQS) FIFO queue as a target for AWS CloudTrail logs. Create an AWS Lambda function to send an alert to an Amazon Simple Notification Service (Amazon SNS) topic when a CreateImage API call is detected.

**Correct Answer: C**

Community vote distribution

C (77%)

A (15%)

5%

## Comments

**[Removed]** **Highly Voted** 3 years, 6 months ago

**Selected Answer: C**

I'm team C.

<https://docs.aws.amazon.com/AWSEC2/latest/WindowsGuide/monitor-ami-events.html#:~:text=For%20example%2C%20you%20can%20create%20an%20EventBridge%20rule%20that%20detects%20when%20the%20AMI%20creation%20process%20has%20completed%20and%20then%20invokes%20an%20Amazon%20SNS%20topic%20to%20send%20an%20email%20notification%20to%20you.>

upvoted 23 times

**MutiverseAgent** 2 years, 10 months ago

C is correct > <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/monitor-ami-events.html>

upvoted 2 times

**JayBee65** 3 years, 5 months ago

That link contains the exact use case and explains how C can be used.

Option B requires you to send logs to S3 and use Athena, 2 additional services that are not required, so this does not meet the "LEAST operational overhead?" requirement, since these are extra services requiring management.

upvoted 6 times

**Wajif** **Highly Voted** 3 years, 5 months ago

**Selected Answer: A**

Why not A? API calls are already logged in Cloudtrail.

upvoted 15 times

**pentium75** 2 years, 5 months ago

"Least operational overhead" is when the event triggers an action, not when you run a scheduled task that searches logs for the event.

upvoted 8 times

**lofzee** **Most Recent** 2 years ago

**Selected Answer: C**

C all day. Trust me



upvoted 3 times

**ManikRoy** 2 years, 1 month ago

**Selected Answer: C**

It can be done with option A but you'll have to write a lambda function. option C is least operational overhead.

upvoted 3 times

**zinabu** 2 years, 1 month ago

**Selected Answer: C**

Monitor AMI events using Amazon EventBridge is possible and here is the link:

<https://docs.aws.amazon.com/AWSEC2/latest/WindowsGuide/monitor-ami-events.html>

upvoted 2 times

**Mahmoudddddddd** 2 years, 2 months ago

Just took the exam today, most of the questions were from here wish I saw them all to be honest before entering the exam. Anyways, this question was at the exam, I picked option A because as the question stated it wanted two things not one thing only an application that CAPTURES API calls and SEND ALERTS WHENEVER CreateImage API call is made, OPTION C CLEARLY STATES THAT IN THIS CASE IT WILL ONLYYY LOOK FOR CREATEIMAGE API CALL it will not capture other API calls like the lambda in option A would! Am i the only one that is thinks that or what? TBH I am not sure about anything in this question but that is why I did not pick option C during the exam.

upvoted 3 times

**JA2018** 1 year, 6 months ago

Hi Mahmoudddddddd, care to share which answer did you actually choose in your exam? Thanks.

upvoted 1 times

**bujuman** 2 years, 4 months ago

**Selected Answer: D**

On of the requirements is LEAST operational overhead

CloudTrail sends a notification when log files are written to the Amazon S3 bucket. An active account can generate a large number of notifications. If you subscribe with email or SMS, you can receive a large volume of messages. We recommend that you subscribe using Amazon Simple Queue Service (Amazon SQS), which lets you handle notifications programmatically. For more information, see Subscribing a Queue to an Amazon SNS Topic in the Amazon Simple Queue Service Developer Guide.

upvoted 1 times

**Wang87** 2 years, 5 months ago

**Selected Answer: C**

Answer is c.

upvoted 2 times

**farnamjam** 2 years, 5 months ago

**Selected Answer: C**

C is correct

upvoted 2 times

**master9** 2 years, 5 months ago

**Selected Answer: D**

AWS CloudTrail primarily focuses on auditing and recording API calls made in your AWS account. It logs all API requests made via the AWS Management Console, AWS SDKs, command line tools, and other AWS services. This includes the identity of the caller, the time of the API call, the source IP address of the caller, the request parameters, and the response elements returned by the AWS service. This information is useful for security analysis, resource change tracking, and troubleshooting.

upvoted 1 times

**pentium75** 2 years, 5 months ago

But this is not about "auditing and recording", you don't want to create reports who created images during the last year, you want an instant alert when someone creates an image. Thus CloudWatch Events.

upvoted 2 times

**rlamberti** 2 years, 7 months ago

**Selected Answer: C**

"LEAST operational overhead"

Option A involves coding a Lamba. Not good!

Option C seems to be the correct.

upvoted 3 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: C**

Event bridge was built specifically to handle this kind of scenario:

CreateImage API call (Event Source) -> Event bus -> Rules -> Amazon SNS (Event target)

upvoted 7 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: C**

C. Create an Amazon EventBridge (Amazon CloudWatch Events) rule for the CreateImage API call. Configure the target as an Amazon Simple Notification Service (Amazon SNS) topic to send an alert when a CreateImage API call is detected

upvoted 4 times

**Nava702** 2 years, 9 months ago

**Selected Answer: A**

A look like the least overhead option to capture an API call.

upvoted 2 times

**Mia2009687** 2 years, 11 months ago

**Selected Answer: B**

The company needs to design an application that captures AWS API calls and sends alerts whenever the Amazon EC2 CreateImage API operation is called within the company's account.

With option C, it won't "The company needs to design an application that captures AWS API calls". it only sends the "CreateImage API " event. We need to store the AWS API calls as well.

upvoted 1 times

**cookieMr** 2 years, 11 months ago

EventBridge (formerly CloudWatch Events) is a fully managed event bus service that allows you to monitor and respond to events within your AWS environment. By creating an EventBridge rule specifically for the CreateImage API call, you can easily detect and capture this event. Configuring the target as an SNS topic allows you to send an alert whenever a CreateImage API call occurs. This solution requires minimal operational overhead as EventBridge and SNS are fully managed services.

A. While using an Lambda to query CloudTrail logs and send an alert can achieve the desired outcome, it introduces additional operational overhead compared to using EventBridge and SNS directly.

B. Configuring CloudTrail with an SNS notification and using Athena to query on CreateImage API calls would require more setup and maintenance compared to using EventBridge and SNS.

D. Configuring an SQS FIFO queue as a target for CloudTrail logs and using a function to send an alert to an SNS topic adds unnecessary complexity to the solution and increases operational overhead. Using EventBridge and SNS directly is a simpler and more efficient approach.

upvoted 9 times

**pisica134** 2 years, 11 months ago

D makes no sense, FIFO is not required, SQS is not used for sending notifications...C all the way

upvoted 2 times

A company owns an asynchronous API that is used to ingest user requests and, based on the request type, dispatch requests to the appropriate microservice for processing. The company is using Amazon API Gateway to deploy the API front end, and an AWS Lambda function that invokes Amazon DynamoDB to store user requests before dispatching them to the processing microservices.

The company provisioned as much DynamoDB throughput as its budget allows, but the company is still experiencing availability issues and is losing user requests.

What should a solutions architect do to address this issue without impacting existing users?

- A. Add throttling on the API Gateway with server-side throttling limits.
- B. Use DynamoDB Accelerator (DAX) and Lambda to buffer writes to DynamoDB.
- C. Create a secondary index in DynamoDB for the table with the user requests.
- D. Use the Amazon Simple Queue Service (Amazon SQS) queue and Lambda to buffer writes to DynamoDB. **Most Voted**

**Correct Answer: D**

Community vote distribution

D (99%)

## Comments

**pentium75** **Highly Voted** 1 year, 11 months ago

**Selected Answer: D**

A does not meet the "without impacting existing users" requirement

B does not help with writing (DAX caches reads)

C does not help with writing (index could increase read performance only)

D decouples writing from front-end, which is acceptable because it is "an asynchronous API" anyway

upvoted 14 times

**nder** **Highly Voted** 2 years, 9 months ago

**Selected Answer: D**

The key here is "Losing user requests" sqs messages will stay in the queue until it has been processed

upvoted 12 times

**lofzee** **Most Recent** 1 year, 6 months ago

**Selected Answer: D**

D bro. Believe

upvoted 3 times

**Guru4Cloud** 2 years, 3 months ago

**Selected Answer: D**

This solution can handle bursts of incoming requests more effectively and reduce the chances of losing requests due to DynamoDB capacity limitations. The Lambda can be configured to retrieve messages from the SQS and write them to DynamoDB at a controlled rate, allowing DynamoDB to handle the requests within its provisioned capacity. This approach provides resilience to spikes in traffic and ensures that requests are not lost during periods of high demand.

upvoted 5 times

**cookieMr** 2 years, 5 months ago

**Selected Answer: D**

This solution can handle bursts of incoming requests more effectively and reduce the chances of losing requests due to DynamoDB capacity limitations. The Lambda can be configured to retrieve messages from the SQS and write them to DynamoDB at a controlled rate, allowing DynamoDB to handle the requests within its provisioned capacity. This approach provides resilience to spikes in traffic and ensures that requests are not lost during periods of high demand.

A. It limits can help control the request rate, but it may lead to an increase in errors and affect the user experience. Throttling alone may not be sufficient to address the availability issues and prevent the loss of requests.

B. It can improve read performance but does not directly address the availability issues and loss of requests. It focuses on optimizing read operations rather than buffering writes.

C. It may help with querying the user requests efficiently, but it does not directly solve the availability issues or prevent the loss of requests. It is more focused on data retrieval rather than buffering writes.

upvoted 4 times

**studynoplay** 2 years, 6 months ago

**Selected Answer: D**

DAX is for reads

upvoted 4 times

**smartegnine** 2 years, 6 months ago

DAX is not ideal for the following types of applications:

Applications that require strongly consistent reads (or that cannot tolerate eventually consistent reads).

Applications that do not require microsecond response times for reads, or that do not need to offload repeated read activity from underlying tables

Applications that are write-intensive, or that do not perform much read activity.

Applications that are already using a different caching solution with DynamoDB, and are using their own client-side logic for working with that caching solution.

upvoted 3 times

**dark\_firzen** 2 years, 10 months ago

**Selected Answer: D**

D because SQS is the cheapest way. First 1,000,000 requests are free each month.

Question states: "The company provisioned as much DynamoDB throughput as its budget allows"

upvoted 4 times

**Wajif** 2 years, 11 months ago

**Selected Answer: D**

D is more likely to fix this problem as SQS queue has the ability to wait (buffer) for consumer to notify that the request or message has been processed.

upvoted 2 times

**Buruguduystunstugudunstuy** 2 years, 11 months ago

**Selected Answer: D**

To address the issue of lost user requests and improve the availability of the API, the solutions architect should use the Amazon Simple Queue Service (Amazon SQS) queue and Lambda to buffer writes to DynamoDB. Option D (correct answer)

By using an SQS queue and Lambda, the solutions architect can decouple the API front end from the processing microservices and improve the overall scalability and availability of the system. The SQS queue acts as a buffer, allowing the API front end to continue accepting user requests even if the processing microservices are experiencing high workloads or are temporarily unavailable. The Lambda function can then retrieve requests from the SQS queue and write them to DynamoDB, ensuring that all user requests are stored and processed. This approach allows the company to scale the processing microservices independently from the API front end, ensuring that the API remains available to users even during periods of high demand.

upvoted 7 times

**alect096** 2 years, 11 months ago

**Selected Answer: B**

I would go to B : <https://aws.amazon.com/es/blogs/database/amazon-dynamodb-accelerator-dax-a-read-throughwrite-through-cache-for-dynamodb/>

upvoted 1 times

**ruqui** 2 years, 5 months ago

That's wrong. The document you mentioned explained it very clearly:

"Whereas both read-through and write-through caches address read-heavy workloads, a write-back (or write-behind) cache is designed to address write-heavy workloads. Note that DAX is not a write-back cache currently"

upvoted 3 times

**BENICE** 2 years, 11 months ago

D is correct answer

upvoted 2 times

**NikaCZ** 2 years, 11 months ago

**Selected Answer: D**

D. Use the Amazon Simple Queue Service (Amazon SQS) queue and Lambda to buffer writes to DynamoDB.

upvoted 2 times

**career360guru** 2 years, 11 months ago

**Selected Answer: D**

Option D is right answer

upvoted 2 times

**alexfk** 2 years, 12 months ago

Why not B? DAX.

"When you're developing against DAX, instead of pointing your application at the DynamoDB endpoint, you point it at the DAX endpoint, and DAX handles the rest. As a read-through/write-through cache, DAX seamlessly intercepts the API calls that an application normally makes to DynamoDB so that both read and write activity are reflected in the DAX cache."

<https://aws.amazon.com/es/blogs/database/amazon-dynamodb-accelerator-dax-a-read-throughwrite-through-cache-for-dynamodb/>

upvoted 1 times

**ruqui** 2 years, 5 months ago

B is wrong because of this:

"Whereas both read-through and write-through caches address read-heavy workloads, a write-back (or write-behind) cache is designed to address write-heavy workloads. Note that DAX is not a write-back cache currently"

upvoted 2 times

**AgboolaKun** 2 years, 8 months ago

It is not DAX because of the company's budget restriction associated with the DynamoDB. This is a requirement in the question. DynamoDB charge for DAX capacity by the hour and your DAX instances run with no long-term commitments. Please refer to:

[https://aws.amazon.com/dynamodb/pricing/provisioned/#.E2.80.A2\\_DynamoDB\\_Accelerator\\_.28DAX.29](https://aws.amazon.com/dynamodb/pricing/provisioned/#.E2.80.A2_DynamoDB_Accelerator_.28DAX.29)

upvoted 3 times

**akosigengen** 3 years ago

yeah I though the answer is also DAX.

upvoted 1 times

**leonn** 3 years ago

**Selected Answer: D**

Using SQS should be the answer.

upvoted 4 times

**nVizz** 3 years ago

Why not DAX? Could somebody explain?

upvoted 1 times

**Rameez1** 3 years ago

DAX helps in reducing the read loads from DynamoDB, here we need a solution to handle write requests, which is well handled by SQS and Lambda to buffer writes on DynamoDB.

upvoted 5 times

**Buruguduystunstugudunstuy** 2 years, 11 months ago

Using DynamoDB Accelerator (DAX) and Lambda to buffer writes to DynamoDB, may improve the write performance of the system, but it does not provide the same level of scalability and availability as using an SQS queue and Lambda.

Hence, Option B is incorrect.

upvoted 2 times

**jambajuice** 3 years ago

**Selected Answer: D**

Answer d

upvoted 3 times

A company needs to move data from an Amazon EC2 instance to an Amazon S3 bucket. The company must ensure that no API calls and no data are routed through public internet routes. Only the EC2 instance can have access to upload data to the S3 bucket.

Which solution will meet these requirements?

A. Create an interface VPC endpoint for Amazon S3 in the subnet where the EC2 instance is located. Attach a resource policy to the S3 bucket to only allow the EC2 instance's IAM role for access. **Most Voted**

B. Create a gateway VPC endpoint for Amazon S3 in the Availability Zone where the EC2 instance is located. Attach appropriate security groups to the endpoint. Attach a resource policy to the S3 bucket to only allow the EC2 instance's IAM role for access.

C. Run the nslookup tool from inside the EC2 instance to obtain the private IP address of the S3 bucket's service API endpoint. Create a route in the VPC route table to provide the EC2 instance with access to the S3 bucket. Attach a resource policy to the S3 bucket to only allow the EC2 instance's IAM role for access.

D. Use the AWS provided, publicly available ip-ranges.json file to obtain the private IP address of the S3 bucket's service API endpoint. Create a route in the VPC route table to provide the EC2 instance with access to the S3 bucket. Attach a resource policy to the S3 bucket to only allow the EC2 instance's IAM role for access.

**Correct Answer: A**

Community vote distribution

A (56%)

B (44%)

## Comments

**SSASSWS** **Highly Voted** 3 years, 6 months ago

**Selected Answer: A**

I think answer should be A and not B.  
as we cannot "Attach a security groups to a gateway endpoint."  
upvoted 42 times

**A\_New\_Guy** 3 years, 5 months ago

It's possible:

<https://aws.amazon.com/premiumsupport/knowledge-center/connect-s3-vpc-endpoint/>  
upvoted 5 times

**kruasan** 3 years, 1 month ago

No, it's not  
upvoted 3 times

**smartegnine** 3 years ago

Create a security group that allows the resources in your VPC to communicate with the endpoint network interfaces for the VPC endpoint. To ensure that tools such as the AWS CLI can make requests over HTTPS from resources in the VPC to the AWS service, the security group must allow inbound HTTPS traffic.

For Security groups, select the security groups to associate with the endpoint network interfaces for the VPC endpoint. By default, we associate the default security group for the VPC.  
upvoted 2 times

**slackbot** 2 years, 9 months ago

this is valid for interface endpoint, not for gateway endpoint, which option B mentioned  
upvoted 5 times

**markw92** 2 years, 11 months ago

Gateway endpoint must be used as a target in a route table does not use security groups.

upvoted 10 times

**Iconique** 2 years, 8 months ago

Go to console and test it yourself! With Interface Endpoint you can add security groups.

upvoted 2 times

**elmyth** 1 year, 7 months ago

interface VPC endpoint is A))))

upvoted 2 times

**Buruguduystunstugudunstuy** Highly Voted 3 years, 5 months ago

**Selected Answer: B**

The correct solution to meet the requirements is Option B. A gateway VPC endpoint for Amazon S3 should be created in the Availability Zone where the EC2 instance is located. This will allow the EC2 instance to access the S3 bucket directly, without routing through the public internet. The endpoint should also be configured with appropriate security groups to allow access to the S3 bucket. Additionally, a resource policy should be attached to the S3 bucket to only allow the EC2 instance's IAM role for access.

upvoted 38 times

**Buruguduystunstugudunstuy** 3 years, 5 months ago

Option A is incorrect because an interface VPC endpoint for Amazon S3 would not provide a direct connection between the EC2 instance and the S3 bucket.

Option C is incorrect because using the nslookup tool to obtain the private IP address of the S3 bucket's service API endpoint would not provide a secure connection between the EC2 instance and the S3 bucket.

Option D is incorrect because using the ip-ranges.json file to obtain the private IP address of the S3 bucket's service API endpoint is not a secure method to connect the EC2 instance to the S3 bucket.

upvoted 4 times

**mhmt4438** 3 years, 5 months ago

An interface VPC endpoint does provide a direct connection between the EC2 instance and the S3 bucket. It enables private communication between instances in your VPC and resources in other services without requiring an internet gateway, a NAT device, or a VPN connection.

Option A, which recommends creating an interface VPC endpoint for Amazon S3 in the subnet where the EC2 instance is located and attaching a resource policy to the S3 bucket to only allow the EC2 instance's IAM role for access, is the correct solution for the given scenario. It meets the requirement to ensure that no API calls and no data are routed through public internet routes and that only the EC2 instance can have access to upload data to the S3 bucket.

upvoted 7 times

**Omok** 3 years, 4 months ago

In support, see <https://docs.aws.amazon.com/AmazonS3/latest/userguide/privatelink-interface-endpoints.html#types-of-vpc-endpoints-for-s3>

upvoted 8 times

**ChrisG1454** 3 years, 3 months ago

There are two types VPC Endpoint:

Gateway endpoint

Interface endpoint

A Gateway endpoint:

- 1) Helps you to securely connect to Amazon S3 and DynamoDB
- 2) Endpoint serves as a target in your route table for traffic
- 3) Provide access to endpoint (endpoint, identity and resource policies)

An Interface endpoint:

- 1) Help you to securely connect to AWS services EXCEPT FOR Amazon S3 and DynamoDB
- 2) Powered by PrivateLink (keeps network traffic within AWS network)
- 3) Needs a elastic network interface (ENI) (entry point for traffic)

upvoted 35 times

**slackbot** 2 years, 9 months ago

interface endpoint exists for S3 as well

upvoted 10 times

**JustEugen** 9 months, 2 weeks ago

Yes, but only if you need to use the PrivateLink

upvoted 1 times

**JA2018** 1 year, 6 months ago

check this out: <https://www.examttopics.com/discussions/amazon/view/83857-exam-aws-certified-solutions-architect-associate-saa-c02/>

Selected answer: B

So which is the correct answer?!!!!

upvoted 1 times

**111380c** Most Recent 10 months ago

**Selected Answer: B**

interface VPC endpoint 's used for SNS, SQS, KMS, Secrets Manager, but excepted S3

upvoted 1 times

**607d657** 10 months, 2 weeks ago

**Selected Answer: B**

B. Create a gateway VPC endpoint for Amazon S3

upvoted 1 times

**Clouddon** 11 months, 1 week ago

**Selected Answer: B**

B. Create a gateway VPC endpoint for Amazon S3 in the Availability Zone where the EC2 instance is located. Attach appropriate security groups to the endpoint. Attach a resource policy to the S3 bucket to only allow the EC2 instance's IAM role for access.

✓ Why this is the correct solution:

The requirements are:

🔒 No API calls or data should go through the public internet

✓ Only the EC2 instance should be able to upload to S3

🟡 Minimize public exposure and securely control access

upvoted 1 times

**Smith3136** 11 months, 2 weeks ago

**Selected Answer: B**

Only Gateway Endpoint can be attached to S3. Interface Endpoint is not an option with S3

upvoted 1 times

**TEC65** 1 year, 1 month ago

**Selected Answer: B**

Here's why this option is suitable:

Gateway VPC Endpoint for Amazon S3: A gateway VPC endpoint allows you to privately connect your VPC to supported AWS services, such as Amazon S3, without requiring an internet gateway, NAT device, VPN connection, or AWS Direct Connect connection. Traffic between your VPC and the service does not leave the Amazon network.

Resource Policy: By attaching a resource policy to the S3 bucket that only allows access from the EC2 instance's IAM role, you ensure that only authorized instances can access the bucket.

Here's why the other options are not as suitable:

A. Create an interface VPC endpoint for Amazon S3:

Amazon S3 uses gateway VPC endpoints, not interface VPC endpoints. Interface VPC endpoints are used for services that are not supported by gateway endpoints.

upvoted 1 times

**7dcef09** 1 year, 2 months ago

**Selected Answer: B**

Interface Endpoint = AWS PrivateLink

Gateway Endpoint = Supports S3 and DynamoDB

upvoted 1 times

**nadeerm** 1 year, 3 months ago

Reasons for B:

Simplicity: Gateway VPC endpoints are easier to configure and manage compared to interface VPC endpoints.

Cost: Gateway VPC endpoints are free to use, whereas interface VPC endpoints incur costs for each hour and per GB of data processed.

Performance: Gateway VPC endpoints are highly scalable and optimized for S3 and DynamoDB traffic.

upvoted 1 times

**chir827389\_3** 1 year, 4 months ago

**Selected Answer: A**

gateway endpoint does not have any SG.

upvoted 1 times



**certifiedlegend** 1 year, 4 months ago

**Selected Answer: B**

A VPC gateway endpoint is primarily used for accessing specific AWS services like Amazon S3 and DynamoDB privately within your VPC by specifying a route in your route table, while a VPC interface endpoint offers more flexible connectivity to a wider range of AWS services through AWS PrivateLink, allowing access from both within your VPC and from other VPCs using peering or Transit Gateways, typically with a dedicated private IP address within your VPC network; essentially, gateway endpoints are simpler and often free, while interface endpoints provide greater control and may incur additional costs depending on usage.

Gateway endpoints are ideal for simple private access to S3 and DynamoDB, while interface endpoints are better suited for more complex scenarios where you need to access various AWS services from different VPCs or on-premises networks.

upvoted 1 times

**DeliKadir** 1 year, 4 months ago

**Selected Answer: A**

Even though Amazon S3 typically uses a Gateway VPC Endpoint, AWS now provides Interface VPC Endpoints for Amazon S3 as well via AWS PrivateLink....and "Attach a security groups to a gateway endpoint." makes the option B - false

upvoted 1 times

**FlyingHawk** 1 year, 4 months ago

**Selected Answer: A**

B will be correct if removing the "Attach appropriate security group to the endpoint" and gateway point is free of charge.

upvoted 1 times

**MaxMingxing** 1 year, 5 months ago

**Selected Answer: B**

Option A uses an interface VPC endpoint, which is typically used for services that require a private IP address within your VPC. For S3, a gateway VPC endpoint is more appropriate and cost-effective<sup>1</sup>

upvoted 1 times

**hilker1983** 1 year, 5 months ago

**Selected Answer: B**

Interface VPC Endpoint: Interface endpoints are generally used for other AWS services and do not provide the same direct access optimization for S3 that gateway endpoints do.

upvoted 1 times

**LizavetaD** 1 year, 5 months ago

**Selected Answer: B**

In this context, a Gateway VPC Endpoint is the correct choice for S3, as it provides direct, private access to S3 and routes traffic internally within the AWS network

upvoted 1 times

**dipenich** 1 year, 5 months ago

**Selected Answer: B**

Why Option B is Correct:

Gateway VPC Endpoint for Amazon S3:

S3 does not require an interface endpoint; it uses a gateway VPC endpoint.

Gateway endpoints ensure that requests stay within the AWS network, meeting the requirement of no public internet routes.

Resource Policy:

Attaching a bucket policy ensures that only the EC2 instance's IAM role has access to the bucket.

Appropriate Security Controls:

The gateway endpoint can be secured further using policies and security group configurations.

upvoted 1 times

A solutions architect is designing the architecture of a new application being deployed to the AWS Cloud. The application will run on Amazon EC2 On-Demand Instances and will automatically scale across multiple Availability Zones. The EC2 instances will scale up and down frequently throughout the day. An Application Load Balancer (ALB) will handle the load distribution. The architecture needs to support distributed session data management. The company is willing to make changes to code if needed.

What should the solutions architect do to ensure that the architecture supports distributed session data management?

- A. Use Amazon ElastiCache to manage and store session data. **Most Voted**
- B. Use session affinity (sticky sessions) of the ALB to manage session data.
- C. Use Session Manager from AWS Systems Manager to manage the session.
- D. Use the GetSessionToken API operation in AWS Security Token Service (AWS STS) to manage the session.

**Correct Answer: A**

Community vote distribution

A (100%)

## Comments

**Buruguduystunstugudunstuy** **Highly Voted** 2 years, 11 months ago

**Selected Answer: A**

The correct answer is A. Use Amazon ElastiCache to manage and store session data.

In order to support distributed session data management in this scenario, it is necessary to use a distributed data store such as Amazon ElastiCache. This will allow the session data to be stored and accessed by multiple EC2 instances across multiple Availability Zones, which is necessary for a scalable and highly available architecture.

Option B, using session affinity (sticky sessions) of the ALB, would not be sufficient because this would only allow the session data to be stored on a single EC2 instance, which would not be able to scale across multiple Availability Zones.

Options C and D, using Session Manager and the GetSessionToken API operation in AWS STS, are not related to session data management and would not be appropriate solutions for this scenario.

upvoted 34 times

**TariqKipkemei** **Highly Voted** 2 years, 2 months ago

**Selected Answer: A**

Yap agree with go you guys, this is one of the use cases for Amazon ElastiCache.

It was designed to store ephemeral session data to quickly personalize gaming, e-commerce, social media, and online applications with microsecond response times.

<https://aws.amazon.com/elasticache/#:~:text=Store-,ephemeral,-session%20data%20to>

upvoted 5 times

**Cloud\_A** 1 year, 11 months ago

<https://aws.amazon.com/elasticache/#:~:text=Store%20ephemeral%20session%20data%20to%20quickly%20personalize%20gaming%2C%20e%2Dcommerce%2C%20social%20media%2C%20and%20online%20applications%20with%20microsecond%20response%20times>.

upvoted 2 times

**Hkayne** **Most Recent** 1 year, 7 months ago

**Selected Answer: A**

A is the correct answer because it allows to manage distributed sessions

upvoted 3 times

**awsgeek75** 1 year, 11 months ago

**Selected Answer: A**

A is in scope of question as company is willing to make code changes.  
B would have been correct if no code changes were allowed and scaling could be compromised.  
C is wrong technology (cloud management)  
D is also wrong technology (AWS IAM or account management).

upvoted 5 times

**Michael\_Li** 2 years ago

A is correct  
B is not correct as session affinity allow web user stick to a EC2 instance for a period time, that EC2 could go down then the session data will lost, so doesn't fit this use case  
C is wrong as Session Manager is for admins users to manage EC2 CLI access, it's not for web end users  
D is wrong as GetSessionToken API is for use case such as you need to grant user access to a S3 bucket with customized code

upvoted 4 times

**Guru4Cloud** 2 years, 3 months ago

**Selected Answer: A**

The correct answer is A. Use Amazon ElastiCache to manage and store session data.

upvoted 2 times

**cookieMr** 2 years, 5 months ago

**Selected Answer: A**

ElastiCache is a managed in-memory data store service that is well-suited for managing session data in a distributed architecture. It provides high-performance, scalable, and durable storage for session data, allowing multiple EC2 instances to access and share session data seamlessly. By using ElastiCache, the application can offload the session management workload from the EC2 instances and leverage the distributed caching capabilities of ElastiCache for improved scalability and performance.

Option B, using session affinity (sticky sessions) of the ALB, is not the best choice for distributed session data management because it ties each session to a specific EC2 instance. As the instances scale up and down frequently, it can lead to uneven load distribution and may not provide optimal scalability.

Options C and D are not applicable for managing session data. AWS Systems Manager's Session Manager is primarily used for secure remote shell access to EC2 instances, and the AWS STS GetSessionToken API operation is used for temporary security credentials and not session data management.

upvoted 4 times

**cookieMr** 2 years, 5 months ago

ElastiCache is a managed in-memory data store service that is well-suited for managing session data in a distributed architecture. It provides high-performance, scalable, and durable storage for session data, allowing multiple EC2 instances to access and share session data seamlessly. By using ElastiCache, the application can offload the session management workload from the EC2 instances and leverage the distributed caching capabilities of ElastiCache for improved scalability and performance.

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upvoted 4 times

**Abrar2022** 2 years, 5 months ago

**Selected Answer: A**

A. Use Amazon ElastiCache to manage and store session data.  
- Correct. - Session data is managed at the application-layer, and a distributed cache should be used

B. Use session affinity (sticky sessions) of the ALB to manage session data.  
- Wrong. This tightly couples the individual EC2 instances to the session data, and requires additional logic in the ALB. When scale-in happens, the session data stored on individual EC2 instances is destroyed

upvoted 2 times

**techhb** 2 years, 11 months ago

**Selected Answer: A**

correct answer is A as instance are getting up and down.

upvoted 2 times

**inseong** 2 years, 11 months ago

야 근데 210문제는 어딴냐 ..?

upvoted 1 times

**noche** 2 years, 9 months ago

<https://www.examtopycs.com/discussions/amazon/view/94992-exam-aws-certified-solutions-architect-associate-saa-c03/>  
여기 임마

upvoted 1 times

**NikaCZ** 2 years, 12 months ago

**Selected Answer: A**

Amazon ElastiCache to manage and store session data.

upvoted 2 times

**k1kavi1** 2 years, 12 months ago

**Selected Answer: A**

<https://www.examttopics.com/discussions/amazon/view/46412-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 2 times

**Shasha1** 2 years, 12 months ago

A

Amazon ElastiCache to manage and store session data. This solution will allow the application to automatically scale across multiple Availability Zones without losing session data, as the session data will be stored in a cache that is accessible from any EC2 instance. Additionally, using Amazon ElastiCache will enable the company to easily manage and scale the cache as needed, without requiring any changes to the application code. Option C is not correct because, Session Manager from AWS Systems Manager will not provide the necessary support for distributed session data management. Session Manager is a tool for managing and tracking sessions on EC2 instances, but it does not provide a mechanism for storing and managing session data in a distributed environment.

upvoted 4 times

**TelaO** 3 years ago

better justification found here...

<https://www.examttopics.com/discussions/amazon/view/46412-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 4 times

**kmaneith** 3 years ago

why not C?

upvoted 1 times

**leonnnn** 3 years ago

**Selected Answer: A**

ALB sticky session can keep request accessing to the same backend application. But it says "distributed session management" and company "will to change code", so I think A is better

upvoted 4 times

A company offers a food delivery service that is growing rapidly. Because of the growth, the company's order processing system is experiencing scaling problems during peak traffic hours. The current architecture includes the following:

- A group of Amazon EC2 instances that run in an Amazon EC2 Auto Scaling group to collect orders from the application
- Another group of EC2 instances that run in an Amazon EC2 Auto Scaling group to fulfill orders

The order collection process occurs quickly, but the order fulfillment process can take longer. Data must not be lost because of a scaling event.

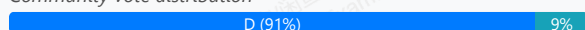
A solutions architect must ensure that the order collection process and the order fulfillment process can both scale properly during peak traffic hours. The solution must optimize utilization of the company's AWS resources.

Which solution meets these requirements?

- A. Use Amazon CloudWatch metrics to monitor the CPU of each instance in the Auto Scaling groups. Configure each Auto Scaling group's minimum capacity according to peak workload values.
- B. Use Amazon CloudWatch metrics to monitor the CPU of each instance in the Auto Scaling groups. Configure a CloudWatch alarm to invoke an Amazon Simple Notification Service (Amazon SNS) topic that creates additional Auto Scaling groups on demand.
- C. Provision two Amazon Simple Queue Service (Amazon SQS) queues: one for order collection and another for order fulfillment. Configure the EC2 instances to poll their respective queue. Scale the Auto Scaling groups based on notifications that the queues send.
- D. Provision two Amazon Simple Queue Service (Amazon SQS) queues: one for order collection and another for order fulfillment. Configure the EC2 instances to poll their respective queue. Create a metric based on a backlog per instance calculation. Scale the Auto Scaling groups based on this metric. **Most Voted**

**Correct Answer: D**

Community vote distribution



## Comments

**TungPham** **Highly Voted** 3 years, 3 months ago

**Selected Answer: D**

When the backlog per instance reaches the target value, a scale-out event will happen. Because the backlog per instance is already 150 messages (1500 messages / 10 instances), your group scales out, and it scales out by five instances to maintain proportion to the target value. Backlog per instance: To calculate your backlog per instance, start with the ApproximateNumberOfMessages queue attribute to determine the length of the SQS queue (number of messages available for retrieval from the queue). Divide that number by the fleet's running capacity, which for an Auto Scaling group is the number of instances in the InService state, to get the backlog per instance. <https://docs.aws.amazon.com/autoscaling/ec2/userguide/as-using-sqs-queue.html>

upvoted 16 times

**n43u435b543ht2b** **Highly Voted** 2 years, 10 months ago

**Selected Answer: D**

C is incorrect as scaling based on the number of "notifications" doesn't make logical sense. This means that both the order collection and fulfillment instances would scale in parallel, but they have clearly said that the collection is processing quickly while the fulfillment is struggling. Therefore, we should scale the pool when there is a backlog building in a respective queue - not just based on the number of incoming requests.

upvoted 14 times

**jerryI** **Most Recent** 1 year, 2 months ago

**Selected Answer: C**

I can understand you guy's explanation on D (the burden is on one queue only not both) but i dont understand how D imply that the metric is monitoring on on side

upvoted 1 times

**satyaamm** 1 year, 3 months ago

**Selected Answer: D**

Since only the order fulfillment requires scaling capacity so D is the most suitable here.

upvoted 1 times

**jatric** 1 year, 11 months ago

**Selected Answer: C**

both have their own queue. Instance processing order will be scale up based on the queue length that collect messages that collected by other queue.

upvoted 1 times

**lofzee** 2 years ago

**Selected Answer: D**

not C as the questions state that only one system is struggling, so C doesnt really solve the problem. D does.

upvoted 2 times

**Uzbekistan** 2 years, 2 months ago

**Selected Answer: D**

Decoupling with Amazon SQS: By using Amazon SQS queues for order collection and order fulfillment, the system can decouple the components, ensuring that orders are not lost, even during scaling events. Orders are queued up and processed in a reliable and scalable manner. Scalability Based on Queue Backlog: By creating a metric based on the backlog per instance calculation, the system can monitor the workload of each instance in the Auto Scaling groups. This allows for dynamic scaling based on the workload, ensuring that additional instances are launched when the backlog increases and terminated when the backlog decreases. Optimization of AWS Resources: This solution optimizes the utilization of AWS resources by dynamically scaling the Auto Scaling groups based on the actual workload, preventing over-provisioning or under-provisioning of instances. It ensures that the system can handle peak traffic efficiently without incurring unnecessary costs.

upvoted 3 times

**bujuman** 2 years, 4 months ago

**Selected Answer: D**

D is the most appropriate response base on <https://docs.aws.amazon.com/autoscaling/ec2/userguide/as-using-sqs-queue.html>

upvoted 2 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: D**

D. Provision two Amazon Simple Queue Service (Amazon SQS) queues: one for order collection and another for order fulfillment. Configure the EC2 instances to poll their respective queue. Create a metric based on a backlog per instance calculation. Scale the Auto Scaling groups based on this metric.

upvoted 5 times

**argl1995** 2 years, 11 months ago

SQS auto-scales by default so I don't think we need to mention it explicitly. Option D should be correct.

upvoted 2 times

**cookieMr** 2 years, 11 months ago

**Selected Answer: D**

A. This approach focuses solely on CPU utilization, which may not accurately reflect the scaling needs of the order collection and fulfillment processes. It does not address the need for decoupling and reliable message processing.

B. While this approach incorporates alarms to trigger additional Auto Scaling groups, it lacks the decoupling and reliable message processing provided by using SQS queues. It may lead to inefficient scaling and potential data loss.

C. Although using SQS queues is a step in the right direction, scaling solely based on queue notifications may not provide optimal resource utilization. It does not consider the backlog per instance and does not allow for fine-grained control over scaling.

Overall, option D, which involves using SQS queues for order collection and fulfillment, creating a metric based on backlog per instance calculation, and scaling the Auto Scaling groups accordingly, is the most suitable solution to address the scaling problems while optimizing resource utilization and ensuring reliable message processing.

upvoted 5 times

**studynoplay** 3 years ago

**Selected Answer: D**

C is incorrect. "based on notifications that the queues send" SQS does not send notification

upvoted 4 times

**mandragon** 3 years, 1 month ago

**Selected Answer: C**

D is not correct because it requires more operational overhead and complexity than option C which is simpler and more cost-effective. It uses the existing queue metrics that are provided by Amazon SQS and does not require creating or publishing any custom metrics. You can use target tracking scaling policies to automatically maintain a desired backlog per instance ratio without having to calculate or monitor it yourself.

upvoted 2 times

**pentium75** 2 years, 5 months ago

"You can use target tracking scaling policies" but you don't with option C. What is "scaling based on notifications that the queues send"? Where do they send these notifications to?

upvoted 1 times

**JayBee65** 3 years, 4 months ago

**Selected Answer: D**

Scale based on queue length

upvoted 3 times

**Rudraman** 3 years, 4 months ago

answer is D.

read question again

upvoted 3 times

**LuckyAro** 3 years, 4 months ago

**Selected Answer: D**

The number of instances in your Auto Scaling group can be driven by how long it takes to process a message and the acceptable amount of latency (queue delay).

The solution is to use a backlog per instance metric with the target value being the acceptable backlog per instance to maintain.

upvoted 2 times

**Aseem8888** 3 years, 4 months ago

**Selected Answer: D**

D is correct

upvoted 2 times

A company hosts multiple production applications. One of the applications consists of resources from Amazon EC2, AWS Lambda, Amazon RDS, Amazon Simple Notification Service (Amazon SNS), and Amazon Simple Queue Service (Amazon SQS) across multiple AWS Regions. All company resources are tagged with a tag name of "application" and a value that corresponds to each application. A solutions architect must provide the quickest solution for identifying all of the tagged components.

Which solution meets these requirements?

- A. Use AWS CloudTrail to generate a list of resources with the application tag.
- B. Use the AWS CLI to query each service across all Regions to report the tagged components.
- C. Run a query in Amazon CloudWatch Logs Insights to report on the components with the application tag.
- D. Run a query with the AWS Resource Groups Tag Editor to report on the resources globally with the application tag.

Most Voted

**Correct Answer: D**

Community vote distribution

D (100%)

## Comments

cookieMr **Highly Voted** 2 years, 11 months ago

**Selected Answer: D**

A is not the quickest solution because CloudTrail primarily focuses on capturing and logging API activity. While it can provide information about resource changes, it may not provide a comprehensive and quick way to identify all the tagged components across multiple services and Regions.

B involves manually querying each service using the AWS CLI, which can be time-consuming and cumbersome, especially when dealing with multiple services and Regions. It is not the most efficient solution for quickly identifying tagged components.

C is focused on analyzing logs rather than directly identifying the tagged components. While CloudWatch Logs Insights can help extract information from logs, it may not provide a straightforward and quick way to gather a consolidated list of all tagged components across different services and Regions.

D is the quickest solution as it leverages the Resource Groups Tag Editor, which is specifically designed for managing and organizing resources based on tags. It offers a centralized and efficient approach to generate a report of tagged components across multiple services and Regions.

upvoted 26 times

satyaamm **Most Recent** 1 year, 3 months ago

**Selected Answer: D**

D is the most suitable here.

upvoted 1 times

JA2018 1 year, 6 months ago

**Selected Answer: D**

To use the AWS Resource Groups Tag Editor to locate resources with specific tags, you can:

1. Open the Tag Editor console
2. Select the AWS regions to search in
3. Choose a resource type
4. Enter a tag key or key and value pair in the Tags fields
5. Select Add or press Enter to finish

<https://docs.aws.amazon.com/tag-editor/latest/userguide/find-resources-to-tag.html#:~:text=To%20find%20resources%20to%20tag,are%20returned%20by%20the%20query.>

upvoted 2 times

TariqKipkemei 2 years, 8 months ago



**Selected Answer: D**

Tags are key and value pairs that act as metadata for organizing your AWS resources  
upvoted 3 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: D**

D. Run a query with the AWS Resource Groups Tag Editor to report on the resources globally with the application tag  
upvoted 4 times

**Bmarodi** 3 years ago

**Selected Answer: D**

A solutions architect can provide the quickest solution for identifying all of the tagged components by running a query with the AWS Resource Groups Tag Editor to report on the resources globally with the application tag, hence the option D is right answer.  
upvoted 3 times

**Dondozy** 3 years, 2 months ago

**Selected Answer: D**

The answer is D  
upvoted 3 times

**sh0811** 3 years, 4 months ago

**Selected Answer: D**

D가 맞습니다.  
upvoted 2 times

**Training4aBetterLife** 3 years, 4 months ago

**Selected Answer: D**

<https://docs.aws.amazon.com/tag-editor/latest/userguide/tagging.html>  
upvoted 4 times

**Rudraman** 3 years, 4 months ago

Answer is D.  
upvoted 2 times

**techhb** 3 years, 4 months ago

**Selected Answer: D**

validated  
<https://docs.aws.amazon.com/tag-editor/latest/userguide/tagging.html>  
upvoted 2 times

**kbaru** 3 years, 4 months ago

**Selected Answer: D**

D is correct  
upvoted 2 times

**waiyu9981** 3 years, 4 months ago

**Selected Answer: D**

<https://www.examttopics.com/discussions/amazon/view/51352-exam-aws-certified-solutions-architect-associate-saa-c02/>  
upvoted 2 times

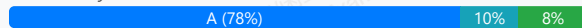
A company needs to export its database once a day to Amazon S3 for other teams to access. The exported object size varies between 2 GB and 5 GB. The S3 access pattern for the data is variable and changes rapidly. The data must be immediately available and must remain accessible for up to 3 months. The company needs the most cost-effective solution that will not increase retrieval time.

Which S3 storage class should the company use to meet these requirements?

- A. S3 Intelligent-Tiering **Most Voted**
- B. S3 Glacier Instant Retrieval
- C. S3 Standard
- D. S3 Standard-Infrequent Access (S3 Standard-IA)

**Correct Answer: A**

Community vote distribution



## Comments

techhb **Highly Voted** 3 years, 4 months ago

**Selected Answer: A**

S3 Intelligent-Tiering monitors access patterns and moves objects that have not been accessed for 30 consecutive days to the Infrequent Access tier and after 90 days of no access to the Archive Instant Access tier.

upvoted 21 times

Devsin2000 3 years, 1 month ago

<https://aws.amazon.com/getting-started/hands-on/getting-started-using-amazon-s3-intelligent-tiering/>

upvoted 6 times

pentium75 **Highly Voted** 2 years, 5 months ago

**Selected Answer: A**

I think it cannot be clearly answered because we know that the 'access pattern is variable and changes rapidly', but ultimately it depends on the total number and volume of accesses. All four options meet the "not increase retrieval time" requirement (even Glacier Instant Retrieval has "the same latency and access time as S3 Standard"). If data would be rarely accessed, B would be cheapest. If it would be constantly accessed, C would be cheapest (we'd pay the Intelligent Tiering fee but it would never move anything to a cheaper tier). Inbetween it would be D.

But I guess the key is Amazon's clear recommendation to use Intelligent Tiering (A) for "unknown or changing access" patterns, which matches the statement in the question.

upvoted 11 times

satyaamm **Most Recent** 1 year, 3 months ago

**Selected Answer: A**

S3 intelligent tiering is the most suitable for unpredictable workloads.

upvoted 1 times

iamroyalty\_k 1 year, 4 months ago

**Selected Answer: A**

S3 Intelligent-Tiering dynamically optimizes storage costs for changing access patterns, ensuring immediate availability and cost-efficiency, which makes it the best fit for the given requirements.

Why not the other options?

B. S3 Glacier Instant Retrieval:

While it provides low retrieval latency, Glacier storage classes are optimized for archival data with infrequent access, not for variable and potentially high-frequency access patterns.

C. S3 Standard:

This provides immediate availability but is more expensive than Intelligent-Tiering for data with unpredictable access patterns.

D. S3 Standard-Infrequent Access (S3 Standard-IA):

S3 Standard-IA is optimized for data that is accessed infrequently but requires immediate availability. However, if the access pattern is variable or frequent, it could lead to higher costs due to retrieval and access fees.

upvoted 1 times

**skybrink** 1 year, 4 months ago

**Selected Answer: D**

A. S3 Intelligent-Tiering:

Intelligent-Tiering automatically moves data between storage tiers based on access patterns. However, it incurs additional monitoring and automation costs. For data with a short, predictable retention period (3 months), S3 Standard-IA is more cost-effective.

B. S3 Glacier Instant Retrieval:

Although Glacier Instant Retrieval offers low-latency access at a lower cost than Standard-IA, it is more suited for archival workloads where retrieval is rare. It is not ideal for workloads with variable access patterns like this one.

C. S3 Standard:

S3 Standard provides low-latency access, but it is more expensive than S3 Standard-IA for infrequently accessed data. It is better suited for frequently accessed data.

upvoted 2 times

**ensbrvsns** 1 year, 9 months ago

**Selected Answer: B**

Instant

upvoted 1 times

**lofzee** 2 years ago

**Selected Answer: A**

unknown / changing access patterns = intelligent tiering. memorise

upvoted 5 times

**ManikRoy** 2 years, 1 month ago

**Selected Answer: A**

Unpredictable access pattern - Intelligent tiering

upvoted 3 times

**MehulKapadia** 2 years, 1 month ago

**Selected Answer: A**

Correct Answer A:

When data access pattern is not known then Intelligent-tiering can help by monitoring data-access pattern and move object internally accordingly and still ensure faster retrieval. Also There is no object retrieval fees/charges for S3 Intelligent Tier(So cost savings).

Option C is not a valid answer because name itself says Infrequent Access(IA): S3 Standard-IA is for data that is accessed less frequently.

upvoted 3 times

**Uzbekistan** 2 years, 2 months ago

**Selected Answer: C**

Immediate Availability: S3 Standard provides immediate access to the data upon upload. This ensures that the exported database is immediately available for other teams to access without any retrieval delays.

Variable Access Pattern: S3 Standard is designed to handle variable access patterns efficiently. It can accommodate rapid changes in access patterns without any impact on performance or latency.

Retention Period: S3 Standard is suitable for storing data that needs to remain accessible for up to 3 months. It does not have any retrieval fees or delays, making it ideal for this scenario where immediate access is required.

upvoted 3 times

**escalibran** 2 years, 3 months ago

Feels like half the scenario or answers are missing. Where's the "remove objects after 90 days"? Intelligent Tiering has an upcharge for the provided convenience - does it even make sense, when objects won't remain long enough to be archived?

Other classes trade storage cost for request costs. Dependent on how often objects are queried, IA might make sense. Even Glacier Instant Retrieval could come out ahead, given minimal access (and it has 90 days minimum storage duration, exact fit for the description).

With no further details provided, this is just throwing darts blindly.

upvoted 1 times

**escalibran** 2 years, 3 months ago

Given just the uncertain access patterns AND limited storage time, I would argue in favor of simple S3 Standard.

If the question mentioned that the pattern of access varies across objects, but is relatively consistent for the individual objects, intelligent tiering may be worth it. Otherwise you just pay more to have objects monitored for Infrequent Access, and then suddenly become popular after being moved.

upvoted 2 times

**MrPCarrot** 2 years, 4 months ago

A is the perfect answer - The S3 access pattern for the data is variable and changes rapidly.

upvoted 2 times

**bujuman** 2 years, 4 months ago

**Selected Answer: A**

With regard to "The S3 access pattern for the data is variable and changes rapidly"

Even though Answer B could fulfill some requirements, Answer A is For long-lived data that have unpredictable access patterns.

upvoted 2 times

**theochan** 2 years, 4 months ago

**Selected Answer: B**

"immediately available" =>

D is not immediately, and for cost  $B < A/C$

upvoted 3 times

**VladanO** 2 years, 7 months ago

Selected Answer: B

<https://aws.amazon.com/s3/storage-classes/glacier/instant-retrieval/>

"Amazon S3 Glacier Instant Retrieval is an archive storage class that delivers the lowest-cost storage for long-lived data that is rarely accessed and requires retrieval in milliseconds"

upvoted 3 times

**ivan\_riqueros12** 2 years, 7 months ago

**Selected Answer: A**

A. El patrón de acceso a los datos es variable y cambia rápidamente = S3 Intelligent-Tiering

upvoted 1 times

**Abdou1604** 2 years, 8 months ago

very important note , S3 Intelligent-Tiering got no retrieval charges

upvoted 4 times

A company is developing a new mobile app. The company must implement proper traffic filtering to protect its Application Load Balancer (ALB) against common application-level attacks, such as cross-site scripting or SQL injection. The company has minimal infrastructure and operational staff. The company needs to reduce its share of the responsibility in managing, updating, and securing servers for its AWS environment.

What should a solutions architect recommend to meet these requirements?

- A. Configure AWS WAF rules and associate them with the ALB. **Most Voted**
- B. Deploy the application using Amazon S3 with public hosting enabled.
- C. Deploy AWS Shield Advanced and add the ALB as a protected resource.
- D. Create a new ALB that directs traffic to an Amazon EC2 instance running a third-party firewall, which then passes the traffic to the current ALB.

**Correct Answer: A**

Community vote distribution

A (78%)

C (22%)

## Comments

**ShinobiGrappler** **Highly Voted** 3 years, 4 months ago

**Selected Answer: C**

C --- Read and understand the question. \*The company needs to reduce its share of responsibility in managing, updating, and securing servers for its AWS environment\* Go with AWS Shield advanced --This is a managed service that includes AWS WAF, custom mitigations, and DDoS insight.

upvoted 19 times

**arjundevops** 3 years, 1 month ago

Brother answer is A, Read the question once again or ask CHATGPT for more in-depth analysis

upvoted 4 times

**rokeus** 2 years, 7 months ago

I agree, both A and C answer the first demand, where A is answer to the technical request, but for reducing responsibility you will need shield advance - meaning C.

upvoted 2 times

**pentium75** 2 years, 5 months ago

No because this is about application-level attacks which requires WAF aka answer A.

upvoted 4 times

**Steve\_4542636** 3 years, 3 months ago

You stated, "This is a managed service that includes AWS WAF, custom mitigations, and DDoS insight." and you are correct. However, the service you would actually have to setup to prevent SQL injection attacks is WAF.

upvoted 11 times

**darn** 3 years, 1 month ago

exactly, that's like saying let's implement Network Firewall Manager to manage WAF, absurd!

upvoted 4 times

**Guru4Cloud** 2 years, 9 months ago

I don't know how this comment gets 11x upvotes.

A. To filter traffic and protect against application attacks like cross-site scripting and SQL injection, the company can use AWS Web Application Firewall with managed rules on the Application Load Balancer. This provides security with minimal infrastructure and operations overhead.

upvoted 30 times

**cookieMr** **Highly Voted** 2 years, 11 months ago

**Selected Answer: A**

By configuring AWS WAF rules and associating them with the ALB, the company can filter and block malicious traffic before it reaches the application. AWS WAF offers pre-configured rule sets and allows custom rule creation to protect against common vulnerabilities like XSS and SQL injection.

Option B does not provide the necessary security and traffic filtering capabilities to protect against application-level attacks. It is more suitable for hosting static content rather than implementing security measures.

Option C is focused on DDoS protection rather than application-level attacks like XSS or SQL injection. While AWS Shield Advanced does not address the specific requirements mentioned in the scenario.

Option D involves maintaining and securing additional infrastructure, which goes against the requirement of reducing responsibility and relying on minimal operational staff.

upvoted 19 times

**satyaamm** **Most Recent** 1 year, 3 months ago

**Selected Answer: A**

AWS WAF is the most suitable for Application Level attacks like Cross Site Scripting and SQL Injection attacks.

upvoted 1 times

**toyaji** 1 year, 9 months ago

**Selected Answer: C**

Most of all, A and C are both available technically, right? So the point of question is not about technical possibility. Its about "share of the responsibility" which is intended to ask of which service provides "Support Plan" - AWS Shield Response Team (SRT)

upvoted 2 times

**ChymKuBoy** 1 year, 11 months ago

**Selected Answer: A**

A for sure

upvoted 2 times

**a7md0** 1 year, 11 months ago

**Selected Answer: A**

AWS Shield Advanced for DDoS Attacks and not SQL injection which is protected by AWS WAF

upvoted 2 times

**ManikRoy** 2 years, 1 month ago

**Selected Answer: A**

AWS WAF with managed rules.

upvoted 2 times

**Solomon2001** 2 years, 1 month ago

Explanation:

Option A:

AWS WAF (Web Application Firewall) provides protection against common web exploits by allowing you to create rules that block common attack patterns such as SQL injection and cross-site scripting (XSS).

By associating AWS WAF rules with the ALB, you can protect your application from these types of attacks without managing, updating, and securing servers yourself.

AWS WAF is a managed service, so it reduces the operational overhead for the company.

Option C:

AWS Shield Advanced provides DDoS protection, but it doesn't include application-level protection like AWS WAF does.

upvoted 4 times

**sandordini** 2 years, 2 months ago

If you read SQL Injection, Cross-site scripting >>> Always look for: WAF

upvoted 2 times

**bujuman** 2 years, 4 months ago

**Selected Answer: A**

This is confusing "The company needs to reduce its share of the responsibility in managing, updating, and securing servers for its AWS environment." But could be achieved when using WAF and AWS managed Rules.

upvoted 3 times

**thewalker** 2 years, 4 months ago

**Selected Answer: A**

A is the answer.

upvoted 2 times

**farnamjam** 2 years, 4 months ago

**Selected Answer: A**

AWS Shield Advanced does not directly protect against XSS (cross-site scripting) or SQL injection attacks. It focuses on defending against Distributed Denial of Service (DDoS) attacks, which aim to overwhelm resources and disrupt availability.

upvoted 2 times

**awsgeek75** 2 years, 5 months ago

**Selected Answer: A**

S makes more sense as Shield Advanced (which actually contains WAF) doesn't provide any additional benefits apart from networks protection. WAF will still have to be configured. So just use WAF to fulfil the requirements.

upvoted 3 times

**pentium75** 2 years, 5 months ago

**Selected Answer: A**

You need to "configure AWS WAF rules and associate them with the ALB" which is A. AWS Shield Advance INTEGRATES with WAF, so you can manage WAF through Shield Advanced, but still you would need to set it up and configure rules, which C does not mention.

upvoted 5 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: A**

AWS WAF helps you protect against common web exploits and bots that can affect availability, compromise security, or consume excessive resources. Protect against vulnerabilities and exploits such as SQL injection or Cross site scripting attacks.

upvoted 6 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: A**

To filter traffic and protect against application attacks like cross-site scripting and SQL injection, the company can use AWS Web Application Firewall with managed rules on the Application Load Balancer. This provides security with minimal infrastructure and operations overhead.

upvoted 4 times

**Undisputed** 2 years, 10 months ago

**Selected Answer: A**

To achieve proper traffic filtering and protect the Application Load Balancer (ALB) against common application-level attacks, such as cross-site scripting (XSS) or SQL injection, while minimizing infrastructure and operational overhead, the company can consider using AWS Web Application Firewall (WAF) with AWS Managed Rules.

upvoted 3 times

A company's reporting system delivers hundreds of .csv files to an Amazon S3 bucket each day. The company must convert these files to Apache Parquet format and must store the files in a transformed data bucket.

Which solution will meet these requirements with the LEAST development effort?

- A. Create an Amazon EMR cluster with Apache Spark installed. Write a Spark application to transform the data. Use EMR File System (EMRFS) to write files to the transformed data bucket.
- B. Create an AWS Glue crawler to discover the data. Create an AWS Glue extract, transform, and load (ETL) job to transform the data. Specify the transformed data bucket in the output step. **Most Voted**
- C. Use AWS Batch to create a job definition with Bash syntax to transform the data and output the data to the transformed data bucket. Use the job definition to submit a job. Specify an array job as the job type.
- D. Create an AWS Lambda function to transform the data and output the data to the transformed data bucket. Configure an event notification for the S3 bucket. Specify the Lambda function as the destination for the event notification.

**Correct Answer: B**

Community vote distribution

B (100%)

## Comments

**Babba** **Highly Voted** 2 years, 11 months ago

**Selected Answer: B**

It looks like AWS Glue allows fully managed CSV to Parquet conversion jobs: <https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/three-aws-glue-etl-job-types-for-converting-data-to-apache-parquet.html>

upvoted 20 times

**awsgeek75** 1 year, 11 months ago

A text book use case: <https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/three-aws-glue-etl-job-types-for-converting-data-to-apache-parquet.html#three-aws-glue-etl-job-types-for-converting-data-to-apache-parquet-epics>

B is the correct answer.

upvoted 2 times

**cookieMr** **Highly Voted** 2 years, 5 months ago

**Selected Answer: B**

AWS Glue is a fully managed ETL service that simplifies the process of preparing and transforming data for analytics. Using AWS Glue requires minimal development effort compared to the other options.

Option A requires more development effort as it involves writing a Spark application to transform the data. It also introduces additional infrastructure management with the EMR cluster.

Option C requires writing and managing custom Bash scripts for data transformation. It requires more manual effort and does not provide the built-in capabilities of AWS Glue for data transformation.

Option D requires developing and managing a custom Lambda for data transformation. While Lambda can handle the transformation, it requires more effort compared to AWS Glue, which is specifically designed for ETL operations.

Therefore, option B provides the easiest and least development effort by leveraging AWS Glue's capabilities for data discovery, transformation, and output to the transformed data bucket.

upvoted 10 times

**satyaamm** **Most Recent** 1 year, 3 months ago

**Selected Answer: B**

AWS Glue is designed for ETL and this scenario.



upvoted 1 times

**iamroyalty\_k** 1 year, 4 months ago

**Selected Answer: B**

AWS Glue offers a serverless, automated, and cost-effective solution with minimal development and operational effort, making it the best choice for this use case.

Why not the other options?

A. Amazon EMR cluster with Apache Spark:

While EMR and Spark can handle this task, it requires more setup, maintenance, and development effort compared to AWS Glue. Managing the cluster introduces operational overhead.

C. AWS Batch with Bash job definition:

Using AWS Batch for this would require creating custom Bash scripts for the transformation and managing jobs. This introduces more complexity and development effort than AWS Glue.

D. AWS Lambda with S3 event notifications:

Lambda is suitable for lightweight, real-time processing. However, converting hundreds of .csv files into Parquet format could exceed Lambda's execution time and resource limits, leading to scalability challenges.

upvoted 1 times

**lofzee** 1 year, 6 months ago

**Selected Answer: B**

AWS Glue and parquet go hand in hand

upvoted 2 times

**zinabu** 1 year, 8 months ago

i will go with answer B cause: You can use AWS Glue to write ETL jobs in a Python shell environment. You can also create both batch and streaming ETL jobs by using Python (PySpark) or Scala in a managed Apache Spark environment.

Apache Parquet is built to support efficient compression and encoding schemes. It can speed up your analytics workloads because it stores data in a columnar fashion. Converting data to Parquet can save you storage space, cost, and time in the longer run.

upvoted 3 times

**Rido4good** 1 year, 11 months ago

D

I think people are forgetting the question says "Low Overhead".

upvoted 1 times

**awsgeek75** 1 year, 10 months ago

Pray tell, how is a Lambda less overhead than B or even A?

upvoted 3 times

**nileeka97** 2 years, 2 months ago

**Selected Answer: B**

Parquet format =====> Amazon Glue

upvoted 4 times

**Guru4Cloud** 2 years, 3 months ago

**Selected Answer: B**

B. Create an AWS Glue crawler to discover the data. Create an AWS Glue extract, transform, and load (ETL) job to transform the data. Specify the transformed data bucket in the output step.

upvoted 3 times

**markw92** 2 years, 5 months ago

Least development effort means lambda. Glue also works but more overhead and cost. A simple lambda like this <https://github.com/ayshaysha/aws-csv-to-parquet-converter/blob/main/csv-parquet-converter.py> can be used to convert as soon as you see files in s3 bucket.

upvoted 4 times

**achevez85** 2 years, 9 months ago

**Selected Answer: B**

<https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/three-aws-glue-etl-job-types-for-converting-data-to-apache-parquet.html>

upvoted 3 times

**Training4aBetterLife** 2 years, 10 months ago

**Selected Answer: B**

S3 provides a single control to automatically encrypt all new objects in a bucket with SSE-S3 or SSE-KMS. Unfortunately, these controls only affect new objects. If your bucket already contains millions of unencrypted objects, then turning on automatic encryption does not make your bucket secure as the unencrypted objects remain.

For S3 buckets with a large number of objects (millions to billions), use Amazon S3 Inventory to get a list of the unencrypted objects, and Amazon S3 Batch Operations to encrypt the large number of old, unencrypted files.

upvoted 2 times

**Training4aBetterLife** 2 years, 10 months ago

Please delete this. I was meaning to place this response on a different question.

upvoted 1 times

**Training4aBetterLife** 2 years, 10 months ago

Versioning:

When you overwrite an S3 object, it results in a new object version in the bucket. However, this will not remove the old unencrypted versions of the object. If you do not delete the old version of your newly encrypted objects, you will be charged for the storage of both versions of the objects.

S3 Lifecycle

If you want to remove these unencrypted versions, use S3 Lifecycle to expire previous versions of objects. When you add a Lifecycle configuration to a bucket, the configuration rules apply to both existing objects and objects added later. C is missing this step, which I believe is what makes B the better choice. B includes the functionality of encrypting the old unencrypted objects via Batch Operations, whereas, Versioning does not address the old unencrypted objects.

upvoted 1 times

**Training4aBetterLife** 2 years, 10 months ago

Please delete this. I was meaning to place this response on a different question.

upvoted 2 times

**Rudraman** 2 years, 10 months ago

ETL = Glue

upvoted 4 times

**Aninina** 2 years, 11 months ago

**Selected Answer: B**

B is the correct answer

upvoted 2 times

**techhb** 2 years, 11 months ago

**Selected Answer: B**

AWS Glue Crawler is for ETL

upvoted 2 times

**kbaru** 2 years, 11 months ago

**Selected Answer: B**

The correct answer is B

upvoted 2 times

**Mamiololo** 2 years, 11 months ago

B is the answer

upvoted 3 times

A company has 700 TB of backup data stored in network attached storage (NAS) in its data center. This backup data need to be accessible for infrequent regulatory requests and must be retained 7 years. The company has decided to migrate this backup data from its data center to AWS. The migration must be complete within 1 month. The company has 500 Mbps of dedicated bandwidth on its public internet connection available for data transfer.

What should a solutions architect do to migrate and store the data at the LOWEST cost?

- A. Order AWS Snowball devices to transfer the data. Use a lifecycle policy to transition the files to Amazon S3 Glacier Deep Archive. **Most Voted**
- B. Deploy a VPN connection between the data center and Amazon VPC. Use the AWS CLI to copy the data from on premises to Amazon S3 Glacier.
- C. Provision a 500 Mbps AWS Direct Connect connection and transfer the data to Amazon S3. Use a lifecycle policy to transition the files to Amazon S3 Glacier Deep Archive.
- D. Use AWS DataSync to transfer the data and deploy a DataSync agent on premises. Use the DataSync task to copy files from the on-premises NAS storage to Amazon S3 Glacier.

**Correct Answer:** A

Community vote distribution

A (98%)

2

## Comments

**voccer** **Highly Voted** 2 years, 11 months ago

**Selected Answer: A**

hundreds of Terabytes => always use Snowball  
upvoted 14 times

**TariqKipkemei** **Highly Voted** 2 years, 8 months ago

**Selected Answer: A**

Terabytes, low costs, limited time = AWS snowball devices  
upvoted 10 times

**sjb009** **Most Recent** 1 year, 2 months ago

**Selected Answer: A**

70TB = AWS Snowball  
upvoted 1 times

**Rcosmos** 1 year, 4 months ago

**Selected Answer: B**

Resumo:  
A Opção B é a solução mais eficiente, pois:

Garante a criptografia automática de todos os objetos futuros com configurações de criptografia padrão no bucket.  
Usa o Inventário do S3 e trabalhos de operações em lote para criptografar objetos existentes sem esforço manual significativo ou movimentação de dados desnecessária.  
upvoted 1 times

**awsgeek75** 2 years, 5 months ago

It took me quite some time to do the mental math for realising that the data can't be transferred in 30 days. Also, note the MBps (Megabits) and not Megabytes. 500Mbps is like 60MBps. That's a lame connection to transfer anything!  
upvoted 7 times

**lofzee** 2 years ago

well.. standard internet connections are measured in Mbps and 500 Mbps is pretty decent to be fair (by english standards). even still at that speed it would take you about 4 months to upload 700 TB. So the only option here is to use snowball.  
Answer is A

upvoted 3 times

**pentium75** 2 years, 5 months ago

**Selected Answer: A**

B and D would use existing 500 mbps Internet connection which cannot transfer more than ca. 160 TB in a month. C would cost a lot, take weeks to deliver, and still not provide more bandwidth. Thus A is the simply the only option, thus also the one with "lowest cost".

upvoted 3 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: A**

A. Order AWS Snowball devices to transfer the data. Use a lifecycle policy to transition the files to Amazon S3 Glacier Deep Archive.

upvoted 2 times

**gosai90786** 2 years, 11 months ago

one DataSync agent can use 10GBps and can setup a bandwidth.

So total time = (700X1000)GB/10GBps = 70000 sec = 19.4 days.

Using Multiple Snowball devices will involve ordering them from AWS, setting them up on your data-center for copy and then incurring the shipping cost for too and fro movement to your AWS cloud.

if time constraint was critical , say 1 week then snowball would have been a viable option. But here we have 30 days, so DataSync will be less costly(takes `19days)

upvoted 2 times

**slackbot** 2 years, 9 months ago

your math is wrong mate, and they have 0.5Gbps connection, not 10GBps

500Mbps = roughly 60MBps

30x24x3600x0.06TB = roughly 155TB

this is way short of 700TB

upvoted 7 times

**JA2018** 1 year, 6 months ago

From Google search: "do I need to pay to ship back aws snow devices back to AWS?"

No, you don't need to pay to ship back an AWS Snowball Edge device to AWS. The device comes with a prepaid UPS shipping label on the E Ink display that contains the correct address for return. You can arrange for UPS to pick up the device or drop it off at a UPS package drop-off facility.

upvoted 1 times

**cookieMr** 2 years, 11 months ago

**Selected Answer: A**

By ordering Snowball devices, the company can transfer the 700 TB of backup data from its data center to AWS. Once the data is transferred to S3, a lifecycle policy can be applied to automatically transition the files from the S3 Standard storage class to the cost-effective Amazon S3 Glacier Deep Archive storage class.

Option B would require continuous data transfer over the public internet, which could be time-consuming and costly given the large amount of data. It may also require significant bandwidth allocation.

Option C would involve additional costs for provisioning and maintaining the dedicated connection, which may not be necessary for a one-time data migration.

Option D could be a viable option, but it may incur additional costs for deploying and managing the DataSync agent.

Therefore, option A is the recommended choice as it provides a secure and efficient data transfer method using Snowball devices and allows for cost optimization through lifecycle policies by transitioning the data to S3 Glacier Deep Archive for long-term storage.

upvoted 4 times

**arjundeovps** 3 years, 1 month ago

A is the correct answer.

even though they have 500mbps internetspeed, it will take around 130days to transfer the data from on premises to AWS

so they have only 1 option which is Snowball devices

upvoted 3 times

**Paras043** 3 years, 2 months ago

**Selected Answer: A**

A is the correct one

upvoted 2 times

**CapJackSparrow** 3 years, 2 months ago

Q: What is AWS Snowball Edge?

AWS Snowball Edge is an edge computing and data transfer device provided by the AWS Snowball service. It has on-board storage and compute power that provides select AWS services for use in edge locations. Snowball Edge comes in two options, Storage Optimized and Compute Optimized to support local data processing and collection in disconnected environments such as ships, windmills, and remote factories. Learn more about its features [here](#).

Q: What happened with the original 50 TB and 80 TB AWS Snowball devices?

The original Snowball devices were transitioned out of service and Snowball Edge Storage Optimized are now the primary devices used for data transfer.

Q: Can I still order the original Snowball 50 TB and 80 TB devices?

No. For data transfer needs now, please select the Snowball Edge Storage Optimized devices.

upvoted 2 times

**vherman** 3 years, 3 months ago

**Selected Answer: A**

Snowball

upvoted 1 times

**KZM** 3 years, 3 months ago

9 Snowball devices are needed to migrate the 700TB of data.

upvoted 3 times

**KZM** 3 years, 3 months ago

700TB of Data can not be transferred through a 500Mbps link within one month.

Total data that can be transferred in one month = bandwidth x time  
= (500 Mbps / 8 bits per byte) x (30 days x 24 hours x 3600 seconds per hour)  
= 648,000 GB or 648 TB

This is calculated theoretically with the maximum available situation. Due to a number of factors, the actual total transferred Data may be less than 645 TB.

upvoted 4 times

**mandragon** 3 years, 1 month ago

Good thinking. Agree with the solution. Only the calculation is wrong. It should give 162tb as a result

upvoted 5 times

**Rudraman** 3 years, 4 months ago

Snow ball Devices the answe is AAAAA.

upvoted 3 times

**wmp7039** 3 years, 4 months ago

A is incorrect as DC is an expensive option. Correct answer should be C as the company already has 500Mbps that can be used for data transfer. By consuming all the available internet bandwidth, data transfer will complete in 3 hours 6 mins - <https://www.omnicalculator.com/other/data-transfer>

upvoted 1 times

**wmp7039** 3 years, 4 months ago

Ignore please, miscalculated time to transfer, it will take 129 days and will breach the 1 month requirement. A is correct.

upvoted 6 times

**kbaru** 3 years, 4 months ago

**Selected Answer: A**

A is correct

upvoted 2 times

A company has a serverless website with millions of objects in an Amazon S3 bucket. The company uses the S3 bucket as the origin for an Amazon CloudFront distribution. The company did not set encryption on the S3 bucket before the objects were loaded. A solutions architect needs to enable encryption for all existing objects and for all objects that are added to the S3 bucket in the future.

Which solution will meet these requirements with the LEAST amount of effort?

- A. Create a new S3 bucket. Turn on the default encryption settings for the new S3 bucket. Download all existing objects to temporary local storage. Upload the objects to the new S3 bucket.
- B. Turn on the default encryption settings for the S3 bucket. Use the S3 Inventory feature to create a .csv file that lists the unencrypted objects. Run an S3 Batch Operations job that uses the copy command to encrypt those objects. **Most Voted**
- C. Create a new encryption key by using AWS Key Management Service (AWS KMS). Change the settings on the S3 bucket to use server-side encryption with AWS KMS managed encryption keys (SSE-KMS). Turn on versioning for the S3 bucket.
- D. Navigate to Amazon S3 in the AWS Management Console. Browse the S3 bucket's objects. Sort by the encryption field. Select each unencrypted object. Use the Modify button to apply default encryption settings to every unencrypted object in the S3 bucket.

**Correct Answer: B**

Community vote distribution

B (92%)

7%

## Comments

**Parsons** **Highly Voted** 2 years, 11 months ago

**Selected Answer: B**

Step 1: S3 inventory to get object list  
Step 2 (If needed): Use S3 Select to filter  
Step 3: S3 object operations to encrypt the unencrypted objects.

On the going object use default encryption.  
upvoted 17 times

**Parsons** 2 years, 11 months ago

Useful ref link: <https://aws.amazon.com/blogs/storage/encrypting-objects-with-amazon-s3-batch-operations/>  
upvoted 10 times

**cookieMr** **Highly Voted** 2 years, 5 months ago

**Selected Answer: B**

By enabling default encryption settings on the S3, all newly added objects will be automatically encrypted. To encrypt the existing objects, the S3 Inventory feature can be used to generate a list of unencrypted objects. Then, an S3 Batch Operations job can be executed to copy those objects while applying encryption.

A. This solution involves creating a new S3 and manually downloading and uploading all existing objects. It requires significant effort and time to transfer millions of objects, making it a less efficient solution.

C. While enabling SSE with AWS KMS is a valid approach to encrypt objects in an S3, it does not address the requirement of encrypting existing objects. It only applies encryption to new objects added to the bucket.

D. Manually modifying each object in the S3 to apply default encryption settings is a labor-intensive and error-prone process. It would require individually selecting and modifying each unencrypted object, which is impractical for a large number of objects.  
upvoted 15 times

**iamroyalty\_k** **Most Recent** 1 year, 4 months ago

**Selected Answer: B**

Option B uses AWS tools like S3 Default Encryption, Inventory, and Batch Operations to enable encryption for both existing and future objects with minimal effort and automation, making it the best solution.

Why not the other options?

A. Create a new S3 bucket and migrate objects:

This involves downloading and re-uploading all objects, which is highly manual, time-consuming, and prone to errors. It also incurs additional data transfer costs.

C. Use SSE-KMS and turn on versioning:

While enabling SSE-KMS for future objects is valid, this does not encrypt the existing objects unless explicitly copied or rewritten, requiring additional manual effort.

D. Browse and modify objects manually:

Manually selecting and encrypting each object is impractical for a bucket with millions of objects, as it is labor-intensive and time-consuming.

upvoted 1 times

**lofzee** 1 year, 6 months ago

**Selected Answer: B**

to be fair all these options take a hell a lot of work to do but i think the least amount of effort is B.

<https://aws.amazon.com/blogs/storage/encrypting-objects-with-amazon-s3-batch-operations/>

upvoted 4 times

**pentium75** 1 year, 11 months ago

**Selected Answer: B**

A - Extreme amount of effort

B - Should work

C - SSE-KMS is not "least amount of effort" compared to SSE-S3; Turning versioning is not required to achieve the result but on the contrary, it will cause the non-encrypted files to remain as old versions even if you encrypt them in the future.

D - Even more effort as A

upvoted 3 times

**foha2012** 1 year, 10 months ago

.csv for millions of objects ?? C looks simpler.

upvoted 2 times

**foha2012** 1 year, 10 months ago

B doesnt look like least amount of effort

upvoted 1 times

**CapJackSparrow** 2 years, 9 months ago

**Selected Answer: B**

B...

<https://catalog.us-east-1.prod.workshops.aws/workshops/05f16f1a-0bbf-45a7-a304-4fcd7fca3d1f/en-US/s3-track/module-2>

You're welcome

upvoted 4 times

**bdp123** 2 years, 10 months ago

**Selected Answer: B**

Amazon S3 now configures default encryption on all existing unencrypted buckets to apply server-side encryption with S3 managed keys (SSE-S3) as the base level of encryption for new objects uploaded to these buckets. Objects that are already in an existing unencrypted bucket won't be automatically encrypted.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/default-encryption-faq.html>

upvoted 4 times

**Yelizaveta** 2 years, 10 months ago

**Selected Answer: B**

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/batch-ops-copy-example-bucket-key.html>

upvoted 2 times

**aakashkumar1999** 2 years, 10 months ago

**Selected Answer: B**

B is the correct answer

upvoted 2 times

**Val182** 2 years, 10 months ago

**Selected Answer: B**

B 100%

<https://spin.atomicobject.com/2020/09/15/aws-s3-encrypt-existing-objects/>

upvoted 2 times

**LuckyAro** 2 years, 10 months ago

**Selected Answer: A**

Why is no one discussing A ? I think A can also achieve the required results. B is the most appropriate answer though.

upvoted 1 times

**pentium75** 1 year, 11 months ago

Downloading and uploading "millions of objects" is surely not "least amount of effort", thus does not meet the requirements.

upvoted 2 times

**Training4aBetterLife** 2 years, 10 months ago

**Selected Answer: B**

S3 provides a single control to automatically encrypt all new objects in a bucket with SSE-S3 or SSE-KMS. Unfortunately, these controls only affect new objects. If your bucket already contains millions of unencrypted objects, then turning on automatic encryption does not make your bucket secure as the unencrypted objects remain.

For S3 buckets with a large number of objects (millions to billions), use Amazon S3 Inventory to get a list of the unencrypted objects, and Amazon S3 Batch Operations to encrypt the large number of old, unencrypted files.

upvoted 4 times

**Training4aBetterLife** 2 years, 10 months ago

Versioning:

When you overwrite an S3 object, it results in a new object version in the bucket. However, this will not remove the old unencrypted versions of the object. If you do not delete the old version of your newly encrypted objects, you will be charged for the storage of both versions of the objects.

S3 Lifecycle

If you want to remove these unencrypted versions, use S3 Lifecycle to expire previous versions of objects. When you add a Lifecycle configuration to a bucket, the configuration rules apply to both existing objects and objects added later. C is missing this step, which I believe is what makes B the better choice. B includes the functionality of encrypting the old unencrypted objects via Batch Operations, whereas, Versioning does not address the old unencrypted objects.

upvoted 2 times

**Training4aBetterLife** 2 years, 10 months ago

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upvoted 2 times

**Training4aBetterLife** 2 years, 10 months ago

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upvoted 1 times

**Training4aBetterLife** 2 years, 10 months ago

Please remove duplicate response as I was meaning to submit a voting comment.

upvoted 1 times

**John\_Zhuang** 2 years, 10 months ago

**Selected Answer: B**

C is wrong. Even though you turn on the SSE-KMS with a new key, the existing objects are still yet to be encrypted. They still need to be manually encrypted by AWS batch

upvoted 3 times



**pentium75** 1 year, 11 months ago

And as in C you "turn on versioning", the old, unencrypted objects will be kept.  
upvoted 2 times

**LuckyAro** 2 years, 10 months ago

**Selected Answer: B**

<https://spin.atomicobject.com/2020/09/15/aws-s3-encrypt-existing-objects/>  
upvoted 2 times

**Aninina** 2 years, 11 months ago

**Selected Answer: C**

C is the answer  
upvoted 2 times

**pentium75** 1 year, 11 months ago

Why? This does not include a step to encrypt existing objects, and by turning on versioning you will keep the unencrypted versions forever.  
upvoted 1 times

**techhb** 2 years, 11 months ago

**Selected Answer: B**

Agree with Parsons  
upvoted 2 times

A company runs a global web application on Amazon EC2 instances behind an Application Load Balancer. The application stores data in Amazon Aurora. The company needs to create a disaster recovery solution and can tolerate up to 30 minutes of downtime and potential data loss. The solution does not need to handle the load when the primary infrastructure is healthy.

What should a solutions architect do to meet these requirements?

A. Deploy the application with the required infrastructure elements in place. Use Amazon Route 53 to configure active-passive failover. Create an Aurora Replica in a second AWS Region. **Most Voted**

B. Host a scaled-down deployment of the application in a second AWS Region. Use Amazon Route 53 to configure active-active failover. Create an Aurora Replica in the second Region.

C. Replicate the primary infrastructure in a second AWS Region. Use Amazon Route 53 to configure active-active failover. Create an Aurora database that is restored from the latest snapshot.

D. Back up data with AWS Backup. Use the backup to create the required infrastructure in a second AWS Region. Use Amazon Route 53 to configure active-passive failover. Create an Aurora second primary instance in the second Region.

**Correct Answer: A**

Community vote distribution

A (72%)

D (28%)

## Comments

**Parsons** **Highly Voted** 3 years, 4 months ago

**Selected Answer: A**

A is correct.

- "The solution does not need to handle the load when the primary infrastructure is healthy." => Should use Route 53 Active-Passive ==> Exclude B, C

- D is incorrect because "Create an Aurora second primary instance in the second Region.", we need to create an Aurora Replica enough.

upvoted 31 times

**Parsons** 3 years, 4 months ago

Ref link: <https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/dns-failover-types.html>

upvoted 6 times

**diabloexodia** **Highly Voted** 2 years, 10 months ago

**Selected Answer: A**

Anything that is not instant recovery is active - passive.

In active -passive we have :

1. Aws Backup(least op overhead) - RTO/RPO = hours
2. Pilot Light ( Basic Infra is already deployed, but needs to be fully implemented) -RTO/RPO = 10's of minutes.
3. Warm Standby- (Basic infra + runs small loads ( might need to add auto scaling) -RTO/RPO= minutes
4. ( ACTIVE -ACTIVE ) : Multi AZ option : instant

here we can tolerate 30 mins

hence B,D are incorrect. AWS backup is in hours, hence D is incorrect .

therefore A

upvoted 24 times

**pentium75** 2 years, 5 months ago

A does not create the infrastructure in the DR region though.

upvoted 2 times

**bora4motion** **Most Recent** 1 year, 1 month ago

**Selected Answer: A**

A seems logic to me. D seems legit as well but it does not guarantee everything will be live in 30 minutes.

upvoted 1 times

**FlyingHawk** 1 year, 6 months ago

**Selected Answer: A**

The Aurora Global Database would be ideal for a production system requiring near-instantaneous failover, however, it was not in the options. Since the option D may not meet the 30-minute tolerance, I choose option A- Aurora Replica-based disaster recovery although replication is more designed for reading traffic, it can still be used for failover based on this doc <https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Replication.html>.

upvoted 2 times

**ensbrvsns** 1 year, 9 months ago

**Selected Answer: D**

D is right

upvoted 1 times

**jatric** 1 year, 11 months ago

**Selected Answer: D**

confused with A and D but D looks more promising when it says doesn't need to handle the load when primary infrastructure is healthy

upvoted 2 times

**lofzee** 2 years ago

**Selected Answer: D**

I went for D as the wording of A is weird....

D seems most plausible

upvoted 2 times

**Jazz888** 2 years ago

A

For those you are choosing D, I have a question for you. How do you guarantee the provisioning of resources will take less than 30 min through AWS Backup?

upvoted 3 times

**ManikRoy** 2 years, 1 month ago

By excluding other options you can choose A but this option is incomplete as it doesn't mention deploying/recovering the application in secondary region.

upvoted 1 times

**MrPCarrot** 2 years, 4 months ago

A is perfect - Active-Passive Failover: Use this failover configuration when you want a primary group of resources to be available the majority of the time and you want a secondary group of resources to be on standby in case all of the primary resources become unavailable.

upvoted 2 times

**MrPCarrot** 2 years, 4 months ago

A is perfect

upvoted 1 times

**farnamjam** 2 years, 4 months ago

**Selected Answer: A**

Here's why the other options aren't as suitable:

B. Active-active failover: Incur higher costs due to running both infrastructures simultaneously and introduces complexity in managing traffic distribution.

C. Restoring from snapshot: Could take longer than 30 minutes to recover, exceeding the company's downtime tolerance.

D. AWS Backup: Dependent on backup and restore times, potentially exceeding the 30-minute recovery window.

upvoted 2 times

**pentium75** 2 years, 5 months ago

**Selected Answer: D**

Not A - does not mention a second region for the infrastructure elements. Also, you cannot really "create an Aurora Replica in a second AWS Region", replicas must be in same region unless using Aurora Global Database (which is not mentioned)

Not B - would send half of the traffic to the DR region

Not C - this could send traffic to the DR instance even when the primary instance is healthy

D - the wording "Aurora second primary instance" is a bit strange, but still a "primary instance" is what we would need in the other region. We would still need to establish replication between the databases (like binlog), or restore a snapshot before failover, but in general this option could meet the 30 minute RTO/RPO requirement.

upvoted 5 times

**lofzee** 2 years ago

the question doesn't state Aurora MySQL but you can set up cross-region replicas for Aurora MySQL, but not PostgreSQL. the question only says Amazon Aurora, so it's left a bit open as Aurora is either MySQL or PostgreSQL. this is without using Global Database.

tbh i think this question and answers are well off... it is a dump after all.

upvoted 2 times

**awsgeek75** 2 years, 5 months ago

**Selected Answer: D**

I agreed with D as the requirements of 30 min downtime and potential data loss and no load consideration when primary instance is healthy. It makes D more feasible than A. Aurora-Replica is normally used for active-active failovers. Be frugal!

upvoted 4 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: A**

'Can tolerate up to 30 minutes of downtime and potential data loss' rules out any option with 'active-active'. Leaves D and A. D is convoluted. Leaving A.

upvoted 6 times

**cookieMr** 2 years, 11 months ago

**Selected Answer: A**

A. involves deploying the application and infrastructure elements in the primary Region. An Aurora Replica is created in a second Region to serve as the standby database. Route 53 is configured with active-passive failover, directing traffic to the primary Region by default. In the event of a disaster, Route 53 can automatically redirect traffic to the standby Region, minimizing downtime. Data loss may occur up to the point of the last replication to the standby Region, which can be within the defined tolerance of 30 minutes.

Option B, is not necessary in this case as the solution does not need to handle the load when the primary infrastructure is healthy, and it may involve higher complexity and costs.

Option C, may introduce additional complexity and potential data loss, as the standby database might not be up-to-date with the primary database.

Option D, may be suitable for backup and recovery scenarios but may not provide the required failover and downtime tolerance specified in the requirements.

upvoted 3 times

**antropaws** 3 years ago

**Selected Answer: D**

I vote D, because option A is not highly available. In option A, you can't configure active-passive failover because you haven't created a backup infrastructure.

upvoted 2 times

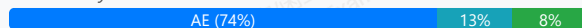
A company has a web server running on an Amazon EC2 instance in a public subnet with an Elastic IP address. The default security group is assigned to the EC2 instance. The default network ACL has been modified to block all traffic. A solutions architect needs to make the web server accessible from everywhere on port 443.

Which combination of steps will accomplish this task? (Choose two.)

- A. Create a security group with a rule to allow TCP port 443 from source 0.0.0.0/0. **Most Voted**
- B. Create a security group with a rule to allow TCP port 443 to destination 0.0.0.0/0.
- C. Update the network ACL to allow TCP port 443 from source 0.0.0.0/0.
- D. Update the network ACL to allow inbound/outbound TCP port 443 from source 0.0.0.0/0 and to destination 0.0.0.0/0.
- E. Update the network ACL to allow inbound TCP port 443 from source 0.0.0.0/0 and outbound TCP port 32768-65535 to destination 0.0.0.0/0. **Most Voted**

**Correct Answer:** AE

Community vote distribution



## Comments

**Parsons** **Highly Voted** 3 years, 4 months ago

**Selected Answer: AE**

A, E is perfect the combination. To be more precise, We should add outbound with "outbound TCP port 32768-65535 to destination 0.0.0.0/0." as an ephemeral port due to the stateless of NACL.

upvoted 20 times

**oguzbeliren** 2 years, 10 months ago

What is the main reason that you are using the TCP port 32768-65535? In the question, it doesn't ask you any requirement about it.

upvoted 5 times

**MohammadTofic8787** 2 years, 8 months ago

i Think AD because acl is stateless we must open the port outbound and inbound , in option c we only open 443 on inbound

upvoted 3 times

**bora4motion** 1 year, 1 month ago

with d you only allow replies using port 443 which is wrong.

upvoted 1 times

**MohammadTofic8787** 2 years, 8 months ago

i Think AD because acl is stateless we must open the port outbound and inbound , in option E we only open 443 on inbound

upvoted 3 times

**pentium75** **Highly Voted** 2 years, 5 months ago

**Selected Answer: E**

For me it's grammatically unclear whether "port 443" and "port 32768-65535" in answers D and E are referring to the source or destination ports of the outbound traffic. If source ports then it would be D. If destination ports (which seems more likely) then it's E.

"On Windows, the ephemeral port range is usually from 49152 to 65535.

On Linux, it is often from 32768 to 61000."

Thus 32768-65535 would cover both Windows and Linux.

upvoted 10 times

**Omariox** **Most Recent** 1 year, 8 months ago

**Selected Answer: AD**

Option A: Creating a security group that allows inbound traffic on TCP port 443 from all sources (0.0.0.0/0) ensures that the web server can accept incoming HTTPS requests.

Option D: Updating the network ACL to allow inbound traffic on TCP port 443 from all sources (0.0.0.0/0) allows the requests to reach the EC2 instance. Additionally, it is necessary to allow outbound traffic on TCP port 443 to enable responses to clients, which is crucial for HTTPS communication.

upvoted 3 times

**bora4motion** 1 year, 1 month ago

D is wrong: when the ec2 replies back it will use a random port not the same port (443) for the incoming traffic. AE.

upvoted 1 times

**srinibas.velumuri** 1 year, 8 months ago

Higher priority NACL to allow inbound and outbound traffic on 443 with take the precedence over default blocked NACL

upvoted 1 times

**ChinthaGurumurthi** 1 year, 10 months ago

**Selected Answer: AD**

AD

How can E be the answer. How can we assure that the port range is definitely from the given port range in the option E?

upvoted 1 times

**lofzee** 2 years ago

**Selected Answer: AE**

Security group only needs inbound rules.

ACL needs inbound and outbound.. Outbound traffic is going to be dynamic ports. Answer is A and E

upvoted 4 times

**sidharthwader** 2 years, 3 months ago

AE

Security group is a stateful resource and can understand to allow traffic from source 0.0.0.0/0 with port 443 but ACL is stateless so traffic that is allowed inside the network we must configure the same to go outside the network as well.

upvoted 2 times

**awsgeek75** 2 years, 5 months ago

**Selected Answer: AE**

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html#nacl-basics>

"NACLs are stateless, which means that information about previously sent or received traffic is not saved. If, for example, you create a NACL rule to allow specific inbound traffic to a subnet, responses to that traffic are not automatically allowed. This is in contrast to how security groups work.

Security groups are stateful, which means that information about previously sent or received traffic is saved. If, for example, a security group allows inbound traffic to an EC2 instance, responses are automatically allowed regardless of outbound security group rules."

A fulfils the security group requirement

E is the only option that explicitly covers outbound traffic and ports.

D covers outbound destination but given that all traffic is blocked (as per the question) this won't work

upvoted 6 times

**[Removed]** 2 years, 6 months ago

**Selected Answer: AC**

For typical web server scenarios, such as serving content over HTTPS (port 443), you generally do not need to explicitly open outbound ports in the network ACL (NACL) for the return traffic.

upvoted 1 times

**pentium75** 2 years, 5 months ago

But NACLs are stateless. "The default network ACL has been modified to block all traffic"; if you don't allow any outbound traffic then the web server won't be able to reply to clients.

upvoted 2 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: AE**

ACL is stateless. you have to define both inbound and outbound rules.

upvoted 3 times

**MohammadTofic8787** 2 years, 8 months ago

i Think AD because acl is stateless we must open the port outbound and inbound , in option c we only open 443 on inbound

upvoted 2 times

**MohammadTofic8787** 2 years, 8 months ago

i Think AD because acl is stateless we must open the port outbound and inbound , in option D we only open 443 on inbound  
upvoted 1 times

**MohammadTofic8787** 2 years, 8 months ago  
please admin delete this , sorry  
upvoted 1 times

**MohammadTofic8787** 2 years, 8 months ago  
please admin delete this , sorry  
upvoted 1 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: AE**

A, E is perfect the combination. To be more precise, We should add outbound with "outbound TCP port 32768-65535 to destination 0.0.0.0/0." as an ephemeral port due to the stateless of NACL.  
upvoted 4 times

**beginnercloud** 2 years, 9 months ago

**Selected Answer: AE**

AE is the best answer here, but in reality, E is not good enough. Here, it says that the client chooses the ephemeral port, and it can start from 1024. Only Linux clients have the range starting at 32768 <https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html#nacl-ephemeral-ports>. Unless the destination advertises the ephemeral ports, which I don't think is the case  
upvoted 2 times

**pentium75** 2 years, 5 months ago

"On Windows, the ephemeral port range is usually from 49152 to 65535.  
On Linux, it is often from 32768 to 61000."  
Combined: 32768 - 65535 ...  
upvoted 3 times

**Thornessen** 2 years, 10 months ago

**Selected Answer: AE**

AE is the best answer here, but in reality, E is not good enough.

Here, it says that the client chooses the ephemeral port, and it can start from 1024. Only Linux clients have the range starting at 32768 <https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html#nacl-ephemeral-ports>

Unless the destination advertises the ephemeral ports, which I don't think is the case  
upvoted 3 times

**Abrar2022** 3 years ago

32768-65535 ports Allows outbound IPv4 responses to clients on the internet (for example, serving webpages to people visiting the web servers in the subnet).  
upvoted 2 times

**WherecanIstart** 3 years, 3 months ago

**Selected Answer: AE**

NACL blocks outgoing traffic since it is infact stateless..Option E allows outbound traffic from ephemeral ports going outside of the VPC back to the web.  
upvoted 3 times

**Brak** 3 years, 3 months ago

It can't be C, since the current NACL blocks all traffic, including outbound. Need to allow outbound traffic through the NACL.  
But E is a bad answer, since ephemeral ports start at 1024, not 32768.  
upvoted 2 times

A company's application is having performance issues. The application is stateful and needs to complete in-memory tasks on Amazon EC2 instances. The company used AWS CloudFormation to deploy infrastructure and used the M5 EC2 instance family. As traffic increased, the application performance degraded. Users are reporting delays when the users attempt to access the application.

Which solution will resolve these issues in the MOST operationally efficient way?

- A. Replace the EC2 instances with T3 EC2 instances that run in an Auto Scaling group. Make the changes by using the AWS Management Console.
- B. Modify the CloudFormation templates to run the EC2 instances in an Auto Scaling group. Increase the desired capacity and the maximum capacity of the Auto Scaling group manually when an increase is necessary.
- C. Modify the CloudFormation templates. Replace the EC2 instances with R5 EC2 instances. Use Amazon CloudWatch built-in EC2 memory metrics to track the application performance for future capacity planning.
- D. Modify the CloudFormation templates. Replace the EC2 instances with R5 EC2 instances. Deploy the Amazon CloudWatch agent on the EC2 instances to generate custom application latency metrics for future capacity planning. **Most Voted**

**Correct Answer: D**

Community vote distribution

D (94%)

4%

## Comments

**Parsons** **Highly Voted** 3 years, 4 months ago

**Selected Answer: D**

D is the correct answer.

"in-memory tasks" => need the "R" EC2 instance type to archive memory optimization. So we are concerned about C & D. Because EC2 instances don't have built-in memory metrics to CW by default. As a result, we have to install the CW agent to archive the purpose.  
upvoted 39 times

**Babba** **Highly Voted** 3 years, 4 months ago

**Selected Answer: D**

It's D, EC2 do not provide by default memory metrics to CloudWatch and require the CloudWatch Agent to be installed on the monitored instances : <https://aws.amazon.com/premiumsupport/knowledge-center/cloudwatch-memory-metrics-ec2/>  
upvoted 14 times

**surajkrishnamurthy** **Most Recent** 1 year, 3 months ago

**Selected Answer: D**

Answer is most likely "D" because of the following ...

- T3 EC2 instances > low cost burstable general purpose instance type that provide a baseline level of CPU performance with the ability to burst CPU usage
- M5 EC2 instance > M5 instances offer a balance of compute, memory, and networking resources for a broad range of workload.
- R5 EC2 instances > Amazon EC2 R5 instances are the next generation of memory optimized instances for the Amazon Elastic Compute Cloud. R5 instances are well suited for memory intensive applications such as high-performance databases etc

M5 EC2 instance is already in use & having performance issues so there is no point in provisioning a lower cost , general purpose instance type like T3 EC2.

So as per the given Answer option choice the best one is Answer D

upvoted 1 times

**jatric** 1 year, 11 months ago

**Selected Answer: A**



how future capacity planning and just do vertical scaling will improve the performance. Question doesn't specify if these EC2 are behind auto scaling so it means they are not. out of all A seems more close to the solution

upvoted 3 times

**a7md0** 1 year, 11 months ago

**Selected Answer: B**

B will reduce operational overhead and better solution than keep changing the family. Also, I don't think the exam will require you to remember instance families like M5 and R5

upvoted 2 times

**lofzee** 2 years ago

**Selected Answer: D**

D .....

upvoted 2 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: D**

R5 instances are better optimized for the in-memory workload than M5.

Auto Scaling alone doesn't handle stateful applications well, manual capacity adjustments would still be needed.

Custom latency metrics give better visibility than built-in metrics for capacity planning.

upvoted 7 times

**cookieMr** 2 years, 11 months ago

**Selected Answer: D**

By replacing the M5 instances with R5 instances, which are optimized for memory-intensive workloads, the application can benefit from increased memory capacity and performance.

In addition, deploying the CloudWatch agent on the EC2 instances allows for the generation of custom application latency metrics, which can provide valuable insights into the application's performance.

This solution addresses the performance issues efficiently by leveraging the appropriate instance types and collecting custom application metrics for better monitoring and future capacity planning.

A. Replacing with T3 instances may not provide enough memory capacity for in-memory tasks.

B. Manually increasing the capacity of the ASG does not directly address the performance issues.

C. Relying solely on built-in EC2 memory metrics may not provide enough granularity for optimizing in-memory tasks.

The most efficient solution is to modify the CloudFormation templates, replace with R5 instances, and deploy the CloudWatch agent for custom metrics.

upvoted 6 times

**Bmarodi** 3 years ago

**Selected Answer: D**

Option D is the correct answer.

upvoted 2 times

**BABU97** 3 years, 2 months ago

will go for C

upvoted 1 times

**Aninina** 3 years, 4 months ago

**Selected Answer: D**

Would go with D

upvoted 2 times

**mhmt4438** 3 years, 4 months ago

**Selected Answer: D**

I think D

upvoted 2 times

A solutions architect is designing a new API using Amazon API Gateway that will receive requests from users. The volume of requests is highly variable; several hours can pass without receiving a single request. The data processing will take place asynchronously, but should be completed within a few seconds after a request is made.

Which compute service should the solutions architect have the API invoke to deliver the requirements at the lowest cost?

- A. An AWS Glue job
- B. An AWS Lambda function **Most Voted**
- C. A containerized service hosted in Amazon Elastic Kubernetes Service (Amazon EKS)
- D. A containerized service hosted in Amazon ECS with Amazon EC2

**Correct Answer: B**

Community vote distribution

B (97%)

3%

## Comments

**Parsons** **Highly Voted** 3 years, 4 months ago

**Selected Answer: B**

B is the correct answer.

API Gateway + Lambda is the perfect solution for modern applications with serverless architecture.

upvoted 12 times

**cookieMr** **Highly Voted** 2 years, 11 months ago

**Selected Answer: B**

Lambda is a serverless compute service that can be triggered by API Gateway to process requests asynchronously. It automatically scales based on the incoming request volume and allows for cost optimization by charging only for the actual compute time used to process the requests.

A. Glue is a fully managed ETL service. It is designed for data processing and transformation tasks rather than serving API requests. It may not be suitable for handling variable request volumes and delivering responses within a few seconds.

C. While EKS provides scalability and flexibility, it may introduce additional complexity and overhead for managing and scaling the infrastructure for handling variable API request volumes.

D. Similar to the previous option, using ECS with EC2 would require additional effort for infrastructure management and scaling, which may not be necessary for handling intermittent and variable API request volumes.

upvoted 6 times

**a7md0** **Most Recent** 1 year, 11 months ago

**Selected Answer: B**

Screaming Lambda

upvoted 4 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: B**

data processing should be completed within a few seconds = An AWS Lambda function

upvoted 2 times

**JA2018** 1 year, 6 months ago

definitely well within Lambda's maximum runtime limit of 15 minutes.

upvoted 1 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: B**

B. An AWS Lambda function

upvoted 2 times

**ukivanlamipi** 2 years, 9 months ago

**Selected Answer: D**

lambda is expensive than running ECS on EC2

upvoted 1 times

**pentium75** 2 years, 5 months ago

"Several hours can pass without receiving a single request", during which Lambda costs 0.00.

upvoted 5 times

**Undisputed** 2 years, 10 months ago

**Selected Answer: B**

Lambda all the way.

upvoted 2 times

**Bmarodi** 3 years ago

**Selected Answer: B**

Option B meets the requirements.

upvoted 2 times

**Aninina** 3 years, 4 months ago

**Selected Answer: B**

Lambda !

upvoted 4 times

**mhmt4438** 3 years, 4 months ago

**Selected Answer: B**

<https://www.examttopics.com/discussions/amazon/view/43780-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 2 times

A company runs an application on a group of Amazon Linux EC2 instances. For compliance reasons, the company must retain all application log files for 7 years. The log files will be analyzed by a reporting tool that must be able to access all the files concurrently.

Which storage solution meets these requirements MOST cost-effectively?

- A. Amazon Elastic Block Store (Amazon EBS)
- B. Amazon Elastic File System (Amazon EFS)
- C. Amazon EC2 instance store
- D. Amazon S3 **Most Voted**

**Correct Answer: D**

Community vote distribution

D (100%)

## Comments

**cookieMr** **Highly Voted** 2 years, 11 months ago

**Selected Answer: D**

A. EBS provides block-level storage volumes for use with EC2 instances. While it offers durability and persistence, it is not the most cost-effective solution for long-term retention of log files. Additionally, it does not provide concurrent access to the files, which is a requirement in this scenario.

B. EFS is a scalable file storage service that can be mounted on multiple EC2 instances concurrently. While it provides concurrent access to files, it may not be the most cost-effective option for long-term retention due to its higher pricing compared to S3.

C. The instance store is a temporary storage option that is physically attached to the EC2 instance. It does not provide the durability and long-term retention required for compliance purposes. Additionally, the instance store is not accessible outside of the specific EC2 instance it is attached to, so concurrent access by the reporting tool would not be possible.

Therefore, considering the requirements for long-term retention, concurrent access, and cost-effectiveness, S3 is the most suitable and cost-effective storage solution.

upvoted 14 times

**satyaamm** **Most Recent** 1 year, 3 months ago

**Selected Answer: D**

EFS files cannot be stored for more than 365 days even with lifecycle policies. S3 is the most suitable option here.

upvoted 1 times

**Ruffyit** 2 years, 6 months ago

A. EBS provides block-level storage volumes for use with EC2 instances. While it offers durability and persistence, it is not the most cost-effective solution for long-term retention of log files. Additionally, it does not provide concurrent access to the files, which is a requirement in this scenario.

B. EFS is a scalable file storage service that can be mounted on multiple EC2 instances concurrently. While it provides concurrent access to files, it may not be the most cost-effective option for long-term retention due to its higher pricing compared to S3.

C. The instance store is a temporary storage option that is physically attached to the EC2 instance. It does not provide the durability and long-term retention required for compliance purposes. Additionally, the instance store is not accessible outside of the specific EC2 instance it is attached to, so concurrent access by the reporting tool would not be possible.

upvoted 3 times

**Chiquitabandita** 2 years, 8 months ago

this sounds like an expensive solution but if necessary then S3 would be the best

upvoted 2 times

**TariqKipkemei** 2 years, 8 months ago

most cost effective = Amazon S3

upvoted 2 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: D**

D. Amazon S3

upvoted 2 times

**kapit** 2 years, 11 months ago

s3 < efs < ebs

upvoted 2 times

**Iconique** 2 years, 8 months ago

actually S3 < EBS < EFS, but for EBS you need to pay for the underlying provisioned GB.

If you compare 1 GB then S3 < EBS < EFS but if you have 100GB storage for EBS then EBS is more expensive.

upvoted 2 times

**mattcl** 2 years, 12 months ago

"The log files will be analyzed by a reporting tool that must be able to access all the files concurrently" , so you need to access concurrently to get the logs. So is EFS. Letter B

upvoted 1 times

**northyork** 3 years ago

<https://aws.amazon.com/efs/faq/>

EFS is a file storage service for use with Amazon compute (EC2, containers, serverless) and on-premises servers. EFS provides a file system interface, file system access semantics (such as strong consistency and file locking), and concurrently accessible storage for up to thousands of EC2 instances.

upvoted 1 times

**JA2018** 1 year, 6 months ago

from: <https://docs.aws.amazon.com/AmazonS3/latest/userguide/optimizing-performance.html>

Your applications can easily achieve thousands of transactions per second in request performance when uploading and retrieving storage from Amazon S3. Amazon S3 automatically scales to high request rates. For example, your application can achieve at least

\*\*\* 3,500 PUT/COPY/POST/DELETE requests per second \*\*\* or

\*\*\* 5,500 GET/HEAD requests per second \*\*\* per partitioned Amazon S3 prefix.

upvoted 1 times

**alexandercamachop** 3 years ago

**Selected Answer: D**

Whenever we see long time storage and no special requirements that needs EFS or FSx, then S3 is the way.

upvoted 4 times

**elearningtakai** 3 years, 2 months ago

**Selected Answer: D**

To meet the requirements of retaining application log files for 7 years and allowing concurrent access by a reporting tool, while also being cost-effective, the recommended storage solution would be D: Amazon S3.

upvoted 3 times

**osmk** 3 years, 2 months ago

dddddddddddddddddd

upvoted 3 times

**udo2020** 3 years, 2 months ago

What about the keyword "concurrently"? Doesn't this mean EFS?

upvoted 3 times

**Aninina** 3 years, 4 months ago

**Selected Answer: D**

Cost Effective: S3

upvoted 3 times

**Parsons** 3 years, 4 months ago

**Selected Answer: D**

S3 is enough with the lowest cost perspective.

upvoted 2 times

**mhmt4438** 3 years, 4 months ago

**Selected Answer: D**

<https://www.examttopics.com/discussions/amazon/view/22182-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 2 times

A company has hired an external vendor to perform work in the company's AWS account. The vendor uses an automated tool that is hosted in an AWS account that the vendor owns. The vendor does not have IAM access to the company's AWS account.

How should a solutions architect grant this access to the vendor?

A. Create an IAM role in the company's account to delegate access to the vendor's IAM role. Attach the appropriate IAM policies to the role for the permissions that the vendor requires. **Most Voted**

B. Create an IAM user in the company's account with a password that meets the password complexity requirements. Attach the appropriate IAM policies to the user for the permissions that the vendor requires.

C. Create an IAM group in the company's account. Add the tool's IAM user from the vendor account to the group. Attach the appropriate IAM policies to the group for the permissions that the vendor requires.

D. Create a new identity provider by choosing "AWS account" as the provider type in the IAM console. Supply the vendor's AWS account ID and user name. Attach the appropriate IAM policies to the new provider for the permissions that the vendor requires.

**Correct Answer: A**

Community vote distribution

A (94%)

6%

## Comments

**cookieMr** **Highly Voted** 2 years, 5 months ago

By creating an IAM role and delegating access to the vendor's IAM role, you establish a trust relationship between accounts. This allows the vendor's automated tool to assume the role in the company's account and access the necessary resources.

By attaching the appropriate IAM policies to the role, you can define the precise permissions that the vendor requires for their tool to perform its tasks. This ensures that the vendor has the necessary access without granting them direct IAM access to the company's account.

B is incorrect because creating an IAM user with a password would require sharing the credentials with the vendor, which is not recommended for security reasons.

C is incorrect because adding the vendor's IAM user to an IAM group in the company's account would not provide a direct and controlled way to delegate access to the vendor's tool.

D is incorrect because creating a new identity provider for the vendor's AWS account would not provide a straightforward way to delegate access to the vendor's tool. Identity providers are typically used for federated access using external identity systems.

upvoted 11 times

**mp165** **Highly Voted** 2 years, 11 months ago

**Selected Answer: A**

A is proper

[https://docs.aws.amazon.com/IAM/latest/UserGuide/id\\_roles\\_common-scenarios\\_third-party.html](https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles_common-scenarios_third-party.html)

upvoted 10 times

**satyaamm** **Most Recent** 1 year, 3 months ago

**Selected Answer: A**

IAM roles are the most suitable here.

upvoted 1 times

**Ruffyt** 2 years ago

Create an IAM role in the company's account to delegate access to the vendor's IAM role. Attach the appropriate IAM policies to the role for the permissions that the vendor requires

upvoted 2 times

**TariqKipkemei** 2 years, 2 months ago

**Selected Answer: A**

Create an IAM role in the company's account to delegate access to the vendor's IAM role. Attach the appropriate IAM policies to the role for the permissions that the vendor requires

upvoted 2 times

**Guru4Cloud** 2 years, 3 months ago

**Selected Answer: A**

A. Create an IAM role in the company's account to delegate access to the vendor's IAM role. Attach the appropriate IAM policies to the role for the permissions that the vendor requires.

upvoted 2 times

**Bmarodi** 2 years, 6 months ago

**Selected Answer: A**

Option A fulfill the requirements.

upvoted 2 times

**Aninina** 2 years, 11 months ago

**Selected Answer: A**

IAM role is the answer

upvoted 2 times

**techhb** 2 years, 11 months ago

**Selected Answer: A**

A is correct answer.

upvoted 2 times

**kbaru** 2 years, 11 months ago

**Selected Answer: A**

[https://docs.aws.amazon.com/IAM/latest/UserGuide/id\\_roles\\_common-scenarios\\_third-party.html](https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles_common-scenarios_third-party.html)

upvoted 3 times

**venice1234** 2 years, 11 months ago

**Selected Answer: A**

[https://docs.aws.amazon.com/IAM/latest/UserGuide/id\\_roles\\_create\\_for-user\\_externalid.html](https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles_create_for-user_externalid.html)

upvoted 3 times

**Parsons** 2 years, 11 months ago

**Selected Answer: A**

A is the correct answer.

upvoted 4 times

**Babba** 2 years, 11 months ago

**Selected Answer: D**

My guess is D: [https://docs.aws.amazon.com/IAM/latest/UserGuide/id\\_roles\\_common-scenarios\\_third-party.html](https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles_common-scenarios_third-party.html)

upvoted 2 times

**pentium75** 1 year, 11 months ago

But your link describes A, not D.

upvoted 2 times



A company has deployed a Java Spring Boot application as a pod that runs on Amazon Elastic Kubernetes Service (Amazon EKS) in private subnets. The application needs to write data to an Amazon DynamoDB table. A solutions architect must ensure that the application can interact with the DynamoDB table without exposing traffic to the internet.

Which combination of steps should the solutions architect take to accomplish this goal? (Choose two.)

- A. Attach an IAM role that has sufficient privileges to the EKS pod. **Most Voted**
- B. Attach an IAM user that has sufficient privileges to the EKS pod.
- C. Allow outbound connectivity to the DynamoDB table through the private subnets' network ACLs.
- D. Create a VPC endpoint for DynamoDB. **Most Voted**
- E. Embed the access keys in the Java Spring Boot code.

**Correct Answer: AD**

Community vote distribution

AD (100%)

## Comments

iamroyalty\_k 1 year, 4 months ago

**Selected Answer: AD**

A and D are the best choices because they ensure secure authentication with IAM roles and private connectivity to DynamoDB via a VPC endpoint.

B. Attach an IAM user that has sufficient privileges to the EKS pod:  
IAM users are intended for individuals, not applications.

C. Allow outbound connectivity to the DynamoDB table through the private subnets' network ACLs:  
While this ensures outbound traffic can reach DynamoDB, it doesn't eliminate the need for internet access unless a VPC endpoint is used.

E. Embed the access keys in the Java Spring Boot code:  
Hardcoding access keys in the application code is a security risk.  
upvoted 2 times

Burrito69 1 year, 8 months ago

After seeing D, I didn't even look at option E. its AD correct  
upvoted 2 times

awsgeek75 1 year, 10 months ago

**Selected Answer: AD**

B: Wrong, cannot be a user for EKS  
C: Not possible as NACL need destination CIDR/ports etc. This is not correct way to connect to DynamoDB  
E: Not secure  
AD is correct because you need roles for allowing service permissions and accessing DynamoDB with VPC endpoint is the correct way  
upvoted 3 times

Ruffyt 2 years ago

The application needs to write data to an Amazon DynamoDB table = Attach an IAM role that has write privileges to the EKS pod  
Without exposing traffic to the internet = VPC endpoint for DynamoDB  
upvoted 2 times

TariqKipkemei 2 years, 2 months ago

**Selected Answer: AD**

The application needs to write data to an Amazon DynamoDB table = Attach an IAM role that has write privileges to the EKS pod  
Without exposing traffic to the internet = VPC endpoint for DynamoDB  
upvoted 4 times

**Guru4Cloud** 2 years, 3 months ago

**Selected Answer: AD**

A. By attaching an IAM role to the EKS pod, you can grant the necessary permissions for the pod to access DynamoDB. The IAM role should have appropriate policies allowing access to the DynamoDB table.

D. Creating a VPC endpoint for DynamoDB allows the EKS pod to access DynamoDB privately within the VPC, without the need for internet connectivity. The VPC endpoint provides a direct and secure connection to DynamoDB, eliminating the need for traffic to flow over the internet.  
upvoted 3 times

**cookieMr** 2 years, 5 months ago

**Selected Answer: AD**

A. By attaching an IAM role to the EKS pod, you can grant the necessary permissions for the pod to access DynamoDB. The IAM role should have appropriate policies allowing access to the DynamoDB table.

D. Creating a VPC endpoint for DynamoDB allows the EKS pod to access DynamoDB privately within the VPC, without the need for internet connectivity. The VPC endpoint provides a direct and secure connection to DynamoDB, eliminating the need for traffic to flow over the internet.

B is incorrect because attaching an IAM user to the pod is not a recommended approach. IAM users are meant for accessing AWS services through the AWS Management Console or AP.

C is incorrect because configuring outbound connectivity through network ACLs would not provide a secure and direct connection to DynamoDB.

E is incorrect because embedding access keys in the code is not a recommended security practice. It can lead to potential security vulnerabilities. It is better to use IAM roles or other secure mechanisms for providing access to AWS services.  
upvoted 3 times

**Bmarodi** 2 years, 6 months ago

**Selected Answer: AD**

A & D options fulfill the requirements.  
upvoted 2 times

**LuckyAro** 2 years, 10 months ago

**Selected Answer: AD**

Definitely  
upvoted 2 times

**Aninina** 2 years, 11 months ago

**Selected Answer: AD**

A D are the correct options  
upvoted 2 times

**venice1234** 2 years, 11 months ago

**Selected Answer: AD**

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/vpc-endpoints-dynamodb.html>  
<https://aws.amazon.com/about-aws/whats-new/2019/09/amazon-eks-adds-support-to-assign-iam-permissions-to-kubernetes-service-accounts/>  
upvoted 3 times

**Parsons** 2 years, 11 months ago

**Selected Answer: AD**

A, D is the correct answer.  
upvoted 3 times

**mhmt4438** 2 years, 11 months ago

**Selected Answer: AD**

The correct answer is A,D  
upvoted 2 times

A company recently migrated its web application to AWS by rehosting the application on Amazon EC2 instances in a single AWS Region. The company wants to redesign its application architecture to be highly available and fault tolerant. Traffic must reach all running EC2 instances randomly.

Which combination of steps should the company take to meet these requirements? (Choose two.)

- A. Create an Amazon Route 53 failover routing policy.
- B. Create an Amazon Route 53 weighted routing policy.
- C. Create an Amazon Route 53 multivalue answer routing policy. **Most Voted**
- D. Launch three EC2 instances: two instances in one Availability Zone and one instance in another Availability Zone.
- E. Launch four EC2 instances: two instances in one Availability Zone and two instances in another Availability Zone. **Most Voted**

**Correct Answer:** CE

Community vote distribution

CE (76%)

BE (24%)

## Comments

**cookieMr** **Highly Voted** 2 years, 11 months ago

**Selected Answer: CE**

C. A multivalue answer routing policy in Route 53 allows you to configure multiple values for a DNS record, and Route 53 responds to DNS queries with multiple random values. This enables the distribution of traffic randomly among the available EC2 instances.

E. By launching EC2 instances in different AZs, you achieve high availability and fault tolerance. Launching four instances (two in each AZ) ensures that there are enough resources to handle the traffic load and maintain the desired level of availability.

A. Failover routing is designed to direct traffic to a backup resource or secondary location only when the primary resource or location is unavailable.

B. Although a weighted routing policy allows you to distribute traffic across multiple EC2 instances, it does not ensure random distribution.

D. While launching instances in multiple AZs is important for fault tolerance, having only three instances does not provide an even distribution of traffic. With only three instances, the traffic may not be evenly distributed, potentially leading to imbalanced resource utilization.

upvoted 20 times

**Steve\_4542636** **Highly Voted** 3 years, 3 months ago

**Selected Answer: BE**

I went back and rewatched the lectures from Udemy on Weighted and Multi-Value. The lecturer said that Multi-value is \*not\* as substitute for ELB and he stated that DNS load balancing is a good use case for Weighted routing policies

upvoted 9 times

**smartegnine** 3 years ago

Weighted routing based on weight assigned, it can not do randomly choose, please see last sentence of the question choose randomly.

upvoted 9 times

**foha2012** 2 years, 4 months ago

what about 50 50 weighted ?

upvoted 2 times

**satyaamm** **Most Recent** 1 year, 3 months ago

**Selected Answer: CE**

Multi value routing will offer random distribution of traffic and 4 EC2 instances in different regions will offer HA

upvoted 1 times

**iamroyalty\_k** 1 year, 4 months ago

**Selected Answer: CE**

To achieve high availability, fault tolerance, and random traffic distribution, the company should use Route 53 multivalued answer routing and launch four EC2 instances across two Availability Zones.

- A. Failover routing policy: This is for active-passive setups, not for distributing traffic randomly.
  - B. Weighted routing policy: This allows control over traffic distribution proportions but doesn't ensure randomness or health checks.
  - D. Three EC2 instances in unbalanced zones: This setup lacks symmetry, reducing fault tolerance if the AZ with two instances fails.
- upvoted 2 times

**Ode7d1b** 1 year, 6 months ago

**Selected Answer: BE**

By combining these two strategies, the company can achieve high availability and fault tolerance:

Weighted Routing Policy: This Route 53 policy distributes traffic across multiple endpoints based on weights assigned to each endpoint. By assigning equal weights to all EC2 instances, traffic can be distributed evenly across them.

Multiple Availability Zones: Deploying EC2 instances across multiple Availability Zones ensures that the application can continue to function even if one Availability Zone experiences an outage.<https://www.examtopycs.com/exams/amazon/aws-certified-solutions-architect-associate-saa-c03/view/23/#>

upvoted 1 times

**bignatov** 1 year, 8 months ago

**Selected Answer: CE**

C and E are the correct answers.

upvoted 2 times

**rohitph** 2 years ago

**Selected Answer: CE**

CE is correct

upvoted 1 times

**bujuman** 2 years, 4 months ago

**Selected Answer: CE**

CE seems good to me due to "highly available and fault tolerant" and following explanation:  
<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy-multivalued.html>

upvoted 2 times

**bujuman** 2 years, 4 months ago

Specifically with this requirement: "Traffic must reach all running EC2 instances randomly"

upvoted 2 times

**awsgeek75** 2 years, 4 months ago

**Selected Answer: CE**

E: For HA

C: Random routing can only be created with multivalued answer routing policy.

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy-multivalued.html>

"To route traffic approximately randomly to multiple resources, such as web servers, you create one multivalued answer record for each resource and, optionally, associate a Route 53 health check with each record."

upvoted 4 times

**Ruffyit** 2 years, 6 months ago

C. A multivalued answer routing policy in Route 53 allows you to configure multiple values for a DNS record, and Route 53 responds to DNS queries with multiple random values. This enables the distribution of traffic randomly among the available EC2 instances.

E. By launching EC2 instances in different AZs, you achieve high availability and fault tolerance. Launching four instances (two in each AZ) ensures that there are enough resources to handle the traffic load and maintain the desired level of availability.

A. Failover routing is designed to direct traffic to a backup resource or secondary location only when the primary resource or location is unavailable.

B. Although a weighted routing policy allows you to distribute traffic across multiple EC2 instances, it does not ensure random distribution.

upvoted 2 times

**mohamoha** 2 years, 7 months ago

First I thought it was weighted but after research C is the correct answer:

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy.html>

upvoted 2 times

**daniel33** 2 years, 8 months ago

**Selected Answer: CE**

Multivalue routing can do random load balancing according to the AWS website:

To route traffic approximately randomly to multiple resources, such as web servers, you create one multivalue answer record for each resource and, optionally, associate a Route 53 health check with each record.

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy-multivalue.html>

upvoted 5 times

**Tom123456ac** 2 years, 8 months ago

This questions is so wired , 3 instances nothing wrong with it

upvoted 1 times

**pentium75** 2 years, 5 months ago

I guess it's the "highly available AND fault tolerant" requirement. If AZ 1 fails, you have only a single server left in the other region.

upvoted 3 times

**Techi47** 2 years, 8 months ago

Option CE Correct:

To route traffic roughly and randomly to multiple resources, such as web servers, you create a multi-value response record for each resource and optionally associate a Route 53 health check with each record.

<https://disaster-recovery.workshop.aws/en/services/networking/route53/routing-policies/routing-multiple-answer.html>

upvoted 2 times

**kwang312** 2 years, 8 months ago

**Selected Answer: CE**

CE is correct

upvoted 2 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: CE**

Highly available and fault tolerant = two instances in two AZs

Route traffic randomly = Amazon Route 53 multivalue answer routing policy

upvoted 2 times

**LazyTs** 2 years, 9 months ago

**Selected Answer: CE**

Multivalue answer routing policy – Use when you want Route 53 to respond to DNS queries with up to eight healthy records selected at random. You can use multivalue answer routing to create records in a private hosted zone.

Weighted routing policy – Use to route traffic to multiple resources in proportions that you specify. You can use weighted routing to create records in a private hosted zone.

upvoted 2 times

**foha2012** 2 years, 4 months ago

what about 50 50 weighted routing ?

upvoted 1 times

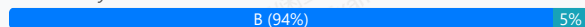
A media company collects and analyzes user activity data on premises. The company wants to migrate this capability to AWS. The user activity data store will continue to grow and will be petabytes in size. The company needs to build a highly available data ingestion solution that facilitates on-demand analytics of existing data and new data with SQL.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Send activity data to an Amazon Kinesis data stream. Configure the stream to deliver the data to an Amazon S3 bucket.
- B. Send activity data to an Amazon Kinesis Data Firehose delivery stream. Configure the stream to deliver the data to an Amazon Redshift cluster. **Most Voted**
- C. Place activity data in an Amazon S3 bucket. Configure Amazon S3 to run an AWS Lambda function on the data as the data arrives in the S3 bucket.
- D. Create an ingestion service on Amazon EC2 instances that are spread across multiple Availability Zones. Configure the service to forward data to an Amazon RDS Multi-AZ database.

**Correct Answer: B**

Community vote distribution



## Comments

**beginnercloud** **Highly Voted** 2 years, 9 months ago

**Selected Answer: B**

Petabyte scale- Redshift  
upvoted 13 times

**Berny** **Highly Voted** 3 years, 4 months ago

**Selected Answer: B**

Data ingestion through Kinesis data streams will require manual intervention to provide more shards as data size grows. Kinesis firehose will ingest data with the least operational overhead.  
upvoted 8 times

**Neeraja2023** **Most Recent** 7 months, 1 week ago

**Selected Answer: C**

C. Place activity data in an Amazon S3 bucket. Configure Amazon S3 to run an AWS Lambda function on the data as the data arrives in the S3 bucket  
S3 + Lambda provides serverless ingestion, no servers to manage.  
S3 can store petabytes of data cheaply.  
Querying the data can be done with Amazon Athena, which allows SQL queries directly on S3 objects.  
Fully managed → least operational overhead.  
Supports highly available ingestion, as S3 is multi-AZ and Lambda scales automatically.  
upvoted 1 times

**Clouddon** 11 months ago

**Selected Answer: A**

Why Option A works best:  
Amazon Kinesis Data Streams provides a durable, scalable, and highly available ingestion layer.  
Data is streamed to an Amazon S3 bucket, which:  
Supports virtually unlimited storage  
Is cost-effective for storing petabytes of data  
Once in S3, you can use:  
Amazon Athena to run on-demand SQL queries  
Amazon Redshift Spectrum or Amazon EMR for advanced analytics  
Fully managed, minimal maintenance, and scales automatically

Why the other options are less ideal:

B. Kinesis Data Firehose to Amazon Redshift

- ✗ Amazon Redshift is more expensive and suited for structured data with known schema
- ✗ Petabyte-scale data could require frequent manual scaling and maintenance
- ✗ Less cost-effective than S3 + Athena for large-scale, intermittent querying

upvoted 1 times

**Clouddon** 11 months ago

Why this is the best solution:

The requirements are:

Highly available data ingestion, Supports petabyte-scale data, Enables on-demand analytics with SQL and Minimal operational overhead

upvoted 1 times

**scar0909** 2 years, 3 months ago

**Selected Answer: B**

Kinesis data stream cannot detined to s3

upvoted 3 times

**Ruffyyit** 2 years, 6 months ago

1- Kinesis Data Stream provides a fully managed platform for custom data processing and analysis. Or we can say that used for custom data processing and analysis which required more manual intervention.

2- Kinesis Data Firehose simplifies the delivery of streaming data to various destinations without the need for complex transformations.

Option B is more suitable for the given scenario.

upvoted 6 times

**Rhydian25** 1 year, 12 months ago

Copy-paste from A1975's answer

upvoted 1 times

**David\_Ang** 2 years, 7 months ago

**Selected Answer: B**

always if you have a service that is meant for a specific job, it the correct answer, is logic. "A" is not good enough for this situation

upvoted 3 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: B**

B. Send activity data to an Amazon Kinesis Data Firehose delivery stream. Configure the stream to deliver the data to an Amazon Redshift cluster.

upvoted 3 times

**NVenkatS** 2 years, 9 months ago

**Selected Answer: B**

Petabyte scale- Redshift

upvoted 5 times

**A1975** 2 years, 10 months ago

**Selected Answer: B**

1- Kinesis Data Stream provides a fully managed platform for custom data processing and analysis. Or we can say that used for custom data processing and analysis which required more manual intervention.

2- Kinesis Data Firehose simplifies the delivery of streaming data to various destinations without the need for complex transformations.

Option B is more suitable for the given scenario.

upvoted 3 times

**sickcow** 2 years, 11 months ago

**Selected Answer: B**

Petabyte Scale sounds like Redshift!

upvoted 3 times

**cookieMr** 2 years, 11 months ago

**Selected Answer: B**

B provides a fully managed and scalable solution for data ingestion and analytics. KDF simplifies the data ingestion process by automatically scaling to handle large volumes of streaming data. It can directly load the data into an Redshift cluster, which is a powerful and fully managed data warehousing solution.

A. While Kinesis can handle streaming data, it requires additional processing to load the data into an analytics solution.

C. Although S3 and Lambda can handle the storage and processing of data, it requires more manual configuration and management compared to the fully managed solution offered by KDF and Redshift.

D. This option involves more operational overhead, as it requires managing and scaling the EC2 instances and RDS database infrastructure manually.

Therefore, option B with KDF delivering the data to Redshift cluster offers the most streamlined and operationally efficient solution for ingesting and analyzing the user activity data in the given scenario.

upvoted 3 times

**pisica134** 2 years, 11 months ago

petabytes in size => redshift

upvoted 3 times

**mattcl** 2 years, 12 months ago

It's A. Data Stream is better in this case, and you can query data in S3 with Athena

upvoted 2 times

**Yadav\_Sanjay** 2 years, 11 months ago

Data Stream Can't write to S3. That's why B is only left correct answer.

upvoted 1 times

**baba365** 2 years, 11 months ago

Answer A... key phrase 'least operational overhead'

KDF can write to S3 ... <https://docs.aws.amazon.com/firehose/latest/dev/what-is-this-service.html>

upvoted 1 times

**JoeGuan** 2 years, 9 months ago

<https://aws.amazon.com/streaming-data/> a good explanation of either option. firehose appears to be an option for Least operational overhead, as the streams product requires some building of apps etc.

upvoted 2 times

**Bmarodi** 3 years ago

**Selected Answer: B**

Option B is correct answer.

upvoted 2 times

**kruasan** 3 years, 1 month ago

**Selected Answer: B**

This solution meets the requirements as follows:

- Kinesis Data Firehose can scale to ingest and process multiple terabytes per hour of streaming data. This can easily handle the petabyte-scale data volumes.
- Firehose can deliver the data to Redshift, a petabyte-scale data warehouse, enabling on-demand SQL analytics of the data.
- Redshift is a fully managed service, minimizing operational overhead. Firehose is also fully managed, handling scalability, availability, and durability of the streaming data ingestion.

upvoted 7 times

**gold4otas** 3 years, 2 months ago

**Selected Answer: B**

B: The answer is certainly option "B" because ingesting user activity data can easily be handled by Amazon Kinesis Data streams. The ingested data can then be sent into Redshift for Analytics.

Amazon Redshift is a fully managed, petabyte-scale data warehouse service in the cloud. Amazon Redshift Serverless lets you access and analyze data without all of the configurations of a provisioned data warehouse.

<https://docs.aws.amazon.com/redshift/latest/mgmt/welcome.html>

upvoted 3 times



A company collects data from thousands of remote devices by using a RESTful web services application that runs on an Amazon EC2 instance. The EC2 instance receives the raw data, transforms the raw data, and stores all the data in an Amazon S3 bucket. The number of remote devices will increase into the millions soon. The company needs a highly scalable solution that minimizes operational overhead.

Which combination of steps should a solutions architect take to meet these requirements? (Choose two.)

- A. Use AWS Glue to process the raw data in Amazon S3. **Most Voted**
- B. Use Amazon Route 53 to route traffic to different EC2 instances.
- C. Add more EC2 instances to accommodate the increasing amount of incoming data.
- D. Send the raw data to Amazon Simple Queue Service (Amazon SQS). Use EC2 instances to process the data.
- E. Use Amazon API Gateway to send the raw data to an Amazon Kinesis data stream. Configure Amazon Kinesis Data Firehose to use the data stream as a source to deliver the data to Amazon S3. **Most Voted**

**Correct Answer:** AE

Community vote distribution

AE (100%)

## Comments

**Parsons** **Highly Voted** 3 years, 4 months ago

**Selected Answer: AE**

A, E is the correct answer

"RESTful web services" => API Gateway.

"EC2 instance receives the raw data, transforms the raw data, and stores all the data in an Amazon S3 bucket" => GLUE with (Extract - Transform - Load)

upvoted 15 times

**cookieMr** **Highly Voted** 2 years, 11 months ago

**Selected Answer: AE**

A. It automatically discovers the schema of the data and generates ETL code to transform it.

E. API Gateway can be used to receive the raw data from the remote devices via RESTful web services. It provides a scalable and managed infrastructure to handle the incoming requests. The data can then be sent to an Amazon Kinesis data stream, which is a highly scalable and durable real-time data streaming service. From there, Amazon Kinesis Data Firehose can be configured to use the data stream as a source and deliver the transformed data to Amazon S3. This combination of services allows for the seamless ingestion and processing of data while minimizing operational overhead.

B. It does not directly address the need for scalable data processing and storage. It focuses on managing DNS and routing traffic to different endpoints.

C. Adding more EC2 can lead to increased operational overhead in terms of managing and scaling the instances.

D. Using SQS and EC2 for processing data introduces more complexity and operational overhead.

upvoted 5 times

**Noveo** **Most Recent** 1 year, 9 months ago

AE breaks the original workflow "receive raw data - process - store" to "receive - store- process - store again" which leads to additional storage consuming (and thus money consuming).

upvoted 2 times

**Ruffyt** 2 years, 6 months ago

A - Use AWS Glue to process the raw data in Amazon S3

E - Use Amazon API Gateway to send the raw data to an Amazon Kinesis data stream. Configure Amazon Kinesis Data Firehose to use the data stream as a source to deliver the data to Amazon S3

upvoted 2 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: AE**

E then A no doubt.

upvoted 2 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: AE**

A. It automatically discovers the schema of the data and generates ETL code to transform it.

E. API Gateway can be used to receive the raw data from the remote devices via RESTful web services. It provides a scalable and managed infrastructure to handle the incoming requests. The data can then be sent to an Amazon Kinesis data stream, which is a highly scalable and durable real-time data streaming service. From there, Amazon Kinesis Data Firehose can be configured to use the data stream as a source and deliver the transformed data to Amazon S3. This combination of services allows for the seamless ingestion and processing of data while minimizing operational overhead.

upvoted 2 times

**ibu007** 2 years, 9 months ago

**Selected Answer: AE**

A - Use AWS Glue to process the raw data in Amazon S3

E - Use Amazon API Gateway to send the raw data to an Amazon Kinesis data stream. Configure Amazon Kinesis Data Firehose to use the data stream as a source to deliver the data to Amazon S3

upvoted 3 times

**GCB1990** 2 years, 9 months ago

Correct answer: D and E

upvoted 1 times

**wRhIH** 2 years, 11 months ago

Why not BC?

upvoted 1 times

**AnnieTran\_91** 2 years, 12 months ago

Why it not CE?

Add more EC2 instances to accommodate the increasing amount of incoming data?

upvoted 1 times

**TTaws** 2 years, 11 months ago

EC2 is not server-less. they want to minimize overhead

upvoted 1 times

**studynoplay** 3 years ago

**Selected Answer: AE**

minimizes operational overhead = Serverless

Glue, Kinesis Datastream, S3 are serverless

upvoted 3 times

**1e22522** 1 year, 10 months ago

facts!!

upvoted 1 times

**KZM** 3 years, 3 months ago

How about "C" to increase EC2 instances for the increased devices soon?

upvoted 1 times

**Aninina** 3 years, 4 months ago

**Selected Answer: AE**

Glue and API

upvoted 3 times

**mhmt4438** 3 years, 4 months ago

**Selected Answer: AE**

<https://www.examtopycs.com/discussions/amazon/view/83387-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 2 times



A company needs to retain its AWS CloudTrail logs for 3 years. The company is enforcing CloudTrail across a set of AWS accounts by using AWS Organizations from the parent account. The CloudTrail target S3 bucket is configured with S3 Versioning enabled. An S3 Lifecycle policy is in place to delete current objects after 3 years.

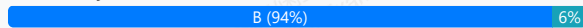
After the fourth year of use of the S3 bucket, the S3 bucket metrics show that the number of objects has continued to rise. However, the number of new CloudTrail logs that are delivered to the S3 bucket has remained consistent.

Which solution will delete objects that are older than 3 years in the MOST cost-effective manner?

- A. Configure the organization's centralized CloudTrail trail to expire objects after 3 years.
- B. Configure the S3 Lifecycle policy to delete previous versions as well as current versions. **Most Voted**
- C. Create an AWS Lambda function to enumerate and delete objects from Amazon S3 that are older than 3 years.
- D. Configure the parent account as the owner of all objects that are delivered to the S3 bucket.

**Correct Answer: B**

Community vote distribution



## Comments

**Guru4Cloud** **Highly Voted** 2 years, 3 months ago

**Selected Answer: B**

This is the most cost-effective option because:

- Versioning has caused the number of objects to increase over time, even as current objects are deleted after 3 years. By deleting previous versions as well, this will clean up old object versions and reduce storage costs.
- An S3 Lifecycle policy incurs no additional charges and requires no additional resources to configure and run. It is a native S3 tool for managing object lifecycles cost-effectively.

upvoted 11 times

**cookieMr** **Highly Voted** 2 years, 5 months ago

**Selected Answer: B**

By configuring the S3 Lifecycle policy to delete previous versions as well as current versions, the older versions of the CloudTrail logs will be deleted. This ensures that objects older than 3 years are removed from the S3 bucket, reducing the object count and controlling storage costs.

A. This option is not directly related to managing objects in the S3. It focuses on configuring the expiration of CloudTrail trails, which may not address the need to delete objects from the S3 bucket.

C. While it is technically possible to create a Lambda to delete objects older than 3 years, this approach would introduce additional complexity and operational overhead.

D. Changing the ownership of the objects in the S3 bucket does not directly address the need to delete objects older than 3 years. Ownership does not affect the deletion behavior of the objects.

upvoted 5 times

**Mikado211** **Most Recent** 2 years ago

**Selected Answer: B**

I did something similar recently : Lifecycle is triggered more or less each 24 hours, in my case it removed hundreds of gigabytes and millions of small files in one shot. Using another mechanism like a script would have taken days if not weeks.

upvoted 3 times

**Ruffyt** 2 years ago

This is the most cost-effective option because:

- Versioning has caused the number of objects to increase over time, even as current objects are deleted after 3 years. By deleting previous versions as well, this will clean up old object versions and reduce storage costs.

• An S3 Lifecycle policy incurs no additional charges and requires no additional resources to configure and run. It is a native S3 tool for managing object lifecycles cost-effectively.

upvoted 3 times

**TariqKipkemei** 2 years, 2 months ago

**Selected Answer: B**

Ensure to delete previous versions as well.

upvoted 2 times

**Bmarodi** 2 years, 6 months ago

**Selected Answer: B**

I go for option B.

upvoted 2 times

**ruqui** 2 years, 6 months ago

I don't think it's possible to configure an S3 lifecycle policy to delete all versions of an object, so B is wrong ... I think the question is improperly worded

upvoted 3 times

**Rahulbit34** 2 years, 7 months ago

• Versioning has caused the number of objects to increase over time, even as current objects are deleted after 3 years. By deleting previous versions as well, this will clean up old object versions and reduce storage costs. • An S3 Lifecycle policy incurs no additional charges and requires no additional resources to configure and run. It is a native S3 tool for managing object lifecycles cost-effectively.

upvoted 2 times

**kruasan** 2 years, 7 months ago

**Selected Answer: B**

This is the most cost-effective option because:

• Versioning has caused the number of objects to increase over time, even as current objects are deleted after 3 years. By deleting previous versions as well, this will clean up old object versions and reduce storage costs.

• An S3 Lifecycle policy incurs no additional charges and requires no additional resources to configure and run. It is a native S3 tool for managing object lifecycles cost-effectively.

upvoted 4 times

**kruasan** 2 years, 7 months ago

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/DeletingObjectVersions.html>

upvoted 3 times

**bullrem** 2 years, 10 months ago

**Selected Answer: C**

A more cost-effective solution would be to configure the organization's centralized CloudTrail trail to expire objects after 3 years. This would ensure that all objects, including previous versions, are deleted after the specified retention period.

Another option would be to create an AWS Lambda function to enumerate and delete objects from Amazon S3 that are older than 3 years, this would allow you to have more control over the deletion process and to write a custom logic that best fits your use case.

upvoted 3 times

**pentium75** 1 year, 11 months ago

As long as versioning on the S3 bucket is enabled, any deletion, whether performed by CloudTrail or by your custom Lambda function, will simply add a new version with a deletion marker but will not delete the previous version.

upvoted 3 times

**JayBee65** 2 years, 10 months ago

**Selected Answer: B**

The question clearly says "An S3 Lifecycle policy is in place to delete current objects after 3 years". This implies that previous versions are not deleted since this is a separate setting, and since logs are constantly changed, it would seem to make sense to delete previous versions so, so B. D is wrong, since the parent account (the management account) will already be the owner of all objects delivered to the S3 bucket, "All accounts in the organization can see MyOrganizationTrail in their list of trails, but member accounts cannot remove or modify the organization trail. Only the management account or delegated administrator account can change or delete the trail for the organization.", see

<https://docs.aws.amazon.com/awscloudtrail/latest/userguide/creating-trail-organization.html>

upvoted 3 times

**John\_Zhuang** 2 years, 10 months ago

**Selected Answer: B**

B is the right answer. Ref: <https://docs.aws.amazon.com/awscloudtrail/latest/userguide/best-practices-security.html#:~:text=The%20CloudTrail%20trail,time%20has%20passed.>

Option A is wrong. No way to expire the cloudtrail logs

upvoted 4 times

**techhb** 2 years, 10 months ago

**Selected Answer: B**

Configure the S3 Lifecycle policy to delete previous versions  
upvoted 3 times

**Aninina** 2 years, 11 months ago

**Selected Answer: B**

B. Configure the S3 Lifecycle policy to delete previous versions as well as current versions.  
upvoted 2 times

**Aninina** 2 years, 11 months ago

B. Configure the S3 Lifecycle policy to delete previous versions as well as current versions.  
upvoted 2 times

**Parsons** 2 years, 11 months ago

**Selected Answer: B**

B is correct answer  
upvoted 3 times

**AHUI** 2 years, 11 months ago

Ans: A  
<https://docs.aws.amazon.com/awscloudtrail/latest/userguide/creating-trail-organization.html>  
When you create an organization trail, a trail with the name that you give it is created in every AWS account that belongs to your organization. Users with CloudTrail permissions in member accounts can see this trail when they log into the AWS CloudTrail console from their AWS accounts, or when they run AWS CLI commands such as describe-trail. However, users in member accounts do not have sufficient permissions to delete the organization trail, turn logging on or off, change what types of events are logged, or otherwise change the organization trail in any way.  
upvoted 1 times

**AHUI** 2 years, 11 months ago

correction: Ans D is the answer.  
<https://docs.aws.amazon.com/awscloudtrail/latest/userguide/creating-trail-organization.html>  
upvoted 1 times

A company has an API that receives real-time data from a fleet of monitoring devices. The API stores this data in an Amazon RDS DB instance for later analysis. The amount of data that the monitoring devices send to the API fluctuates. During periods of heavy traffic, the API often returns timeout errors.

After an inspection of the logs, the company determines that the database is not capable of processing the volume of write traffic that comes from the API. A solutions architect must minimize the number of connections to the database and must ensure that data is not lost during periods of heavy traffic.

Which solution will meet these requirements?

- A. Increase the size of the DB instance to an instance type that has more available memory.
- B. Modify the DB instance to be a Multi-AZ DB instance. Configure the application to write to all active RDS DB instances.
- C. Modify the API to write incoming data to an Amazon Simple Queue Service (Amazon SQS) queue. Use an AWS Lambda function that Amazon SQS invokes to write data from the queue to the database. **Most Voted**
- D. Modify the API to write incoming data to an Amazon Simple Notification Service (Amazon SNS) topic. Use an AWS Lambda function that Amazon SNS invokes to write data from the topic to the database.

**Correct Answer: C**

Community vote distribution

C (100%)

## Comments

**toyaji** **Highly Voted** 1 year, 9 months ago

**Selected Answer: C**

You need to also use AWS RDS Proxy because lambda will increase parallel and it will cause connection error  
upvoted 5 times

**mwwt2022** **Most Recent** 2 years, 5 months ago

**Selected Answer: C**

//minimize the number of connections to the database and must ensure that data is not lost during periods of heavy traffic//

I go for C

upvoted 3 times

**Ruffyit** 2 years, 6 months ago

Decouple the API and the DB with Amazon Simple Queue Service (Amazon SQS) queue.  
upvoted 3 times

**oluolope** 2 years, 7 months ago

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-configure-lambda-function-trigger.html>  
SQS can invoke lambda indeed. Initially I picked D because I wasn't sure it was possible but, this article shows it is. It makes this question even more confusing for me as it is also possible to trigger lambda from SNS:  
<https://docs.aws.amazon.com/sns/latest/dg/sns-lambda-as-subscriber.html>  
I don't know which option between C and D makes more sense. I still have a preference for D as it seems less hacky than C.  
upvoted 1 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: C**

Decouple the API and the DB with Amazon Simple Queue Service (Amazon SQS) queue.  
upvoted 4 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: C**

C. Modify the API to write incoming data to an Amazon Simple Queue Service (Amazon SQS) queue. Use an AWS Lambda function that Amazon SQS invokes to write data from the queue to the database.

upvoted 2 times

**cookieMr** 2 years, 11 months ago

**Selected Answer: C**

By leveraging SQS as a buffer and using an Lambda to process and write data from the queue to the database, the solution provides scalability, decoupling, and reliability while minimizing the number of connections to the database. This approach handles fluctuations in traffic and ensures data integrity during high-traffic periods.

A. Increasing the size of the DB instance may provide more memory, but it does not address the issue of handling high write traffic efficiently and minimizing connections to the database.

B. Modifying the DB instance to be a Multi-AZ instance and writing to all active instances can improve availability but does not address the issue of efficiently handling high write traffic and minimizing connections to the database.

D. Using SNS and an Lambda can provide decoupling and scalability, but it is not suitable for handling heavy write traffic efficiently and minimizing connections to the database.

upvoted 3 times

**Moccorso** 2 years, 11 months ago

I think D, "Use an AWS Lambda function that Amazon SQS invokes to write data from the queue to the database" SQS can't invoke Lambda because SQS is pull.

upvoted 3 times

**pentium75** 2 years, 5 months ago

"Invoke a Lambda function from an Amazon SQS trigger"

[https://docs.aws.amazon.com/lambda/latest/dg/example\\_serverless\\_SQS\\_Lambda\\_section.html](https://docs.aws.amazon.com/lambda/latest/dg/example_serverless_SQS_Lambda_section.html)

upvoted 2 times

**shivamrulz** 2 years, 12 months ago

Why not B

upvoted 2 times

**pentium75** 2 years, 5 months ago

a) You can't write to multiple instances at the same time

b) If you could, it would probably not increase performance

c) if it would increase performance then it would probably double performance, which might not be enough

Decoupling is the way to go - let clients submit data via APIs, and write it asynchronously to the database. Don't let clients wait until data has been written.

upvoted 2 times

**Russs99** 3 years, 2 months ago

C is in deed the correct answer for the use case

upvoted 2 times

**kaushald** 3 years, 3 months ago

**Selected Answer: C**

C is correct

upvoted 2 times

**Steve\_4542636** 3 years, 3 months ago

**Selected Answer: C**

C is correct

upvoted 2 times

**maciekmaciek** 3 years, 3 months ago

**Selected Answer: C**

C looks ok

upvoted 2 times

**iamjaehyuk** 3 years, 4 months ago

why not D?

upvoted 1 times

**Parsons** 3 years, 4 months ago



**Selected Answer: C**

C is correct.

upvoted 2 times

**mhmt4438** 3 years, 4 months ago

**Selected Answer: C**

C. Modify the API to write incoming data to an Amazon Simple Queue Service (Amazon SQS) queue. Use an AWS Lambda function that Amazon SQS invokes to write data from the queue to the database.

To minimize the number of connections to the database and ensure that data is not lost during periods of heavy traffic, the company should modify the API to write incoming data to an Amazon SQS queue. The use of a queue will act as a buffer between the API and the database, reducing the number of connections to the database. And the use of an AWS Lambda function invoked by SQS will provide a more flexible way of handling the data and processing it. This way, the function will process the data from the queue and insert it into the database in a more controlled way.

upvoted 3 times

**Aninina** 3 years, 4 months ago

Did you use ChatGPT?

upvoted 6 times

**Nguyen25183** 3 years, 3 months ago

same question as you :D

upvoted 1 times

A company manages its own Amazon EC2 instances that run MySQL databases. The company is manually managing replication and scaling as demand increases or decreases. The company needs a new solution that simplifies the process of adding or removing compute capacity to or from its database tier as needed. The solution also must offer improved performance, scaling, and durability with minimal effort from operations.

Which solution meets these requirements?

- A. Migrate the databases to Amazon Aurora Serverless for Aurora MySQL. **Most Voted**
- B. Migrate the databases to Amazon Aurora Serverless for Aurora PostgreSQL.
- C. Combine the databases into one larger MySQL database. Run the larger database on larger EC2 instances.
- D. Create an EC2 Auto Scaling group for the database tier. Migrate the existing databases to the new environment.

**Correct Answer:** A

Community vote distribution

A (100%)

## Comments

**cookieMr** **Highly Voted** 1 year, 11 months ago

**Selected Answer: A**

Migrating the databases to Aurora Serverless provides automated scaling and replication capabilities. Aurora Serverless automatically scales the capacity based on the workload, allowing for seamless addition or removal of compute capacity as needed. It also offers improved performance, durability, and high availability without requiring manual management of replication and scaling.

B. Incorrect because it suggests migrating to a different database engine, which may introduce compatibility issues and require significant code modifications.

C. Incorrect because consolidating into a larger MySQL database on larger EC2 instances does not provide the desired scalability and automation.

D. Incorrect because using EC2 Auto Scaling groups for the database tier still requires manual management of replication and scaling.  
upvoted 9 times

**satyaamm** **Most Recent** 1 year, 3 months ago

**Selected Answer: A**

Aurora Serverless is the most suitable option here.  
upvoted 1 times

**TariqKipkemei** 1 year, 8 months ago

**Selected Answer: A**

Migrate the databases to Amazon Aurora Serverless for Aurora MySQL  
upvoted 2 times

**Undisputed** 1 year, 10 months ago

**Selected Answer: A**

Aurora MySQL  
upvoted 2 times

**Bmarodi** 2 years ago

**Selected Answer: A**

Option A is right answer.  
upvoted 2 times

**Bhrino** 2 years, 3 months ago

**Selected Answer: A**

A is correct because aurora might be more expensive but its serverless and is much faster  
upvoted 2 times

**mp165** 2 years, 4 months ago

**Selected Answer: A**

A is porper

<https://aws.amazon.com/rds/aurora/serverless/>  
upvoted 4 times

**Aninina** 2 years, 4 months ago

**Selected Answer: A**

Aurora MySQL  
upvoted 2 times

**mhmt4438** 2 years, 4 months ago

**Selected Answer: A**

<https://www.examttopics.com/discussions/amazon/view/51509-exam-aws-certified-solutions-architect-associate-saa-c02/>  
upvoted 2 times

A company is concerned that two NAT instances in use will no longer be able to support the traffic needed for the company's application. A solutions architect wants to implement a solution that is highly available, fault tolerant, and automatically scalable.

What should the solutions architect recommend?

- A. Remove the two NAT instances and replace them with two NAT gateways in the same Availability Zone.
- B. Use Auto Scaling groups with Network Load Balancers for the NAT instances in different Availability Zones.
- C. Remove the two NAT instances and replace them with two NAT gateways in different Availability Zones. **Most Voted**
- D. Replace the two NAT instances with Spot Instances in different Availability Zones and deploy a Network Load Balancer.

**Correct Answer: C**

Community vote distribution

C (97%)

3%

## Comments

**pentium75** **Highly Voted** 1 year, 11 months ago

**Selected Answer: C**

See <https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-comparison.html>

upvoted 7 times

**Bhrino** **Highly Voted** 2 years, 9 months ago

**Selected Answer: C**

fyi yall in most cases nat instances are a bad thing because their customer managed while nat gateways are AWS Managed. So in this case I already know to get rid of the nat instances the reason its c is because it wants high availability meaning different AZs

upvoted 7 times

**bora4motion** **Most Recent** 1 year, 1 month ago

**Selected Answer: C**

c is correct, nat instances are obsolete anyways

upvoted 1 times

**satyaamm** 1 year, 3 months ago

**Selected Answer: C**

NAT Gateways in 2 different AZ's is the most suitable option here.

upvoted 1 times

**MiniYang** 2 years ago

**Selected Answer: B**

Highly available, fault tolerant and automatically scalable=> Autoscaling and Diffrent AZ

upvoted 1 times

**pentium75** 1 year, 11 months ago

NAT instances are legacy technology. "If you're already using a NAT instance, we recommend that you replace it with a NAT gateway." Thus C.

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-comparison.html>

upvoted 3 times

**TariqKipkemei** 2 years, 2 months ago

**Selected Answer: C**

Highly available, fault tolerant, and automatically scalable = two NAT gateways in different Availability Zones

upvoted 3 times

**Undisputed** 2 years, 4 months ago

**Selected Answer: C**

Remove the two NAT instances and replace them with two NAT gateways in different Availability Zones

upvoted 2 times

**cookieMr** 2 years, 5 months ago

**Selected Answer: C**

This recommendation ensures high availability and fault tolerance by distributing the NAT gateways across multiple AZs. NAT gateways are managed AWS services that provide scalable and highly available outbound NAT functionality. By deploying NAT gateways in different AZs, the company can achieve redundancy and avoid a single point of failure. This solution also provides automatic scaling to handle increasing traffic without manual intervention.

Option A is incorrect because placing both NAT gateways in the same Availability Zone does not provide fault tolerance.

Option B is incorrect because using Auto Scaling groups with Network Load Balancers is not the recommended approach for NAT instances.

Option D is incorrect because Spot Instances are not suitable for critical infrastructure components like NAT instances.

upvoted 5 times

**Axeashes** 2 years, 6 months ago

**Selected Answer: C**

HA: <https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-comparison.html>

Scalability: <https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-gateway.html>

upvoted 2 times

**Theodorz** 2 years, 10 months ago

Could anybody teach me why the B cannot be correct answer? This solution also seems providing Scalability(Auto Scaling Group), High Availability(different AZ), and Fault Tolerance(NLB & AZ).

I honestly think that C is not enough, because each NAT gateway can provide a few scalability, but the bandwidth limit is clearly explained in the document. The C exactly mentioned "two NAT gateways" so the number of NAT is fixed, which will reach its limit soon.

upvoted 3 times

**KZM** 2 years, 9 months ago

Option B proposes to use an Auto Scaling group with Network Load Balancers to continue using the existing two NAT instances. However, NAT instances do not support automatic failover without a script, unlike NAT gateways which provide this functionality. Additionally, using Network Load Balancers to balance traffic between NAT instances adds more complexity to the solution.

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-comparison.html>

upvoted 3 times

**mwwt2022** 1 year, 11 months ago

Thx for your explanation!

upvoted 1 times

**JayBee65** 2 years, 10 months ago

C. If you have resources in multiple Availability Zones and they share one NAT gateway, and if the NAT gateway's Availability Zone is down, resources in the other Availability Zones lose internet access. To create an Availability Zone-independent architecture, create a NAT gateway in each Availability Zone and configure your routing to ensure that resources use the NAT gateway in the same Availability Zone.

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-gateway.html#nat-gateway-basics>

upvoted 3 times

**techhb** 2 years, 10 months ago

**Selected Answer: C**

Replace NAT Instances with Gateway

upvoted 2 times

**mhmt4438** 2 years, 11 months ago

**Selected Answer: C**

Correct answer is C

upvoted 3 times

An application runs on an Amazon EC2 instance that has an Elastic IP address in VPC A. The application requires access to a database in VPC B. Both VPCs are in the same AWS account.

Which solution will provide the required access MOST securely?

- A. Create a DB instance security group that allows all traffic from the public IP address of the application server in VPC A.
- B. Configure a VPC peering connection between VPC A and VPC B. **Most Voted**
- C. Make the DB instance publicly accessible. Assign a public IP address to the DB instance.
- D. Launch an EC2 instance with an Elastic IP address into VPC B. Proxy all requests through the new EC2 instance.

**Correct Answer: B**

Community vote distribution

B (92%)

8%

## Comments

**DUBURA** **Highly Voted** 2 years, 6 months ago

**Selected Answer: B**

B. Configure a VPC peering connection between VPC A and VPC B.

The most secure solution is to configure a VPC peering connection between the two VPCs. This allows private communication between the application server and the database, without exposing resources to the public internet.

Option A exposes the database to the public internet by allowing inbound traffic from a public IP address.

Option C makes the database instance itself public, which is insecure.

Option D adds complexity with a proxy that is not needed when a VPC peering connection can enable private communication between VPCs.

So option B is the most secure while allowing the necessary connectivity between the application server and the database in the separate VPCs.  
upvoted 14 times

**JayBee65** **Highly Voted** 3 years, 4 months ago

A is correct. B will work but is not the most secure method, since it will allow everything in VPC A to talk to everything in VPC B and vice versa, not at all secure. A on the other hand will only allow the application (since you select it's IP address) to talk to the application server in VPC A - you are allowing only the required connectivity. See the link for this exact use case:

<https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Overview.RDSSecurityGroups.html>

upvoted 14 times

**mhmt4438** 3 years, 4 months ago

" allows all traffic from the public IP address" Nice bro niceee This is absolutely the most secure method at all. :)))

upvoted 18 times

**datz** 3 years, 2 months ago

he must be the security engineer lolol :D

"Jaybee" - Please dont ever say that traffic over the public internet is secure :D

upvoted 5 times

**swakan** 1 year, 6 months ago

<https://aws.amazon.com/vpc/faqs/>

upvoted 1 times

**graveend** 2 years, 10 months ago

Both VPCs are in the "SAME AWS ACCOUNT" and the requirement specifies allowing traffic from the \*PUBLIC IP of the APPLICATION SERVER\*. In this case the traffic remains inside the AWS infrastructure or will it go through the public internet?

upvoted 2 times

**pentium75** 2 years, 5 months ago

Answer A (not "the requirement") specifies "allowing traffic from the public IP", which is for sure NOT the "most secure" option.

upvoted 2 times

**FlyingHawk** **Most Recent** 1 year, 4 months ago

**Selected Answer: B**

It should be B, plus the DB security group A. group that allows all traffic from the security group of the application server in VPC A

upvoted 1 times

**Jazz888** 2 years ago

Well this is a tricky one!!! Are we going to assume that the database in VPC B is in private subnet? In that case configuring security group to allow the traffic coming from Elastic IP of VPC A will not work. And if we use peering, the resources that live in the same subnet as the EC2 instance in VPC A will have access to the database? So what would we say to this? Is moving traffic through the public AWS space is safer than allowing access to the DB to other resources in VPC A?..... I don't know what to think

upvoted 2 times

**Ruffyt** 2 years, 6 months ago

When you establish peering relationships between VPCs across different AWS Regions, resources in the VPCs (for example, EC2 instances and Lambda functions) in different AWS Regions can communicate with each other using private IP addresses, without using a gateway, VPN connection, or network appliance. The traffic remains in the private IP space. All inter-Region traffic is encrypted with no single point of failure, or bandwidth bottleneck. Traffic always stays on the global AWS backbone, and never traverses the public internet, which reduces threats, such as common exploits, and DDoS attacks. Inter-Region VPC peering provides a simple and cost-effective way to share resources between regions or replicate data for geographic redundancy.

upvoted 2 times

**rlamberti** 2 years, 7 months ago

**Selected Answer: B**

Most secure = not leaving AWS network.  
VPC peering is the way.

upvoted 3 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: B**

VPC to VPC comms = VPC peering

upvoted 2 times

**Sutariya** 2 years, 9 months ago

B is correct : Setup VPC peering and connect Application from VPC A to connect with VPC B in private subnet so DB instance always secure with internet.

upvoted 2 times

**\_dlrk\_** 2 years, 9 months ago

Am I missing something or simply A is wrong because, without VPC peering (or other inter-connection sharing mechanisms such as Transit Gateway or VPN), VPC A and VPC B cannot communicate each other?

upvoted 1 times

**jacob\_ho** 2 years, 9 months ago

can use vpc endpoints but no option use that

upvoted 1 times

**A1975** 2 years, 10 months ago

**Selected Answer: B**

When you establish peering relationships between VPCs across different AWS Regions, resources in the VPCs (for example, EC2 instances and Lambda functions) in different AWS Regions can communicate with each other using private IP addresses, without using a gateway, VPN connection, or network appliance. The traffic remains in the private IP space. All inter-Region traffic is encrypted with no single point of failure, or bandwidth bottleneck. Traffic always stays on the global AWS backbone, and never traverses the public internet, which reduces threats, such as common exploits, and DDoS attacks. Inter-Region VPC peering provides a simple and cost-effective way to share resources between regions or replicate data for geographic redundancy.

<https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html>

upvoted 4 times

**animefan1** 2 years, 11 months ago

**Selected Answer: B**

With peering, we EC2 can communicate with RDS. RDS SG can have inbound from EC2 IP rather than VPC CIDR for more security

upvoted 2 times

**maggie135** 2 years, 11 months ago

**Selected Answer: B**

VPC peering uses AWS network.

upvoted 2 times

**cookieMr** 2 years, 11 months ago

**Selected Answer: B**

By configuring a VPC peering connection between VPC A and VPC B, you can establish private and secure communication between the EC2 instance in VPC A and the database in VPC B. VPC peering allows traffic to flow between the two VPCs using private IP addresses, without the need for public IP addresses or exposing the database to the internet.

Option A is not the best solution as it requires allowing all traffic from the public IP address of the application server, which can be less secure.

Option C involves making the DB instance publicly accessible, which introduces security risks by exposing the database directly to the internet.

Option D adds unnecessary complexity by launching an additional EC2 instance in VPC B and proxying all requests through it, which is not the most efficient and secure approach in this scenario.

upvoted 5 times

**joechen2023** 2 years, 12 months ago

**Selected Answer: B**

I don't like A because the security group setting is wrong as it set up to allow all public IP addresses. If the security group setting is correct, then I will go for A

I don't like B because it need to set up security group as well on top of peering.

for exam purpose only, I will go with the least worst choice which is B

upvoted 1 times

**Bmarodi** 2 years, 12 months ago

**Selected Answer: A**

The keywords are: "access MOST securely", hence the option A meets these requirements.

upvoted 1 times

**smartegnine** 2 years, 12 months ago

**Selected Answer: A**

Each VPC security group rule makes it possible for a specific source to access a DB instance in a VPC that is associated with that VPC security group. The source can be a range of addresses (for example, 203.0.113.0/24), or another VPC security group.

<https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide>

upvoted 1 times

**MostafaWardany** 3 years ago

**Selected Answer: B**

Most secure = VPC peering

upvoted 2 times

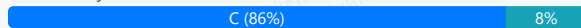


A company runs demonstration environments for its customers on Amazon EC2 instances. Each environment is isolated in its own VPC. The company's operations team needs to be notified when RDP or SSH access to an environment has been established.

- A. Configure Amazon CloudWatch Application Insights to create AWS Systems Manager OpsItems when RDP or SSH access is detected.
- B. Configure the EC2 instances with an IAM instance profile that has an IAM role with the AmazonSSMManagedInstanceCore policy attached.
- C. Publish VPC flow logs to Amazon CloudWatch Logs. Create required metric filters. Create an Amazon CloudWatch metric alarm with a notification action for when the alarm is in the ALARM state. **Most Voted**
- D. Configure an Amazon EventBridge rule to listen for events of type EC2 Instance State-change Notification. Configure an Amazon Simple Notification Service (Amazon SNS) topic as a target. Subscribe the operations team to the topic.

**Correct Answer: C**

Community vote distribution



## Comments

**cookieMr** **Highly Voted** 2 years, 5 months ago

**Selected Answer: C**

By publishing VPC flow logs to CloudWatch Logs and creating metric filters to detect RDP or SSH access, the operations team can configure an CloudWatch metric alarm to notify them when the alarm is triggered. This will provide the desired notification when RDP or SSH access to an environment is established.

Option A is incorrect because CloudWatch Application Insights is not designed for detecting RDP or SSH access.

Option B is also incorrect because configuring an IAM instance profile with the AmazonSSMManagedInstanceCore policy does not directly address the requirement of notifying the operations team when RDP or SSH access occurs.

Option D is wrong because configuring an EventBridge rule to listen for EC2 Instance State-change Notification events and using an SNS topic as a target will notify the operations team about changes in the instance state, such as starting or stopping instances. However, it does not specifically detect or notify when RDP or SSH access is established, which is the requirement stated in the question.

upvoted 17 times

**Vickysss** **Highly Voted** 2 years, 10 months ago

**Selected Answer: C**

<https://aws.amazon.com/blogs/security/how-to-monitor-and-visualize-failed-ssh-access-attempts-to-amazon-ec2-linux-instances/>

upvoted 9 times

**NitiATOS** 2 years, 10 months ago

<https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs-records-examples.html#flow-log-example-accepted-rejected>

Adding this to support that VPC flow logs can be used to capture Accepted or Rejected SSH and RDP traffic.

upvoted 4 times

**ruqui** 2 years, 6 months ago

I don't think C would be an acceptable solution ... the request is to be notified WHEN a SSH and/or RDP connection is established so it requires real-time monitoring and that is something the C solution does not provide ... I would select A as a correct answer

upvoted 1 times

**pentium75** **Most Recent** 1 year, 11 months ago

**Selected Answer: C**

C sounds complex, but is the only answer that can work.

Not A - Application Insights has nothing to do with SSH/RDP access to the OS; also we need a notification, not an OpsItem

Not B - Just attaching a role does not create a notification  
Not D - Establishing SSH/RDP access is not a "state change" that would trigger this

upvoted 3 times

**pentium75** 1 year, 11 months ago

**Selected Answer: C**

A bit clueless here. AWS-recommended approach involves the CloudWatch Logs Agent on each EC2 instance, but that is not involved in any of the answers.

A: Sounds good at first read, but "CloudWatch Application Insights" cannot detect RDP or SSH access.

B: Would allow RDP or SSH access via Systems Manager, but would NOT prevent access without Systems Manager; also we'd need to configure notifications in Systems Manager which is not mentioned here.

upvoted 3 times

**pentium75** 1 year, 11 months ago

C: Could work but it seems overkill to capture VPC flow logs just to detect SSH and RDP traffic. Also it is not real-time, and it's unclear how and when exactly the state transitions and notifications will be triggered. At best you'd get notification few minutes AFTER (not "when") "access has been established". Still, it has most similarity with the recommended approach to detect failed connections:

<https://aws.amazon.com/tr/blogs/security/how-to-monitor-and-visualize-failed-ssh-access-attempts-to-amazon-ec2-linux-instances/>

D: Won't work because establishment of a connection is not an instance state change.

upvoted 3 times

**Ruffyit** 2 years ago

<https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs-records-examples.html#flow-log-example-accepted-rejected>

Adding this to support that VPC flow logs can be used to capture Accepted or Rejected SSH and RDP traffic.

upvoted 2 times

**TariqKipkemei** 2 years, 2 months ago

**Selected Answer: C**

Publish VPC flow logs to Amazon CloudWatch Logs. Create required metric filters. Create an Amazon CloudWatch metric alarm with a notification action for when the alarm is in the ALARM state

upvoted 2 times

**Bmarodi** 2 years, 6 months ago

**Selected Answer: C**

VPC Flow Logs is a feature that enables you to capture information about the IP traffic going to and from network interfaces in your VPC. Flow log data can be published to the following locations: Amazon CloudWatch Logs, Amazon S3, or Amazon Kinesis Data Firehose. After you create a flow log, you can retrieve and view the flow log records in the log group, bucket, or delivery stream that you configured.

Flow logs can help you with a number of tasks, such as:

Diagnosing overly restrictive security group rules

Monitoring the traffic that is reaching your instance

Determining the direction of the traffic to and from the network interfaces

Ref link: <https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs.html>

upvoted 4 times

**cokutan** 2 years, 6 months ago

**Selected Answer: C**

seems like c:

<https://aws.amazon.com/tr/blogs/security/how-to-monitor-and-visualize-failed-ssh-access-attempts-to-amazon-ec2-linux-instances/>

upvoted 2 times

**pentium75** 1 year, 11 months ago

This link does not mention VPC flow logs at all.

upvoted 1 times

**ChrisAn** 2 years, 6 months ago

**Selected Answer: D**

D. Configure an Amazon EventBridge rule to listen for events of type EC2 Instance State-change Notification. Configure an Amazon Simple Notification Service (Amazon SNS) topic as a target. Subscribe the operations team to the topic. This setup allows the EventBridge rule to capture instance state change events, such as when RDP or SSH access is established. The rule can then send notifications to the specified SNS topic, which is subscribed by the operations team.

upvoted 2 times

**markw92** 2 years, 5 months ago

D is wrong. EC2 instance state change is only for pending, running etc. <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/monitoring-instance-state-changes.html> you can't have state change of ssh or rdp.

upvoted 1 times

**datz** 2 years, 8 months ago

**Selected Answer: C**

C:

<https://www.youtube.com/watch?v=KAe3Eju59OU>

upvoted 2 times

**Abhineet9148232** 2 years, 9 months ago

**Selected Answer: C**

<https://aws.amazon.com/blogs/security/how-to-monitor-and-visualize-failed-ssh-access-attempts-to-amazon-ec2-linux-instances/>

upvoted 2 times

**bullrem** 2 years, 10 months ago

**Selected Answer: A**

A. Configuring Amazon CloudWatch Application Insights to create AWS Systems Manager OpsItems when RDP or SSH access is detected would be the most appropriate solution in this scenario. This would allow the operations team to be notified when RDP or SSH access has been established and provide them with the necessary information to take action if needed. Additionally, Amazon CloudWatch Application Insights would allow for monitoring and troubleshooting of the system in real-time.

upvoted 1 times

**Training4aBetterLife** 2 years, 10 months ago

**Selected Answer: C**

EC2 Instance State-change Notifications are not the same as RDP or SSH established connection notifications. Use Amazon CloudWatch Logs to monitor SSH access to your Amazon EC2 Linux instances so that you can monitor rejected (or established) SSH connection requests and take action.

upvoted 5 times

**alexleely** 2 years, 10 months ago

**Selected Answer: A**

The Answer can be A or C depending on the requirement if it requires real-time notification.

A: Allows the operations team to be notified in real-time when access is established, and also provides visibility into the access events through the OpsItems.

C: The logs will need to be analyzed and metric filters applied to detect access, and then the alarm will trigger based on that analysis. This method could have a delay in providing notifications. Thus, not the best solution if real-time notification is required.

Why not D: RDP or SSH access does not cause an EC2 instance to have a state change. The state change events that Amazon EventBridge can listen for include stopping, starting, and terminated instances, which do not apply to RDP or SSH access. But RDP or SSH connection to an EC2 instance does generate an event in the system, such as a log entry which can be used to notify the Operation team. Since its a log, you would require a service that monitors logs like CloudTrail, VPC Flow logs, or AWS Systems Manager Session Manager.

upvoted 3 times

**JayBee65** 2 years, 10 months ago

I completely agree with the logic here, but I'm thinking C, since I believe you will need to "Create required metric filters" in order to detect RDP or SSH access, and this is not specified in the question, see <https://docs.aws.amazon.com/systems-manager/latest/userguide/OpsCenter-create-OpsItems-from-CloudWatch-Alarms.html>

upvoted 2 times

**owlminus** 2 years, 10 months ago

**Selected Answer: C**

It's C fam. RDP or SSH connections won't change the state of the EC2 instance, so D doesn't make sense.

upvoted 5 times

**forzadejan** 2 years, 11 months ago

D. Configure an Amazon EventBridge rule to listen for events of type EC2 Instance State-change Notification. Configure an Amazon Simple Notification Service (Amazon SNS) topic as a target. Subscribe the operations team to the topic.

EC2 instances sends events to the EventBridge when state change occurs, such as when a new RDP or SSH connection is established, you can use EventBridge to configure a rule that listens for these events and trigger an action, like sending an email or SMS, when the connection is detected. The operations team can be notified by subscribing to the Amazon Simple Notification Service (Amazon SNS) topic, which can be configured as the target of the EventBridge rule.

upvoted 3 times

**alanp** 2 years, 11 months ago

Are state changes pending:

running

stopping

stopped  
shutting-down  
terminated

<https://aws.amazon.com/blogs/security/how-to-monitor-and-visualize-failed-ssh-access-attempts-to-amazon-ec2-linux-instances/>  
upvoted 2 times

**mhmt4438** 2 years, 11 months ago

**Selected Answer: D**

Configure an Amazon EventBridge rule to listen for events of type EC2 Instance State-change Notification. Configure an Amazon Simple Notification Service (Amazon SNS) topic as a target. Subscribe the operations team to the topic. This approach allows you to set up a rule that listens for state change events on the EC2 instances, specifically for when RDP or SSH access is established, and trigger a notification via Amazon SNS to the operations team. This way they will be notified when RDP or SSH access to an environment has been established.

upvoted 3 times

**CapJackSparrow** 2 years, 9 months ago

um, isn't "EC2 Instance State-change" like running, terminated, or stopped?

upvoted 2 times

A solutions architect has created a new AWS account and must secure AWS account root user access.

Which combination of actions will accomplish this? (Choose two.)

- A. Ensure the root user uses a strong password. **Most Voted**
- B. Enable multi-factor authentication to the root user. **Most Voted**
- C. Store root user access keys in an encrypted Amazon S3 bucket.
- D. Add the root user to a group containing administrative permissions.
- E. Apply the required permissions to the root user with an inline policy document.

**Correct Answer:** AB

Community vote distribution

AB (86%)

10%

## Comments

**cookieMr** **Highly Voted** 2 years, 5 months ago

**Selected Answer: AB**

- A. Setting a strong password for the root user is an essential security measure to prevent unauthorized access.
  - B. Enabling MFA adds an extra layer of security by requiring an additional authentication factor, such as a code from a mobile app or a hardware token, in addition to the password.
  - C. Root user access keys should be avoided whenever possible, and it is best to use IAM users with restricted permissions instead.
  - D. The root user already has unrestricted access to all resources and services in the account, so granting additional administrative permissions could increase the risk of unauthorized actions.
  - E. Instead, it is recommended to create IAM users with appropriate permissions and use those users for day-to-day operations, while keeping the root user secured and only using it for necessary administrative tasks.
- upvoted 9 times

**Ruffyt** **Most Recent** 2 years ago

Ensure the root user uses a strong password. Enable multi-factor authentication to the root user.

upvoted 2 times

**TariqKipkemei** 2 years, 2 months ago

**Selected Answer: AB**

Ensure the root user uses a strong password. Enable multi-factor authentication to the root user.

upvoted 2 times

**DiscussionMonke** 2 years, 5 months ago

**Selected Answer: AB**

Options A & B are the CORRECT answers.

upvoted 2 times

**Bmarodi** 2 years, 6 months ago

**Selected Answer: AB**

Options A & B are the right answers.

upvoted 2 times

**luisgu** 2 years, 7 months ago

**Selected Answer: AB**

See <https://docs.aws.amazon.com/SetUp/latest/UserGuide/best-practices-root-user.html>

upvoted 2 times

**Kunj7** 2 years, 8 months ago

**Selected Answer: AB**

A and B are the correct answers:

Option A: A strong password is always required for any AWS account you create, and should not be shared or stored anywhere as there is always a risk.

Option B: This is following AWS best practice, by enabling MFA on your root user which provides another layer of security on the account and unauthorised access will be denied if the user does not have the correct password and MFA.

upvoted 2 times

**WheracanIstart** 2 years, 9 months ago

**Selected Answer: AB**

AB are the right answers.

upvoted 2 times

**fkie4** 2 years, 9 months ago

This is probably the hardest question in AWS history

upvoted 4 times

**ProfXsamson** 2 years, 10 months ago

**Selected Answer: AB**

AB is the only feasible answer here.

upvoted 4 times

**bullrem** 2 years, 10 months ago

**Selected Answer: BE**

B. Enabling multi-factor authentication for the root user provides an additional layer of security to ensure that only authorized individuals are able to access the root user account.

E. Applying the required permissions to the root user with an inline policy document ensures that the root user only has the necessary permissions to perform the necessary tasks, and not any unnecessary permissions that could potentially be misused.

upvoted 2 times

**bullrem** 2 years, 10 months ago

[https://docs.aws.amazon.com/IAM/latest/UserGuide/access\\_policies.html](https://docs.aws.amazon.com/IAM/latest/UserGuide/access_policies.html)

upvoted 1 times

**bullrem** 2 years, 10 months ago

The other options are not sufficient to secure the root user access because:

A. A strong password alone is not enough to protect against potential security threats such as phishing or brute force attacks.

C. Storing the root user access keys in an encrypted S3 bucket does not address the root user's authentication process.

D. Adding the root user to a group with administrative permissions does not address the root user's authentication process and does not provide an additional layer of security.

upvoted 1 times

**pentium75** 1 year, 11 months ago

E is wrong because you can't attach permissions or policies to the root user.

A is right because MFA alone won't help too much if the password is "123".

upvoted 3 times

**Pindol** 2 years, 10 months ago

**Selected Answer: AB**

AB obviously

upvoted 2 times

**david76x** 2 years, 10 months ago

**Selected Answer: AB**

Root user already has admin, so D is not correct

upvoted 1 times

**Aninina** 2 years, 11 months ago

**Selected Answer: AB**

AB are correct

upvoted 2 times

**wmp7039** 2 years, 11 months ago

**Selected Answer: AB**

D is incorrect as root user already has full admin access.

upvoted 2 times

**swolfgang** 2 years, 11 months ago

**Selected Answer: AB**

D. Add the root user to a group containing administrative permissions. >>its not about security,actually its unsecure so >> a&B

upvoted 2 times

**raf123123** 2 years, 11 months ago

**Selected Answer: BD**

BD is correct

upvoted 2 times

**pentium75** 1 year, 11 months ago

D is wrong because the root user is outside of IAM, thus you can't put him into a group. Also he does not need "administrative permissions" as he has those anyway.

upvoted 2 times

**awsgeek75** 1 year, 11 months ago

root user is literally the super admin account. What more permissions could you possible give to the root user by adding it to admin group?

upvoted 1 times

A company is building a new web-based customer relationship management application. The application will use several Amazon EC2 instances that are backed by Amazon Elastic Block Store (Amazon EBS) volumes behind an Application Load Balancer (ALB). The application will also use an Amazon Aurora database. All data for the application must be encrypted at rest and in transit.

Which solution will meet these requirements?

- A. Use AWS Key Management Service (AWS KMS) certificates on the ALB to encrypt data in transit. Use AWS Certificate Manager (ACM) to encrypt the EBS volumes and Aurora database storage at rest.
- B. Use the AWS root account to log in to the AWS Management Console. Upload the company's encryption certificates. While in the root account, select the option to turn on encryption for all data at rest and in transit for the account.
- C. Use AWS Key Management Service (AWS KMS) to encrypt the EBS volumes and Aurora database storage at rest. Attach an AWS Certificate Manager (ACM) certificate to the ALB to encrypt data in transit. **Most Voted**
- D. Use BitLocker to encrypt all data at rest. Import the company's TLS certificate keys to AWS Key Management Service (AWS KMS) Attach the KMS keys to the ALB to encrypt data in transit.

**Correct Answer: C**

Community vote distribution

C (100%)

## Comments

cookieMr **Highly Voted** 2 years, 5 months ago

**Selected Answer: C**

AWS KMS can be used to encrypt the EBS and Aurora database storage at rest.

ACM can be used to obtain an SSL/TLS certificate and attach it to the ALB. This encrypts the data in transit between the clients and the ALB.

A is incorrect because it suggests using ACM to encrypt the EBS, which is not the correct service for encrypting EBS.

B is incorrect because relying on the AWS root account and selecting an option in the AWS Management Console to enable encryption for all data at rest and in transit is not a valid approach.

D is incorrect because BitLocker is not a suitable solution for encrypting data in AWS services. It is primarily used for encrypting data on Windows-based operating systems. Additionally, importing TLS certificate keys to AWS KMS and attaching them to the ALB is not the recommended approach for encrypting data in transit.

upvoted 11 times

Ruffyt **Most Recent** 2 years ago

To encrypt data at rest, AWS Key Management Service (AWS KMS) can be used to encrypt EBS volumes and Aurora database storage.

To encrypt data in transit, an AWS Certificate Manager (ACM) certificate can be attached to the Application Load Balancer (ALB) to enable HTTPS and TLS encryption.

upvoted 1 times

TariqKipkemei 2 years, 2 months ago

**Selected Answer: C**

Use AWS Key Management Service (AWS KMS) to encrypt the EBS volumes and Aurora database storage at rest. Attach an AWS Certificate Manager (ACM) certificate to the ALB to encrypt data in transit

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

**Selected Answer: C**

C is the best answer.

To encrypt data at rest, AWS Key Management Service (AWS KMS) can be used to encrypt EBS volumes and Aurora database storage.



To encrypt data in transit, an AWS Certificate Manager (ACM) certificate can be attached to the Application Load Balancer (ALB) to enable HTTPS and TLS encryption.

upvoted 3 times

**MAMADOUG** 2 years, 6 months ago

**Selected Answer: C**

Option C it's correct

upvoted 2 times

**Bmarodi** 2 years, 6 months ago

**Selected Answer: C**

Option C fulfills the requirements.

upvoted 2 times

**techhb** 2 years, 10 months ago

**Selected Answer: C**

C is correct ,A REVERSES the work of each service.

upvoted 4 times

**Aninina** 2 years, 11 months ago

**Selected Answer: C**

C is correct!

upvoted 4 times

**mhmt4438** 2 years, 11 months ago

**Selected Answer: C**

c is correct answer

upvoted 3 times

A company is moving its on-premises Oracle database to Amazon Aurora PostgreSQL. The database has several applications that write to the same tables. The applications need to be migrated one by one with a month in between each migration. Management has expressed concerns that the database has a high number of reads and writes. The data must be kept in sync across both databases throughout the migration.

What should a solutions architect recommend?

- A. Use AWS DataSync for the initial migration. Use AWS Database Migration Service (AWS DMS) to create a change data capture (CDC) replication task and a table mapping to select all tables.
- B. Use AWS DataSync for the initial migration. Use AWS Database Migration Service (AWS DMS) to create a full load plus change data capture (CDC) replication task and a table mapping to select all tables.
- C. Use the AWS Schema Conversion Tool with AWS Database Migration Service (AWS DMS) using a memory optimized replication instance. Create a full load plus change data capture (CDC) replication task and a table mapping to select all tables.
- D. Use the AWS Schema Conversion Tool with AWS Database Migration Service (AWS DMS) using a compute optimized replication instance. Create a full load plus change data capture (CDC) replication task and a table mapping to select the largest tables.

Most Voted

Correct Answer: C

Community vote distribution

C (92%)

8%

## Comments

**aakashkumar1999** Highly Voted 2 years, 10 months ago

**Selected Answer: C**

C : because we need SCT to convert from Oracle to PostgreSQL, and we need memory optimized machine for databases not compute optimized.  
upvoted 16 times

**hissein** 2 years, 3 months ago

why it is memory optimized and not compute optimized machine ?  
upvoted 8 times

**Guru4Cloud** 2 years, 3 months ago

A memory-optimized replication instance is recommended because the database has a high number of reads and writes. Memory-optimized instances are designed to deliver fast performance for workloads that process large data sets in memory.  
upvoted 23 times

**hissein** 2 years, 2 months ago

thank you  
upvoted 2 times

**pentium75** 1 year, 11 months ago

Maybe it doesn't matter, but in D we create a table mapping only for "the largest tables" while obviously we need "all tables" as in C.  
upvoted 9 times

**ansagr** Highly Voted 2 years ago

**Selected Answer: C**

Another reason to rule out D is because it states "a table mapping to select the largest tables", whereas selecting all tables (as stated in option C) in the table mapping is necessary to ensure a comprehensive migration.  
upvoted 5 times

**TechStuff** Most Recent 1 year, 9 months ago

B. Use AWS DataSync for the initial migration. Use AWS Database Migration Service (AWS DMS) to create a full load plus change data capture (CDC) replication task and a table mapping to select all tables.

This approach leverages AWS DataSync for the initial data migration, which is optimized for high-speed transfer of large amounts of data. Then, AWS DMS is used to create a replication task with full load plus change data capture (CDC), ensuring ongoing synchronization between the on-premises Oracle database and Amazon Aurora PostgreSQL. By selecting all tables, the migration process ensures that all applications can continue to read from and write to the database without interruption during the migration period.

upvoted 2 times

**MrPCarrot** 1 year, 10 months ago

Answer is C - AWS Schema Conversion Tool (AWS SCT) supports heterogeneous database migrations by automatically converting the source database schema and a majority of the custom code to a format compatible with the target database.

upvoted 1 times

**awsgeek75** 1 year, 10 months ago

**Selected Answer: C**

Has to be SCT + DMS for all the tables so C is the choice. Why do you need SCT? Read this:

<https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/migrate-data-from-an-on-premises-oracle-database-to-aurora-postgresql.html>

upvoted 2 times

**awsgeek75** 1 year, 10 months ago

AB-> DataSync has nothing to do with DB Migration (<https://aws.amazon.com/datasync/>)

D: Only migrates the largest table

I think these questions are more for seeing how much of AWS product catalogue can you remember efficiently with associated features. Afterall AWS SA is also a salesperson for the right product! I now look at product cheat-sheets to look them up at work.

upvoted 2 times

**pentium75** 1 year, 11 months ago

**Selected Answer: C**

Oracle -> PostgreSQL, we need SCT, thus A and B are out.

D maps only "the largest tables" but we need all tables

upvoted 4 times

**Ruffyyit** 2 years ago

because we need SCT to convert from Oracle to PostgreSQL, and we need memory optimized machine for databases not compute optimized.

A memory-optimized replication instance is recommended because the database has a high number of reads and writes. Memory-optimized instances are designed to deliver fast performance for workloads that process large data sets in memory.

upvoted 2 times

**Po\_chih** 2 years, 2 months ago

**Selected Answer: C**

because we need SCT to convert from Oracle to PostgreSQL, and we need memory optimized machine for databases not compute optimized.

<https://repost.aws/zh-Hant/knowledge-center/dms-optimize-aws-sct-performance>

upvoted 2 times

**TariqKipkemei** 2 years, 2 months ago

**Selected Answer: C**

Oracle database to Amazon Aurora PostgreSQL = AWS Schema Conversion Tool

High number of reads and writes = memory optimized replication instance

upvoted 3 times

**Guru4Cloud** 2 years, 3 months ago

**Selected Answer: C**

A memory-optimized replication instance is recommended because the database has a high number of reads and writes. Memory-optimized instances are designed to deliver fast performance for workloads that process large data sets in memory.

upvoted 2 times

**\_d1rk\_** 2 years, 3 months ago

**Selected Answer: C**

DataSync is for file-level synch, so A and B can be excluded. C is better than D because memory-optimized instances are recommended to handle the high number of reads and writes

upvoted 2 times

**ukivanlamlpi** 2 years, 3 months ago

**Selected Answer: A**

why not a? only capture the change is sufficient

upvoted 2 times

**pentium75** 1 year, 11 months ago

Oracle -> PostgreSQL requires SCT

upvoted 1 times

**Mmmmmmkkkk** 2 years, 5 months ago

Bbbbbbb

upvoted 1 times

**cookieMr** 2 years, 5 months ago

**Selected Answer: C**

The AWS SCT is used to convert the schema and code of the Oracle database to be compatible with Aurora PostgreSQL. AWS DMS is utilized to migrate the data from the Oracle database to Aurora PostgreSQL. Using a memory-optimized replication instance is recommended to handle the high number of reads and writes during the migration process.

By creating a full load plus CDC replication task, the initial data migration is performed, and ongoing changes in the Oracle database are continuously captured and applied to the Aurora PostgreSQL database. Selecting all tables for table mapping ensures that all the applications writing to the same tables are migrated.

Option A & B are incorrect because using AWS DataSync alone is not sufficient for database migration and data synchronization.

Option D is incorrect because using a compute optimized replication instance is not the most suitable choice for handling the high number of reads and writes.

upvoted 3 times

**omoakin** 2 years, 6 months ago

BBBBBBBBBBBBBB

upvoted 2 times

**SimiTik** 2 years, 7 months ago

B chatgpt

upvoted 2 times

**KZM** 2 years, 9 months ago

DMS+SCT for Oracle to Aurora PostgreSQL migration

<https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/migrate-an-oracle-database-to-aurora-postgresql-using-aws-dms-and-aws-sct.html>

upvoted 2 times

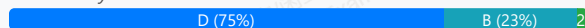
A company has a three-tier application for image sharing. The application uses an Amazon EC2 instance for the front-end layer, another EC2 instance for the application layer, and a third EC2 instance for a MySQL database. A solutions architect must design a scalable and highly available solution that requires the least amount of change to the application.

Which solution meets these requirements?

- A. Use Amazon S3 to host the front-end layer. Use AWS Lambda functions for the application layer. Move the database to an Amazon DynamoDB table. Use Amazon S3 to store and serve users' images.
- B. Use load-balanced Multi-AZ AWS Elastic Beanstalk environments for the front-end layer and the application layer. Move the database to an Amazon RDS DB instance with multiple read replicas to serve users' images.
- C. Use Amazon S3 to host the front-end layer. Use a fleet of EC2 instances in an Auto Scaling group for the application layer. Move the database to a memory optimized instance type to store and serve users' images.
- D. Use load-balanced Multi-AZ AWS Elastic Beanstalk environments for the front-end layer and the application layer. Move the database to an Amazon RDS Multi-AZ DB instance. Use Amazon S3 to store and serve users' images. **Most Voted**

**Correct Answer: D**

Community vote distribution



## Comments

**PDR** **Highly Voted** 2 years, 10 months ago

**Selected Answer: B**

B and D very similar with D being the 'best' solution but it is not the one that requires the least amount of development changes as the application would need to be changed to store images in S3 instead of DB

upvoted 15 times

**pentium75** 1 year, 11 months ago

B is wrong because single "RDS DB instance" is not HA.

No one says that the images are currently stored in S3. Also the requirement is "least amount of change [not "no change"] to the application".

upvoted 10 times

**Aninina** **Highly Voted** 2 years, 11 months ago

**Selected Answer: D**

for "Highly available": Multi-AZ &  
for "least amount of changes to the application": Elastic Beanstalk automatically handles the deployment, from capacity provisioning, load balancing, auto-scaling to application health monitoring

upvoted 13 times

**SirDNS** **Most Recent** 1 year, 2 months ago

**Selected Answer: D**

When we are talking about sharing static content and options have S3 why would I select any other option.

upvoted 1 times

**satyaamm** 1 year, 3 months ago

**Selected Answer: D**

D is the most suitable option here.

upvoted 1 times

**awsgeek75** 1 year, 11 months ago

**Selected Answer: D**

A: Requires changing EC2 application to Lambda. Seems like a big change

B: RDS DB is not best option for serve images and also single instance isn't HA

C: Memory optimised instance is not HA

D: Multi-AZ EBS is lift and shift for EC2 front-end and app later. RDS Multi AZ is HA. S3 for static images is best performance/scalability/availability.

upvoted 4 times

**ansagr** 2 years ago

**Selected Answer: D**

Using Amazon RDS for serving images might not be the optimal solution, as RDS is more suitable for storing structured data in a relational database rather than BLOBs like images. Storing and serving images can be more efficiently handled by object storage services like Amazon S3.

upvoted 2 times

**Ruffyit** 2 years ago

Use load-balanced Multi-AZ AWS Elastic Beanstalk environments for the front-end layer and the application layer. Move the database to an Amazon RDS Multi-AZ DB instance. Use Amazon S3 to store and serve users' images

upvoted 2 times

**rlamberti** 2 years, 1 month ago

Option B - DB is not a good option to store images. Read replicas won't improve HA for write, only scales reading IO. Therefore no true HA achieved.

D is the goal for me.

upvoted 2 times

**TariqKipkemei** 2 years, 2 months ago

**Selected Answer: D**

Use load-balanced Multi-AZ AWS Elastic Beanstalk environments for the front-end layer and the application layer. Move the database to an Amazon RDS Multi-AZ DB instance. Use Amazon S3 to store and serve users' images

upvoted 3 times

**Guru4Cloud** 2 years, 3 months ago

**Selected Answer: D**

Use Elastic Beanstalk load-balanced environments for the web and app tiers. This provides auto scaling and high availability with minimal effort.

Move the database to RDS Multi-AZ. This handles scaling reads and storage, and provides HA with automated failover.

Use S3 for serving user images. S3 is highly scalable and durable storage.

The application code remains unchanged using this approach.

upvoted 4 times

**Mia2009687** 2 years, 5 months ago

**Selected Answer: A**

AWS Elastic Beanstalk makes it even easier for developers to quickly deploy and manage applications in the AWS Cloud. Developers simply upload their application, and Elastic Beanstalk automatically handles the deployment details of capacity provisioning, load balancing, auto-scaling, and application health monitoring.

I don't quite understand why people choose D.

upvoted 1 times

**cookieMr** 2 years, 5 months ago

**Selected Answer: D**

By using load-balanced Multi-AZ AWS EBS, you achieve scalability and high availability for both layers without requiring significant changes to the application. Moving the DB to an RDS Multi-AZ DB ensures high availability and automatic failover. Storing and serving users' images through S3 provides a scalable and highly available solution.

A is incorrect because using S3 for the front-end layer and Lambda for the application layer would require significant changes to the application architecture. Moving the DB to DynamoDB would require rewriting the DB-related code.

B is incorrect because using load-balanced Multi-AZ AWS EBS environments and an RDS DB with read replicas for serving images would be a more suitable solution. RDS with read replicas can handle the image-serving workload more efficiently than using S3 for this purpose.

C is incorrect because using S3 for the front-end layer and an ASG of EC2 for the application layer would require modifying the application architecture. Storing and serving images from a memory-optimized EC2 type may not be the most efficient and scalable approach compared to using S3.

upvoted 5 times

**markw92** 2 years, 5 months ago

"least amount of change to the application." - A has lots of changes, completely revamping the application and lots of new pieces. D is closest with only addition of s3 to store images which is right move. You do not want images to store in any database anyway.

upvoted 4 times

**aaroncelestin** 2 years, 3 months ago

Thats what I was thinking, but the question doesn't mention anything about storing users' images anywhere. Are we supposed to just assume that they wanted to store the images in a DB even though that is a bad idea?

upvoted 2 times

**Bmarodi** 2 years, 6 months ago

**Selected Answer: D**

Option D meets the requirements.

upvoted 2 times

**Grace83** 2 years, 8 months ago

D is correct

upvoted 3 times

**focus\_23** 2 years, 10 months ago

**Selected Answer: D**

RDS multi AZ.

upvoted 3 times

**wmp7039** 2 years, 11 months ago

**Selected Answer: D**

D is correct as application changes needs to me minimal

upvoted 3 times

An application running on an Amazon EC2 instance in VPC-A needs to access files in another EC2 instance in VPC-B. Both VPCs are in separate AWS accounts. The network administrator needs to design a solution to configure secure access to EC2 instance in VPC-B from VPC-A. The connectivity should not have a single point of failure or bandwidth concerns.

Which solution will meet these requirements?

- A. Set up a VPC peering connection between VPC-A and VPC-B. **Most Voted**
- B. Set up VPC gateway endpoints for the EC2 instance running in VPC-B.
- C. Attach a virtual private gateway to VPC-B and set up routing from VPC-A.
- D. Create a private virtual interface (VIF) for the EC2 instance running in VPC-B and add appropriate routes from VPC-A.

**Correct Answer: A**

Community vote distribution

A (97%)

3%

## Comments

**cookieMr** **Highly Voted** 2 years, 11 months ago

**Selected Answer: A**

A VPC peering connection allows secure communication between instances in different VPCs using private IP addresses without the need for internet gateways, VPN connections, or NAT devices. By setting it up, the application running in VPC-A can directly access the EC2 in VPC-B without going through the public internet or any single point of failure.

B is incorrect because VPC gateway endpoints are used for accessing S3 or DynamoDB from a VPC without going over the internet. They are not designed for establishing connectivity between EC2 instances in different VPCs.

C is incorrect because it would require configuring a VPN connection between the VPCs. This would introduce additional complexity and potential single points of failure.

D is incorrect because creating a private VIF and adding routes would be applicable for establishing a direct connection between on-premises infrastructure and VPC-B using Direct Connect, but it is not suitable for the scenario of communication between EC2 instances in separate VPCs within different AWS accounts.

upvoted 20 times

**LuckyAro** **Highly Voted** 3 years, 4 months ago

**Selected Answer: A**

AWS uses the existing infrastructure of a VPC to create a VPC peering connection; it is neither a gateway nor a VPN connection, and does not rely on a separate piece of physical hardware. There is no single point of failure for communication or a bandwidth bottleneck.

<https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html>

upvoted 19 times

**satyaamm** **Most Recent** 1 year, 3 months ago

**Selected Answer: A**

VPC peering is the most suitable here.

upvoted 1 times

**Faridtnx** 2 years, 2 months ago

**Selected Answer: A**

You can create a VPC peering connection between your own VPCs, or with a VPC in another AWS account. Peering within the same AZ is free of charge.

upvoted 3 times

**lostmagnet001** 2 years, 4 months ago



**Selected Answer: A**

I get a little confused about B and A but, because, with a VPC endpoint in B it will work too access from A.  
upvoted 2 times

**GPFT** 1 year, 9 months ago

vpc gw endpoint just dynamo and s3  
upvoted 2 times

**pentium75** 2 years, 5 months ago

**Selected Answer: A**

B is wrong because "VPC gateway endpoint" is for S3 or DynamoDB, not EC2  
C is overkill, would require a second gateway in VPC-A, not be HA and have limited bandwidth  
D is wrong because VIF is for Direct Connect, has nothing to do with VPC-to-VPC communication  
upvoted 5 times

**Ruffyt** 2 years, 6 months ago

AWS uses the existing infrastructure of a VPC to create a VPC peering connection; it is neither a gateway nor a VPN connection, and does not rely on a separate piece of physical hardware. There is no single point of failure for communication or a bandwidth bottleneck.

<https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html>

upvoted 2 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: A**

A. Set up a VPC peering connection between VPC-A and VPC-B  
upvoted 2 times

**MNotABot** 2 years, 10 months ago

<https://www.bing.com/search?pglt=41&q=can+we+do+VPC+peering+across+AWS+accounts&cvid=48a8ceecc85a429c9ddd698b01055890&aqs=edge..69i57j0l8j69i11004.10897.0j1&FORM=ANNAB1&PC=LCTS>  
upvoted 1 times

**Anmol\_1010** 2 years, 12 months ago

D, VPC PEERINGVIS IN SAME ACCOUNT  
upvoted 1 times

**im6h** 2 years, 11 months ago

No, VPC Peering can use across account.

<https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html>

upvoted 4 times

**omoakin** 3 years ago

DDDDDDDDDDDDDDDD  
upvoted 2 times

**omoakin** 3 years ago

This is the only viable solution  
Create a private virtual interface (VIF) for the EC2 instance running in VPC-B and add appropriate routes from VPC-A  
upvoted 1 times

**michellemeloc** 3 years ago

**Selected Answer: A**

"You can create a VPC peering connection between your own VPCs, or with a VPC in another AWS account."

<https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html>

upvoted 5 times

**PDR** 3 years, 4 months ago

**Selected Answer: A**

correct answer is A and as mentioned by JayBee65 below, key reason being that solution should not have a single point of failure and bandwidth restrictions

the following paragraph is taken from the AWS docs page linked below that backs this up

"AWS uses the existing infrastructure of a VPC to create a VPC peering connection; it is neither a gateway nor a VPN connection, and does not rely on a separate piece of physical hardware. There is no single point of failure for communication or a bandwidth bottleneck."

<https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html>

upvoted 3 times

**LuckyAro** 3 years, 4 months ago

**Selected Answer: B**

A VPC endpoint gateway to the EC2 Instance is more specific and more secure than forming a VPC peering that exposes the whole of the VPC infrastructure just for one connection.

upvoted 2 times

**JayBee65** 3 years, 4 months ago

Your logic is correct but security is not a requirement here - the requirements are "The connectivity should not have a single point of failure or bandwidth concerns." A VPC gateway endpoint" would form a single point of failure, so B is incorrect, (and C and D are incorrect for the same reason, they create single points of failure).

upvoted 4 times

**pentium75** 2 years, 5 months ago

B is about a Gateway endpoint, which can be used to connect to S3 or DynamoDB, NOT to another EC2 instance.

upvoted 2 times

**mhmt4438** 3 years, 4 months ago

**Selected Answer: A**

Correct answer is A

upvoted 3 times

**Aninina** 3 years, 4 months ago

**Selected Answer: A**

VPC peering allows resources in different VPCs to communicate with each other as if they were within the same network. This solution would establish a direct network route between VPC-A and VPC-B, eliminating the need for a single point of failure or bandwidth concerns.

upvoted 2 times

**waiyu9981** 3 years, 4 months ago

**Selected Answer: A**

<https://www.examtopics.com/discussions/amazon/view/27763-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 5 times

A company wants to experiment with individual AWS accounts for its engineer team. The company wants to be notified as soon as the Amazon EC2 instance usage for a given month exceeds a specific threshold for each account.

What should a solutions architect do to meet this requirement MOST cost-effectively?

- A. Use Cost Explorer to create a daily report of costs by service. Filter the report by EC2 instances. Configure Cost Explorer to send an Amazon Simple Email Service (Amazon SES) notification when a threshold is exceeded.
- B. Use Cost Explorer to create a monthly report of costs by service. Filter the report by EC2 instances. Configure Cost Explorer to send an Amazon Simple Email Service (Amazon SES) notification when a threshold is exceeded.
- C. Use AWS Budgets to create a cost budget for each account. Set the period to monthly. Set the scope to EC2 instances. Set an alert threshold for the budget. Configure an Amazon Simple Notification Service (Amazon SNS) topic to receive a notification when a threshold is exceeded. **Most Voted**
- D. Use AWS Cost and Usage Reports to create a report with hourly granularity. Integrate the report data with Amazon Athena. Use Amazon EventBridge to schedule an Athena query. Configure an Amazon Simple Notification Service (Amazon SNS) topic to receive a notification when a threshold is exceeded.

**Correct Answer: C**

Community vote distribution

C (95%)

5%

## Comments

**Aninina** **Highly Voted** 2 years, 4 months ago

**Selected Answer: C**

AWS Budgets allows you to create budgets for your AWS accounts and set alerts when usage exceeds a certain threshold. By creating a budget for each account, specifying the period as monthly and the scope as EC2 instances, you can effectively track the EC2 usage for each account and be notified when a threshold is exceeded. This solution is the most cost-effective option as it does not require additional resources such as Amazon Athena or Amazon EventBridge.

upvoted 15 times

**alexleely** **Highly Voted** 2 years, 4 months ago

C: AWS Budgets allows you to set a budget for costs and usage for your accounts and you can set alerts when the budget threshold is exceeded in real-time which meets the requirement.

Why not B: B would be the most cost-effective if the requirements didn't ask for real-time notification. You would not incur additional costs for the daily or monthly reports and the notifications. But doesn't provide real-time alerts.

upvoted 5 times

**926ba59** **Most Recent** 1 month, 2 weeks ago

**Selected Answer: C**

AWS Budgets acts as a proactive "guardrail" setting limits and sending alerts. AWS Cost Explorer acts as a diagnostic tool for analyzing historical data to understand, visualize, and identify trends in spend. Use both together: Budgets for alerting, Cost Explorer for analysis

upvoted 1 times

**satyaamm** 1 year, 3 months ago

**Selected Answer: C**

AWS budgets allow for creating budgets and hence are the most suitable here.

upvoted 1 times

**Ruffyt** 1 year, 6 months ago

AWS Budgets allows you to create budgets for your AWS accounts and set alerts when usage exceeds a certain threshold. By creating a budget for each account, specifying the period as monthly and the scope as EC2 instances, you can effectively track the EC2 usage for each account and be

notified when a threshold is exceeded. This solution is the most cost-effective option as it does not require additional resources such as Amazon Athena or Amazon EventBridge.

upvoted 1 times

**vijaykamal** 1 year, 8 months ago

**Selected Answer: C**

Option A and Option B suggest using Cost Explorer to create reports and send notifications. While Cost Explorer is useful for analyzing costs, it does not provide the real-time alerting capability that AWS Budgets offers.

Option D suggests using AWS Cost and Usage Reports integrated with Amazon Athena and Amazon EventBridge, which can be a more complex and potentially costlier solution compared to AWS Budgets for this specific use case. It's also more suitable for fine-grained, custom analytics rather than straightforward threshold-based alerts.

upvoted 4 times

**TariqKipkemei** 1 year, 8 months ago

**Selected Answer: C**

AWS Budgets was designed to handle this scenario.

upvoted 3 times

**Undisputed** 1 year, 10 months ago

**Selected Answer: C**

Use AWS Budgets to create a cost budget for each account. Set the period to monthly. Set the scope to EC2 instances. Set an alert threshold for the budget. Configure an Amazon Simple Notification Service (Amazon SNS) topic to receive a notification when a threshold is exceeded.

upvoted 2 times

**cookieMr** 1 year, 11 months ago

**Selected Answer: C**

By creating a cost budget for each account, specifying the period as monthly and scoping it to EC2, you can track and monitor the costs associated with EC2 specifically. Set an alert threshold in the budget, which will trigger a notification when the specified threshold is exceeded. Configure an SNS to receive the notification, which can be subscribed to by the company to receive immediate alerts.

A and B are not the most cost-effective solutions as they involve using Cost Explorer to create reports, which may not provide real-time notifications when the threshold is exceeded. Additionally, A. suggests using a daily report, while B. suggests using a monthly report, which may not provide the desired level of granularity for immediate notifications.

D involves using Cost and Usage Reports with Athena and EventBridge. This solution provides more flexibility and data analysis capabilities, it is more complex and may incur additional costs for using Athena and generating hourly reports.

upvoted 3 times

**Samuel03** 2 years, 3 months ago

**Selected Answer: D**

I go with D. It says "as soon as", "daily" reports seems to be a bit longer time frame to wait in my opinion.

upvoted 2 times

**Samuel03** 2 years, 3 months ago

Actually, I take that back. It clearly says "Cost effective."

upvoted 4 times

**Bofi** 2 years, 3 months ago

Athena can only be used in S3, that is enough to discard D

upvoted 2 times

**mp165** 2 years, 4 months ago

**Selected Answer: C**

Agree...C

upvoted 3 times

**mhmt4438** 2 years, 4 months ago

**Selected Answer: C**

Answer is C

upvoted 2 times

**venice1234** 2 years, 4 months ago

**Selected Answer: C**

<https://aws.amazon.com/getting-started/hands-on/control-your-costs-free-tier-budgets/>

upvoted 2 times

**Morinator** 2 years, 5 months ago

**Selected Answer: C**

AWS budget IMO, it's done for it  
upvoted 3 times

A solutions architect needs to design a new microservice for a company's application. Clients must be able to call an HTTPS endpoint to reach the microservice. The microservice also must use AWS Identity and Access Management (IAM) to authenticate calls. The solutions architect will write the logic for this microservice by using a single AWS Lambda function that is written in Go 1.x.

Which solution will deploy the function in the MOST operationally efficient way?

- A. Create an Amazon API Gateway REST API. Configure the method to use the Lambda function. Enable IAM authentication on the API. **Most Voted**
- B. Create a Lambda function URL for the function. Specify AWS\_IAM as the authentication type.
- C. Create an Amazon CloudFront distribution. Deploy the function to Lambda@Edge. Integrate IAM authentication logic into the Lambda@Edge function.
- D. Create an Amazon CloudFront distribution. Deploy the function to CloudFront Functions. Specify AWS\_IAM as the authentication type.

**Correct Answer: A**

Community vote distribution

A (51%)

B (49%)

## Comments

**mhmt4438** **Highly Voted** 3 years, 4 months ago

**Selected Answer: A**

A. Create an Amazon API Gateway REST API. Configure the method to use the Lambda function. Enable IAM authentication on the API. This option is the most operationally efficient as it allows you to use API Gateway to handle the HTTPS endpoint and also allows you to use IAM to authenticate the calls to the microservice. API Gateway also provides many additional features such as caching, throttling, and monitoring, which can be useful for a microservice.

upvoted 23 times

**FlyingHawk** 1 year, 4 months ago

Simplicity:

Lambda Function URLs provide a direct HTTPS endpoint with minimal configuration. You don't need to set up API Gateway, which reduces complexity.

IAM Authentication:

Lambda Function URLs natively support IAM authentication, eliminating the need for custom logic or additional services.

upvoted 2 times

**OlehKom** **Highly Voted** 1 year, 9 months ago

The question specifically asks for the solution that is most operationally efficient, not necessarily the one with the most features (which is A).

Option B—using a Lambda function URL with AWS\_IAM authentication—is indeed the most operationally efficient because:

- 1) Minimal Configuration: Lambda function URLs are designed to quickly create HTTPS endpoints without the need for additional AWS services like API Gateway.
- 2) Built-in IAM Authentication: You can easily specify AWS\_IAM as the authentication type directly, fulfilling the requirement without extra setup.
- 3) No Extra Overhead: Unlike API Gateway, there's no need to manage complex API configurations, throttling settings, or additional API management features unless specifically required.

upvoted 7 times

**DanRuggles** **Most Recent** 1 month, 1 week ago

**Selected Answer: B**

Lambda Function URL (with AWS\_IAM authentication) is more operationally efficient than an Amazon API Gateway REST API (with Lambda proxy integration + IAM authorization enabled). docs.aws.amazon.com

upvoted 1 times

**SH\_** 3 months, 1 week ago

**Selected Answer: B**

Going with B. Lambda function would be most operationally efficient and does not need any additional service like API gateway or load balancers.

upvoted 1 times

**Clouddon** 11 months ago

**Selected Answer: B**

Correct Answer: B. Create a Lambda function URL for the function. Specify AWS\_IAM as the authentication type.

 Explanation:

AWS Lambda Function URLs provide a dedicated HTTPS endpoint for your Lambda function with built-in AWS IAM authentication — no need for API Gateway, no extra cost, and low operational overhead.

Key benefits:

 Fastest deployment of a web-accessible microservice.

 Built-in IAM-based authentication (AWS\_IAM).

 Fully managed HTTPS endpoint — ideal for simple use cases like a single Lambda function.

For a Go 1.x Lambda function that needs to be exposed via HTTPS and protected by IAM, this is the most operationally efficient approach.

upvoted 2 times

**jagga82** 11 months, 1 week ago

**Selected Answer: B**

As per Gemini

upvoted 2 times

**Charlesvg** 11 months, 3 weeks ago

**Selected Answer: A**

b is not suited for production so not operationnaly efficient

upvoted 1 times

**Yak\_Yeti** 1 year, 1 month ago

**Selected Answer: B**

Lambda function URLs with AWS\_IAM authentication provide the most streamlined and operationally efficient way to deploy a single Go Lambda function behind an HTTPS endpoint with IAM-based authentication for client calls. It minimizes the number of moving parts and configuration required, leading to easier management and lower potential for operational issues.

upvoted 2 times

**FlyingHawk** 1 year, 6 months ago

**Selected Answer: B**

Use Lambda Function URLs for simplicity, low cost, and direct integration with IAM for authentication. If your microservice evolves to require more complex features, you can later integrate with API Gateway as needed.

API Gateway can integrate with Lambda and enable IAM authentication, but it introduces more operational overhead and cost compared to a Lambda Function URL. It's unnecessary unless the application requires additional features like request validation, throttling, or custom authorizers.

upvoted 3 times

**LeonSauveterre** 1 year, 6 months ago

**Selected Answer: B**

Why is B better than A: Both options meet the requirements, but Lambda function URLs are simpler and involve less operational overhead compared to setting up and managing an API Gateway, so B for sure.

upvoted 3 times

**maryam\_sh** 1 year, 10 months ago

**Selected Answer: B**

B is the most operationally efficient solution. It provides the necessary functionality with minimal setup and cost, directly supports IAM authentication, and avoids the additional complexity and overhead associated with other options. This approach is particularly suitable for scenarios where the API requirements are straightforward and do not need the advanced features provided by API Gateway or CloudFront.

upvoted 4 times

**hro** 2 years, 2 months ago

I think from a decoupling and separation of concerns A is the answer. You dont want to have a heavy reliance on the Lambda function with you have specific services for what is being required.

there is operationally efficient incorrect and operationally efficient correct.  
So A is the best answer.

upvoted 3 times

**bujuman** 2 years, 3 months ago

**Selected Answer: B**

According to this statement "MOST operationally efficient way" and the following link related to Lambda Function URL security:  
<https://docs.aws.amazon.com/lambda/latest/dg/urls-auth.html>

upvoted 3 times

**awsgeek75** 2 years, 4 months ago

I originally voted B but after reading this article, I am not sure if A is wrong or is just badly worded.

If A actually said "Configure the [authorization] method to use the Lambda function" then it would be way more logical than B but this could be intentional. Although I think this is AWS test not IELTS so picking right answers based on small word mistakes is not the intention!

<https://docs.aws.amazon.com/apigateway/latest/developerguide/http-api-lambda-authorizer.html>

upvoted 3 times

**upliftinghut** 2 years, 5 months ago

**Selected Answer: B**

A& B are all good. The requirement is most operationally efficient so B is faster. In real life, I won't risk B for production, dev & test makes sense but no production please

upvoted 2 times

**upliftinghut** 2 years, 5 months ago

Reference link why B is here: <https://aws.amazon.com/blogs/aws/announcing-aws-lambda-function-urls-built-in-https-endpoints-for-single-function-microservices/>

upvoted 4 times

**awsgeek75** 2 years, 5 months ago

**Selected Answer: B**

I think this question has 2 answers as both A and B will work. However, B is more operationally efficient due to Lambda function URL and direct support for AWS\_IAM as the auth type for this setup.

<https://docs.aws.amazon.com/lambda/latest/dg/urls-auth.html#urls-auth-iam>

C, D are not operationally efficient and GO is not supported on Lambda@Edge or CloudFront functions. Even if AWS start supporting it, the operational efficiency will increase because of CloudFront

upvoted 2 times

**awsgeek75** 2 years, 5 months ago

\* meant to say "operational efficiency will decrease" because of CloudFront

upvoted 2 times

**pentium75** 2 years, 5 months ago

**Selected Answer: B**

We know that the application provides "an HTTPS endpoint" but we don't even know whether it is a REST API. The question is not mentioning any other requirements besides IAM authentication, which can be handled by Lambda alone.

A would work, but would be an additional processing step (lowering operational efficiency). It would also provide benefits but none of those is asked for in the question.

C and D is wrong because Lambda@Edge does not support Go.

upvoted 3 times



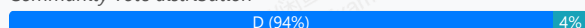
A company previously migrated its data warehouse solution to AWS. The company also has an AWS Direct Connect connection. Corporate office users query the data warehouse using a visualization tool. The average size of a query returned by the data warehouse is 50 MB and each webpage sent by the visualization tool is approximately 500 KB. Result sets returned by the data warehouse are not cached.

Which solution provides the LOWEST data transfer egress cost for the company?

- A. Host the visualization tool on premises and query the data warehouse directly over the internet.
- B. Host the visualization tool in the same AWS Region as the data warehouse. Access it over the internet.
- C. Host the visualization tool on premises and query the data warehouse directly over a Direct Connect connection at a location in the same AWS Region.
- D. Host the visualization tool in the same AWS Region as the data warehouse and access it over a Direct Connect connection at a location in the same Region. **Most Voted**

**Correct Answer: D**

Community vote distribution



## Comments

**AlessandraSAA** **Highly Voted** 3 years, 3 months ago

**Selected Answer: D**

- A. --> No since if you access via internet you are creating egress traffic.
- B. --> It's a good choice to have both DWH and visualization in the same region to lower the egress transfer (i.e. data going egress/out of the region) but if you access over internet you might still have egress transfer.
- C. --> Valid but in this case you send out of AWS 50MB if you query the DWH instead of the visualization tool, D removes this need since puts the visualization tools in AWS with the DWH so reduces data returned out of AWS from 50MB to 500KB
- D. --> Correct, see explanation on answer C

Useful links:

AWS Direct Connect connection create a connection in an AWS Direct Connect location to establish a network connection from your premises to an AWS Region.

<https://docs.aws.amazon.com/directconnect/latest/UserGuide/Welcome.html>

upvoted 16 times

**hro** 2 years, 2 months ago

I thought Direct Connect is exclusively for on-prem to VPC. Why would you use DirectConnect to connect to a service in your own Region in the AWS Cloud?

I believe C is correct.

upvoted 4 times

**chickenmf** 2 years, 2 months ago

Agree with you 100%

upvoted 1 times

**Jacky\_S** 1 year, 11 months ago

It is not quite making sense that the answer D to mention on AWS Direct Connect if the tool is staying with AWS already. That is the main reason i vote on C rather than D.

upvoted 4 times

**mhmt4438** **Highly Voted** 3 years, 4 months ago

**Selected Answer: D**

Correct answer is D

upvoted 5 times

**LeonSauveterre** Most Recent 1 year, 6 months ago

**Selected Answer: D**

Why C is Less Cost-Effective Than D: While Direct Connect minimizes costs in both cases, hosting the visualization tool on premises (C) still incurs some egress costs when pulling large query results to the corporate office. Hosting the visualization tool in the same AWS Region as the data warehouse (D) eliminates most egress costs \*\*entirely\*\* due to AWS's intra-region free data transfer policy.

upvoted 2 times

**Jazz888** 2 years ago

This is question that need a careful reading of the question. Please note: in the question that the company has a DX connection but it doesn't mention the company is utilizing the DX to query the data warehouse (this indicates that the query was using internet). So my verdict would be to place the visualization tool on-prem (which it already is) and use the DX to query the data warehouse to reduce cost of (DTO) so I vote for C

upvoted 3 times

**hro** 2 years, 2 months ago

The answer is C. At no point does the question suggest that the DWH source is out of Region.

upvoted 2 times

**reviewmine** 2 years, 3 months ago

**Selected Answer: D**

It's D. Host the visualization on the same region to avoid egress cost and access the tool via AWS Direct connection.

upvoted 3 times

**upliftinghut** 2 years, 5 months ago

**Selected Answer: D**

Leverage the existing DirectConnect so not incur data transfer charge

upvoted 2 times

**Ruffyt** 2 years, 6 months ago

D. Host the visualization tool in the same AWS Region as the data warehouse and access it over a Direct Connect connection at a location in the same Region.

upvoted 2 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: D**

Host the visualization tool in the same AWS Region as the data warehouse and access it over a Direct Connect connection at a location in the same Region

upvoted 2 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: D**

D. Host the visualization tool in the same AWS Region as the data warehouse and access it over a Direct Connect connection at a location in the same Region.

upvoted 2 times

**jtexam** 2 years, 11 months ago

**Selected Answer: B**

by hosting in same region, you have 500kb transfer charged on internet transfer teir, 50MB charged in inter-region tier.

using direct link, both are charged in direct link tier. direct link tier is not cheap.

so i go for B

upvoted 1 times

**pentium75** 2 years, 5 months ago

We don't want "lowest cost", we want "lowest data transfer egress cost". And "data transfer egress" cost for Direct Connect is WAY lower than for Internet. (Therefor the circuit itself is expensive, but that does exist anyway.)

upvoted 2 times

**Mmmmmmkkkk** 2 years, 11 months ago

Aaaaaaaa

upvoted 1 times

**cookieMr** 2 years, 11 months ago

**Selected Answer: D**

Hosting the visualization tool in the same AWS Region as the data warehouse and accessing it over a Direct Connect connection within the same Region eliminates data transfer fees and ensures low-latency, high-bandwidth connectivity.

A. Hosting the visualization tool on premises and querying the data warehouse over the internet incurs data transfer costs for every query result, as

well as potential latency and bandwidth limitations.

B. Hosting the visualization tool in the same AWS Region as the data warehouse but accessing it over the internet still incurs data transfer costs for each query result.

C. Hosting the visualization tool on premises and querying the data warehouse over a Direct Connect connection within the same AWS Region incurs data transfer costs for every query result and adds complexity by requiring on-premises infrastructure.

upvoted 5 times

**dexpos** 3 years, 4 months ago

**Selected Answer: D**

D let you reduce at minimum the data transfer costs

upvoted 2 times

**alexleely** 3 years, 4 months ago

**Selected Answer: D**

D: Direct Connect connection at a location in the same Region will provide the lowest data transfer egress cost, improved performance, and lower complexity

Why it is not C is because the visualization tool is hosted on-premises, as it's not hosted in the same region as the data warehouse the data transfer between them would occur over the internet, thus, would incur in egress data transfer costs.

upvoted 5 times

**markw92** 2 years, 11 months ago

C option doesnt travel through internet because we have a direct connect. If you are hosting your visualization tool in same region why you need a direct connection which D has? Doesn't make sense. So, C is the right answer.

upvoted 1 times

**Vickysss** 3 years, 4 months ago

**Selected Answer: C**

<https://www.nops.io/reduce-aws-data-transfer-costs-dont-get-stung-by-hefty-egress-fees/>

upvoted 2 times

**JayBee65** 3 years, 4 months ago

Whilst "Direct Connect can help lower egress costs even after taking the installation costs into account. This is because AWS charges lower transfer rates." D removes the need to send the query results out of AWS and instead returns the web page, so reduces data returned from 50MB to 500KB, so D

upvoted 3 times

**Aninina** 3 years, 4 months ago

**Selected Answer: D**

Should be D

<https://aws.amazon.com/directconnect/pricing/>

<https://aws.amazon.com/blogs/aws/aws-data-transfer-prices-reduced/>

upvoted 3 times

An online learning company is migrating to the AWS Cloud. The company maintains its student records in a PostgreSQL database. The company needs a solution in which its data is available and online across multiple AWS Regions at all times.

Which solution will meet these requirements with the LEAST amount of operational overhead?

- A. Migrate the PostgreSQL database to a PostgreSQL cluster on Amazon EC2 instances.
- B. Migrate the PostgreSQL database to an Amazon RDS for PostgreSQL DB instance with the Multi-AZ feature turned on.
- C. Migrate the PostgreSQL database to an Amazon RDS for PostgreSQL DB instance. Create a read replica in another Region. **Most Voted**
- D. Migrate the PostgreSQL database to an Amazon RDS for PostgreSQL DB instance. Set up DB snapshots to be copied to another Region.

**Correct Answer: C**

Community vote distribution

C (82%)

B (18%)

## Comments

**Steve\_4542636** **Highly Voted** 3 years, 3 months ago

**Selected Answer: C**

Multi az is not the same as multi regional  
upvoted 48 times

**alexleely** **Highly Voted** 3 years, 4 months ago

**Selected Answer: B**

B: Amazon RDS Multi-AZ feature automatically creates a synchronous replica in another availability zone and failover to the replica in the event of an outage. This will provide high availability and data durability across multiple AWS regions which fit the requirements.

Though C may sound good, it in fact requires manual management and monitoring of the replication process due to the fact that Amazon RDS read replicas are asynchronous, meaning there is a delay between the primary and read replica. Therefore, there will be a need to ensure that the read replica is constantly up-to-date and someone still has to fix any read replica errors during the replication process which may cause data inconsistency. Lastly, you still have to configure additional steps to make it fail over to the read replica.

upvoted 16 times

**Rehan33** 3 years, 3 months ago

I go with option B because:  
Multi-AZ is for high availability  
Read replicas are for low-latency  
in question they talk about available online  
upvoted 4 times

**awsgeek75** 2 years, 4 months ago

They also ask for multiple regions which is not covered by Multi AZ  
upvoted 3 times

**Mahadeva** 3 years, 4 months ago

But the question is clearly asking for Multiple Regions. Multi-AZ is not across Regions.  
upvoted 22 times

**alexleely** 3 years, 4 months ago

You are right, Multi-AZ is only within one Region. C would be the right answer.  
upvoted 15 times

**smartegnine** 2 years, 12 months ago

<https://aws.amazon.com/rds/features/multi-az/>

smartegnine 0 minutes ago Awaiting moderator approval

Selected Answer: B

In an Amazon RDS Multi-AZ deployment, Amazon RDS automatically creates a primary database (DB) instance and synchronously replicates the data to an instance in a different AZ.

upvoted 1 times

**hasiet** **Most Recent** 1 year, 6 months ago

**Selected Answer: C**

C. Migrate the PostgreSQL database to an Amazon RDS for PostgreSQL DB instance. Create a read replica in another Region.

This option provides cross-region availability with minimal operational overhead, as RDS manages the replication process for you.

upvoted 2 times

**akshay243007** 1 year, 10 months ago

**Selected Answer: C**

Multi AZ is not supported in to diffrent region

read Raplica is supported in to diffrent region

So I am go with the option C

upvoted 2 times

**Uzbekistan** 2 years, 2 months ago

**Selected Answer: B**

Option C involves setting up read replicas in another region, which provides cross-region availability but may introduce additional complexity in managing replication and monitoring.

B. Migrate the PostgreSQL database to an Amazon RDS for PostgreSQL DB instance with Multi-AZ:

High Availability: Amazon RDS Multi-AZ deployments provide high availability and failover support for DB instances. With Multi-AZ, Amazon RDS automatically provisions and maintains a synchronous standby replica in a different Availability Zone (AZ). In the event of a failure, Amazon RDS automatically fails over to the standby replica, ensuring data availability and minimal downtime. + Cross-Region Replication.

upvoted 1 times

**pentium75** 2 years, 5 months ago

**Selected Answer: C**

A - Just no

B - Multi-AZ is multiple AZs in same region, does not meet "Multiple AWS Regions" requirement

C - Meets requirements

D - Does not meet the "online ... at all times" requirement

upvoted 11 times

**Ruffyit** 2 years, 6 months ago

Multi az is not the same as multi regional

upvoted 2 times

**vijaykamal** 2 years, 8 months ago

**Selected Answer: B**

Option C, while providing a read replica in another Region, adds complexity to the architecture and may introduce some additional operational overhead compared to Multi-AZ. Cross-Region replication involves setting up and managing replication between two separate RDS instances.

upvoted 1 times

**pentium75** 2 years, 5 months ago

But data must be online "across multiple AWS regions"!

upvoted 2 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: C**

Migrate the PostgreSQL database to an Amazon RDS for PostgreSQL DB instance. Create a read replica in another Region

upvoted 3 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: C**

Multi-AZ is not the same as Multi-Regional

upvoted 4 times

**Valder21** 2 years, 9 months ago

can someone explain why not D  
upvoted 3 times

**pentium75** 2 years, 5 months ago

"Data must be online (!) across multiple AWS regions at all times (!)". Snapshots are not online.  
upvoted 3 times

**beginnercloud** 2 years, 9 months ago

**Selected Answer: C**

key words "AWS Regions at all times" so C is correct  
upvoted 2 times

**fuzzycr** 2 years, 11 months ago

**Selected Answer: C**

key words "AWS Regions at all times"  
upvoted 2 times

**cookieMr** 2 years, 11 months ago

**Selected Answer: C**

By migrating the PostgreSQL database to an RDS for PostgreSQL DB instance and creating a read replica in another AWS Region, you can achieve data availability and online access across multiple Regions. This solution requires less operational overhead compared to managing a PostgreSQL cluster on EC2 instances (Option A) or setting up manual replication using snapshots (Option D). Additionally, Amazon RDS handles the underlying infrastructure and replication setup, reducing the operational complexity for the company.

Option B, is a valid solution for achieving high availability within a single AWS Region. However, it does not meet the requirement of having the data available and online across multiple AWS Regions at all times, which is specified in the question. The Multi-AZ feature in RDS provides automatic failover within the same Region, but it does not replicate the data to multiple Regions.

upvoted 6 times

**mal1903** 2 years, 12 months ago

**Selected Answer: B**

C and D just specify another single region. This does not translate to multiple regions.

B (Multi-AZ) means the solution will be highly available.

The data will be available in multiple regions for both B and C but B is a better solution!

upvoted 1 times

**Guru4Cloud** 2 years, 8 months ago

its data is available and online across multiple AWS Regions at all times  
upvoted 1 times

**pentium75** 2 years, 5 months ago

"Another single region" plus the original region ARE of course "multiple regions", what else? While B "Multi-AZ" is two AZs in the same region.  
upvoted 2 times

**MrAWSAssociate** 2 years, 12 months ago

**Selected Answer: C**

Answer B is not right, because "RDS Multi-AZ" always span at least two Availability Zones within a single region and the question requirement RDS DB should be available in multiple regions. Therefore, C is the most suitable answer for this question.

upvoted 3 times

**MrAWSAssociate** 2 years, 12 months ago

I would like to change my answer to "B". The question has some distractor words: "its data is available and online across multiple AWS Regions at all times". We agree that AWS is a could service available online around the world in 99 regions. So the option "B" is the most appropriate answer, since multi-AZ focuses on the availability factor and it has the LEAST amount of operational overhead.

upvoted 1 times

**abhishek2021** 2 years, 12 months ago

**Selected Answer: B**

B & C both makes data available. However, B is less overhead.

What I think, the question is asking for data availability across multiple regions not for a DR solution. So, RDS being accessible over public IP will do the trick for data being available across regions.

upvoted 1 times

**Guru4Cloud** 2 years, 8 months ago

Multi-AZ is not the same as Multi-Regional  
upvoted 1 times



A company hosts its web application on AWS using seven Amazon EC2 instances. The company requires that the IP addresses of all healthy EC2 instances be returned in response to DNS queries.

Which policy should be used to meet this requirement?

- A. Simple routing policy
- B. Latency routing policy
- C. Multivalue routing policy **Most Voted**
- D. Geolocation routing policy

**Correct Answer: C**

Community vote distribution

C (97%)

3%

## Comments

**cookieMr** **Highly Voted** 2 years, 5 months ago

The Multivalue routing policy allows Route 53 to respond to DNS queries with multiple healthy IP addresses for the same resource. This is particularly useful in scenarios where multiple instances are serving the same purpose and need to be load balanced or failover capable. With the Multivalue routing policy, Route 53 returns multiple IP addresses in a random order to distribute the traffic across all healthy instances.

Option A (Simple routing policy) would only return a single IP address in response to DNS queries and does not support returning multiple addresses.

Option B (Latency routing policy) is used to route traffic based on the lowest latency to the resource and does not fulfill the requirement of returning all healthy IP addresses.

Option D (Geolocation routing policy) is used to route traffic based on the geographic location of the user and does not fulfill the requirement of returning all healthy IP addresses.

Therefore, the Multivalue routing policy is the most suitable option for returning the IP addresses of all healthy EC2 instances in response to DNS queries.

upvoted 17 times

**LuckyAro** **Highly Voted** 2 years, 10 months ago

**Selected Answer: C**

Use a multivalue answer routing policy to help distribute DNS responses across multiple resources. For example, use multivalue answer routing when you want to associate your routing records with a Route 53 health check. For example, use multivalue answer routing when you need to return multiple values for a DNS query and route traffic to multiple IP addresses.

<https://aws.amazon.com/premiumsupport/knowledge-center/multivalue-versus-simple-policies/>

upvoted 14 times

**DanRuggles** **Most Recent** 1 month, 1 week ago

**Selected Answer: C**

I picked C, but Latency could also work. For each latency record pointing to an EC2 instance (non-alias record): Create a Route 53 health check that monitors the EC2 instance (HTTP, HTTPS, or TCP on a specific port/path). Associate that health check with the latency record.

Route 53 continuously runs these health checks (every 10–30 seconds by default).

upvoted 1 times

**awsgeek75** 1 year, 11 months ago

**Selected Answer: C**

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy.html#routing-policy-multivalue>



"Multivalue answer routing policy – Use when you want Route 53 to respond to DNS queries with up to eight healthy records selected at random. You can use multivalue answer routing to create records in a private hosted zone."

Company requires that the IP addresses of "ALL" healthy EC2 instances be returned so C is the only option.

upvoted 5 times

**Ruffyt** 2 years ago

Use a multivalue answer routing policy to help distribute DNS responses across multiple resources. For example, use multivalue answer routing when you want to associate your routing records with a Route 53 health check. For example, use multivalue answer routing when you need to return multiple values for a DNS query and route traffic to multiple IP addresses.

upvoted 2 times

**TariqKipkemei** 2 years, 2 months ago

**Selected Answer: C**

Use Multivalue answer routing policy when you want Route 53 to respond to DNS queries with up to eight healthy records selected at random.

upvoted 2 times

**Guru4Cloud** 2 years, 3 months ago

**Selected Answer: C**

C. Multivalue routing policy

upvoted 2 times

**Guru4Cloud** 2 years, 3 months ago

**Selected Answer: A**

A. Deploy an AWS Storage Gateway file gateway as a virtual machine (VM) on premises at each clinic

upvoted 1 times

**pentium75** 1 year, 11 months ago

This is for a different question I guess

upvoted 2 times

**animefan1** 2 years, 5 months ago

multivalue supports health checks

upvoted 2 times

**MLCL** 2 years, 8 months ago

IP are returned RANDOMLY for multi-value Routing, is this what we want ?

upvoted 4 times

**WheracanIstart** 2 years, 9 months ago

**Selected Answer: C**

Multivalue answer routing policy ...answer is C

upvoted 2 times

**mhmt4438** 2 years, 11 months ago

**Selected Answer: C**

Answer is C

upvoted 3 times

**Aninina** 2 years, 11 months ago

**Selected Answer: C**

Should be C

upvoted 2 times

**bamishr** 2 years, 11 months ago

**Selected Answer: C**

<https://www.examttopics.com/discussions/amazon/view/46491-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 2 times

**Morinator** 2 years, 11 months ago

**Selected Answer: C**

<https://www.examttopics.com/discussions/amazon/view/46491-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 2 times



A medical research lab produces data that is related to a new study. The lab wants to make the data available with minimum latency to clinics across the country for their on-premises, file-based applications. The data files are stored in an Amazon S3 bucket that has read-only permissions for each clinic.

What should a solutions architect recommend to meet these requirements?

- A. Deploy an AWS Storage Gateway file gateway as a virtual machine (VM) on premises at each clinic **Most Voted**
- B. Migrate the files to each clinic's on-premises applications by using AWS DataSync for processing.
- C. Deploy an AWS Storage Gateway volume gateway as a virtual machine (VM) on premises at each clinic.
- D. Attach an Amazon Elastic File System (Amazon EFS) file system to each clinic's on-premises servers.

**Correct Answer: A**

Community vote distribution

A (97%)

3%

## Comments

**mhmt4438** **Highly Voted** 2 years, 11 months ago

**Selected Answer: A**

A. Deploy an AWS Storage Gateway file gateway as a virtual machine (VM) on premises at each clinic

AWS Storage Gateway is a service that connects an on-premises software appliance with cloud-based storage to provide seamless and secure integration between an organization's on-premises IT environment and AWS's storage infrastructure. By deploying a file gateway as a virtual machine on each clinic's premises, the medical research lab can provide low-latency access to the data stored in the S3 bucket while maintaining read-only permissions for each clinic. This solution allows the clinics to access the data files directly from their on-premises file-based applications without the need for data transfer or migration.

upvoted 24 times

**cookieMr** **Highly Voted** 2 years, 5 months ago

A. It allows the clinics to access the data files stored in the S3 bucket through a file interface. The file gateway caches frequently accessed data locally reducing latency and providing fast access to the data.

B. It involves transferring the data files from the Amazon S3 bucket to each clinic's on-premises applications using AWS DataSync. While this enables data migration, it may not provide real-time access and may introduce additional latency.

C. It is suitable for block-level access to data rather than file-level access. It may not be the most efficient solution for file-based applications.

D. It involves using Amazon EFS, which is a scalable file storage service, to provide file-level access to the data. However, it may introduce additional complexity and latency compared to using a file gateway solution.

upvoted 8 times

**satyaamm** **Most Recent** 1 year, 3 months ago

**Selected Answer: A**

AWS Storage File Gateway is the most suitable here.

upvoted 1 times

**pentium75** 1 year, 11 months ago

**Selected Answer: A**

A: does exactly that is required here

B: "Migrate", as to MOVE the files out from S3, doesn't make sense

C: Volume Gateway provides iSCSI volumes backed by an object in AWS-managed S3, it does not provide access to S3 objects

D: You can do that but it would have high (not "minimum") latency, and the data is not in that EFS volume, it's in S3

upvoted 5 times

**Ruffyt** 2 years ago

AWS Storage Gateway is a service that connects an on-premises software appliance with cloud-based storage to provide seamless and secure integration between an organization's on-premises IT environment and AWS's storage infrastructure. By deploying a file gateway as a virtual machine on each clinic's premises, the medical research lab can provide low-latency access to the data stored in the S3 bucket while maintaining read-only permissions for each clinic. This solution allows the clinics to access the data files directly from their on-premises file-based applications without the need for data transfer or migration.

upvoted 2 times

**TariqKipkemei** 2 years, 2 months ago

**Selected Answer: A**

The Amazon S3 File Gateway enables you to store and retrieve objects in Amazon Simple Storage Service (S3) using file protocols such as Network File System (NFS) and Server Message Block (SMB). Objects written through S3 File Gateway can be directly accessed in S3.

upvoted 4 times

**Guru4Cloud** 2 years, 3 months ago

**Selected Answer: A**

A. Deploy an AWS Storage Gateway file gateway as a virtual machine (VM) on premises at each clinic

upvoted 3 times

**Bmarodi** 2 years, 6 months ago

**Selected Answer: A**

Option A meets the requirements.

upvoted 2 times

**jaswantn** 2 years, 7 months ago

For File-based applications use File Gateway: (Option A)

upvoted 2 times

**Grace83** 2 years, 8 months ago

Definitely A.

Why are there so many wrong answers by Admins?

upvoted 5 times

**maggie135** 2 years, 5 months ago

I guess to force us to read and think, so one can't just memorize the answer and go to exam ?)

upvoted 3 times

**AlessandraSAA** 2 years, 9 months ago

**Selected Answer: A**

Amazon S3 File Gateway enables you to store file data as objects in Amazon S3 cloud storage for data lakes, backups, and Machine Learning workflows. With Amazon S3 File Gateway, each file is stored as an object in Amazon S3 with a one-to-one mapping between a file and an object.

Volume Gateway provides block storage volumes over iSCSI, backed by Amazon S3, and provides point-in-time backups as Amazon EBS snapshots. Volume Gateway integrates with AWS Backup, an automated and centralized backup service, to protect Storage Gateway volumes.

So it's A

upvoted 5 times

**Steve\_4542636** 2 years, 9 months ago

**Selected Answer: A**

A for answer

upvoted 2 times

**bdp123** 2 years, 10 months ago

**Selected Answer: A**

<https://cloud.in28minutes.com/aws-certification-aws-storage-gateway>

upvoted 2 times

**kbaru** 2 years, 10 months ago

**Selected Answer: A**

A. Deploy an AWS Storage Gateway file gateway...

upvoted 2 times

**imisioluwa** 2 years, 10 months ago

**Selected Answer: A**

The correct answer is A.

<https://www.knowledgehut.com/tutorials/aws/aws-storage-gateway#:~:text=AWS%20Storage%20Gateway%20helps%20in%20connecting,as%20well%20as%20providing%20data%20security.&text=AWS%20Storage%20Gateway%20helps,as%20providing%20data%20security.&text=Gateway%20helps%20in%20connecting,as%20well%20as%20providing>  
<https://docs.aws.amazon.com/storagegateway/latest/vgw/WhatIsStorageGateway.html>

upvoted 2 times

**venice1234** 2 years, 11 months ago

**Selected Answer: C**

I think C (Volume Gateway) is correct as it has an option to have Local Storage with Asynchronous sync with S3. This would give low latency access to all local files not just cached/recent files.

upvoted 2 times

**pentium75** 1 year, 11 months ago

Volume gateway provides an iSCSI volume and stores that as a single object in an AWS-managed S3 bucket. It does not provide access to S3 objects.

upvoted 2 times

**laicos** 2 years, 11 months ago

**Selected Answer: A**

<https://aws.amazon.com/storagegateway/file/>

upvoted 2 times

A company is using a content management system that runs on a single Amazon EC2 instance. The EC2 instance contains both the web server and the database software. The company must make its website platform highly available and must enable the website to scale to meet user demand.

What should a solutions architect recommend to meet these requirements?

- A. Move the database to Amazon RDS, and enable automatic backups. Manually launch another EC2 instance in the same Availability Zone. Configure an Application Load Balancer in the Availability Zone, and set the two instances as targets.
- B. Migrate the database to an Amazon Aurora instance with a read replica in the same Availability Zone as the existing EC2 instance. Manually launch another EC2 instance in the same Availability Zone. Configure an Application Load Balancer, and set the two EC2 instances as targets.
- C. Move the database to Amazon Aurora with a read replica in another Availability Zone. Create an Amazon Machine Image (AMI) from the EC2 instance. Configure an Application Load Balancer in two Availability Zones. Attach an Auto Scaling group that uses the AMI across two Availability Zones. **Most Voted**
- D. Move the database to a separate EC2 instance, and schedule backups to Amazon S3. Create an Amazon Machine Image (AMI) from the original EC2 instance. Configure an Application Load Balancer in two Availability Zones. Attach an Auto Scaling group that uses the AMI across two Availability Zones.

**Correct Answer: C**

Community vote distribution

C (98%)

2

## Comments

**mhmt4438** **Highly Voted** 2 years, 11 months ago

**Selected Answer: C**

C. Move the database to Amazon Aurora with a read replica in another Availability Zone. Create an Amazon Machine Image (AMI) from the EC2 instance. Configure an Application Load Balancer in two Availability Zones. Attach an Auto Scaling group that uses the AMI across two Availability Zones.

This approach will provide both high availability and scalability for the website platform. By moving the database to Amazon Aurora with a read replica in another availability zone, it will provide a failover option for the database. The use of an Application Load Balancer and an Auto Scaling group across two availability zones allows for automatic scaling of the website to meet increased user demand. Additionally, creating an AMI from the original EC2 instance allows for easy replication of the instance in case of failure.

upvoted 23 times

**Bmarodi** 2 years, 6 months ago

Very good explanations!

upvoted 2 times

**cookieMr** **Highly Voted** 2 years, 5 months ago

**Selected Answer: C**

Option A does not provide a solution for high availability or scalability. Manually launching another EC2 instance in the same AZ may not ensure high availability, as a failure in that AZ would result in downtime.

Option B improves database performance and provides a level of fault tolerance, it does not address the scalability aspect of the website platform.

Option C provides both high availability and fault tolerance. Creating an AMI allows for easy replication of the EC2 instance across AZs. Configuring an ALB in two AZs and attaching an ASG ensures scalability and load distribution across multiple instances.

Option D does not provide the high availability and scalability required by the company. Scheduled backups to S3 address data protection but do not contribute to website availability or scalability.

upvoted 6 times

**satyaamm** Most Recent 1 year, 3 months ago

**Selected Answer: C**

Aurora with read replica in different region is the most suitable here.

upvoted 1 times

**awsgeek75** 1 year, 11 months ago

**Selected Answer: C**

C is the only option via deduction logic based on the assumption the CMS database is Aurora compatible. Other solutions don't promise scaling as much as Aurora solution in option C does.

upvoted 2 times

**awsgeek75** 1 year, 10 months ago

Just to clarify, the question is vague as we don't know anything about the DB types on the CMS so making an assumption that Aurora will work with the CMS.

upvoted 4 times

**pentium75** 1 year, 11 months ago

**Selected Answer: C**

A and B involve manual steps and do not include scaling (it's just two fixed instances)

D scales the application part but leaves the database on a single EC2 instance which would be neither "highly available" nor "scaleable"

upvoted 5 times

**Ruffyit** 2 years ago

C. Move the database to Amazon Aurora with a read replica in another Availability Zone. Create an Amazon Machine Image (AMI) from the EC2 instance. Configure an Application Load Balancer in two Availability Zones. Attach an Auto Scaling group that uses the AMI across two Availability Zones.

upvoted 2 times

**TariqKipkemei** 2 years, 2 months ago

**Selected Answer: C**

Move the database to Amazon Aurora with a read replica in another Availability Zone. Create an Amazon Machine Image (AMI) from the EC2 instance. Configure an Application Load Balancer in two Availability Zones. Attach an Auto Scaling group that uses the AMI across two Availability Zones.

upvoted 2 times

**Guru4Cloud** 2 years, 3 months ago

**Selected Answer: C**

C. Move the database to Amazon Aurora with a read replica in another Availability Zone. Create an Amazon Machine Image (AMI) from the EC2 instance. Configure an Application Load Balancer in two Availability Zones. Attach an Auto Scaling group that uses the AMI across two Availability Zones.

upvoted 2 times

**MutiverseAgent** 2 years, 4 months ago

**Selected Answer: D**

The question does not say if the current application is using a relational database, so how we can be sure that it can be moved to RDS or Aurora as answers A, B & C states? In my opinion the right answer is D.

upvoted 1 times

**pentium75** 1 year, 11 months ago

In D, you "move the database to a separate EC2 instance" BEFORE creating the AMI for the Auto Scaling group. So you'd still have a single EC2 instance running the database, which would meet neither the availability nor the scalability requirement.

upvoted 2 times

**animefan1** 2 years, 5 months ago

**Selected Answer: C**

has all options needed for HA

upvoted 2 times

**Bmarodi** 2 years, 6 months ago

**Selected Answer: C**

Option C meets the requirements.

upvoted 2 times

**ssoffline** 2 years, 6 months ago

Why not D?

Are we just assuming that there will be no write to the db?

upvoted 1 times

**antropaws** 2 years, 6 months ago

**Selected Answer: C**

Absolutely C.

upvoted 2 times

**Aninina** 2 years, 11 months ago

**Selected Answer: C**

C: This will allow the website platform to be highly available by using Aurora, which provides automatic failover and replication. Additionally, by creating an AMI from the original EC2 instance, the Auto Scaling group can automatically launch new instances in multiple availability zones and use the Application Load Balancer to distribute traffic across them. This way, the website will be able to handle the increased traffic, and will be less likely to go down due to a single point of failure.

upvoted 3 times



A company is launching an application on AWS. The application uses an Application Load Balancer (ALB) to direct traffic to at least two Amazon EC2 instances in a single target group. The instances are in an Auto Scaling group for each environment. The company requires a development environment and a production environment. The production environment will have periods of high traffic.

Which solution will configure the development environment MOST cost-effectively?

- A. Reconfigure the target group in the development environment to have only one EC2 instance as a target.
- B. Change the ALB balancing algorithm to least outstanding requests.
- C. Reduce the size of the EC2 instances in both environments.
- D. Reduce the maximum number of EC2 instances in the development environment's Auto Scaling group. **Most Voted**

**Correct Answer:** D

Community vote distribution

D (62%) A (37%)

## Comments

**mhmt4438** **Highly Voted** 3 years, 4 months ago

**Selected Answer: D**

D. Reduce the maximum number of EC2 instances in the development environment's Auto Scaling group

This option will configure the development environment in the most cost-effective way as it reduces the number of instances running in the development environment and therefore reduces the cost of running the application. The development environment typically requires less resources than the production environment, and it is unlikely that the development environment will have periods of high traffic that would require a large number of instances. By reducing the maximum number of instances in the development environment's Auto Scaling group, the company can save on costs while still maintaining a functional development environment.

upvoted 20 times

**JayBee65** 3 years, 4 months ago

No, it will not reduce the number of instances being used, since a minimum of 2 will be used at all times.

upvoted 10 times

**pentium75** 2 years, 5 months ago

But it will keep the number of instances at two, while the production environment has "AT LEAST two".

upvoted 5 times

**pentium75** **Highly Voted** 2 years, 5 months ago

**Selected Answer: D**

A modifies only the ALB target group (= directs traffic only to one node), but does not affect the number of nodes (and the cost)

B balances load between nodes but does not affect the cost

C impacts the prod environment so that would be unable to handle its "periods of high traffic"

D makes sure that the dev environment will not scale to more than 2 instances, as does the prod environment

upvoted 10 times

**Yak\_Yeti** **Most Recent** 1 year, 1 month ago

**Selected Answer: A**

you cannot get cheaper than one, and D says to reduce the maximum and does not mention the minimum.

upvoted 1 times

**jerry1** 1 year, 2 months ago

**Selected Answer: D**

analysis/explanation by copilot

> A: you reduce the target group instance, doesn't mean the other one is shut down / terminated (sound tricky but could be true, it didn't mention

the other machine is terminated)

> D: that make sure only 1 machine is always up

in that case then D would make more sense (disregard the high availability concern, A is still more costly then)

upvoted 1 times

**SirDNS** 1 year, 2 months ago

**Selected Answer: A**

A. This is the most cost-effective approach for a development environment. Since development environments typically don't need the same level of high availability or performance as production, reducing the number of target EC2 instances to one minimizes costs.

D. Reducing the maximum size of the Auto Scaling group can limit scaling, but it doesn't guarantee cost savings if those instances are running.

Reconfiguring the target group to use only one instance is a more direct way to reduce costs in the development environment.

upvoted 1 times

**zdi561** 1 year, 4 months ago

**Selected Answer: A**

D is not right, if you decrease max number it will not meet the high traffic requirement. Actually it does not matter because the ASG will automatically adjust the running target. A really means set the max, min and desired to one which can save a little bit money, The requirement for two really means the min is 2 in prod. max could be higher than 2.

upvoted 1 times

**XXXXXINN** 1 year, 8 months ago

chatGPT sucks too. it says A, but A obviously just says reduced the number of Target Group, reduce Target Group does not mean the reduce of EC2 instances themselves, so there is no cost saved at all. Thus DDDDDDDDDDD....

upvoted 2 times

**LeonSauveterre** 1 year, 6 months ago

Weird... ChatGPT says to choose D when I asked about this question. It said option A compromises high availability because ALBs require at least two targets for redundancy and fault tolerance, so not a valid option due to the risk of downtime if the single instance becomes unavailable, while option D provides targeted cost optimization for the development environment without compromising availability or production performance.

upvoted 2 times

**babayomi** 1 year, 7 months ago

Answer is A, option A says reduce(rec9nfigure) the instance in target group to one, makes sense. If a group has only implemented instance, it's not autoscaling.

upvoted 1 times

**MatAlves** 1 year, 9 months ago

**Selected Answer: D**

So, in short, the question asks for a way to reduce cost wasted with the dev env, since it's resources are being underused.

(A)Target group vs (B) Auto Scaling Group.

Reducing the target group won't affect the number of "nodes" (instances), cost will stay the same. To eliminate the excess of ec2 instances in the dev env, you actually need to reduce the auto scaling group.

upvoted 3 times

**TheFivePips** 2 years, 3 months ago

**Selected Answer: D**

The application uses an Application Load Balancer (ALB) to direct traffic to at least two Amazon EC2 instances in a single target group

You are required to keep at least two instances in each target group. A sets it to one, which would be more cost effective, but doesn't meet the requirement.

upvoted 4 times

**Priyapani** 2 years, 5 months ago

**Selected Answer: D**

In the question it is said minimum it should have 2 instances in Target group. So in development group we can reduce the the target group.

In option A. It is said it will have only one instance in development group that doesn't match to our question

upvoted 5 times

**awsgeek75** 2 years, 5 months ago

**Selected Answer: D**

B and C don't actually save any cost without impacting performance during high traffic on production.

A and D are basically same thing but A enforces a limit of one EC2 instance which is not acceptable as the question asks: "Application Load Balancer (ALB) to direct traffic to at least two Amazon EC2 instances in a single target group. The instances are in an Auto Scaling group for each environment"

Hence D is the only valid answer.

upvoted 6 times

**ddaanddann** 2 years, 5 months ago

The most cost-effective solution is to reconfigure the target group in the development environment to have only one EC2 instance as a target. This will ensure that the development environment only uses the resources that it needs, which will save the company money.

The other solutions are not as cost-effective. Changing the ALB balancing algorithm to least outstanding requests will not reduce the number of EC2 instances that are used, and it may actually increase the amount of traffic that is directed to each instance. Reducing the size of the EC2 instances will also not reduce the number of instances that are used, and it may actually make the application slower. Reducing the maximum number of EC2 instances in the development environment's Auto Scaling group will only reduce the number of instances that are used when the traffic is high, and it will not reduce the number of instances that are used on average.

Therefore, the most cost-effective solution is to reconfigure the target group in the development environment to have only one EC2 instance as a target.

upvoted 1 times

**pentium75** 2 years, 5 months ago

No, you're confusing "target group" (of the ALB) with "Auto Scaling group". Answer A will direct ALB traffic only to one node, but it does not affect the number of nodes in any way (it will still be "at least two").

upvoted 4 times

**chasingsummer** 2 years, 5 months ago

**Selected Answer: A**

By reconfiguring the target group in the development environment to have only one EC2 instance as a target, it reduces the number of instances handling the development environment's traffic. This ensures the minimum setup required for the development environment's functionality without incurring unnecessary costs associated with multiple instances.

This solution optimizes costs by scaling down the infrastructure specifically in the development environment where lower traffic or fewer resources might be acceptable for testing or development purposes, thus reducing unnecessary expenses related to running multiple instances.

upvoted 1 times

**pentium75** 2 years, 5 months ago

No, you're confusing "target group" (of the ALB) with "Auto Scaling group". Answer A will direct ALB traffic only to one node, but it does not affect the number of nodes in any way (it will still be "at least two").

upvoted 7 times

**MiniYang** 2 years, 5 months ago

**Selected Answer: D**

the correct answer is D. This is from Amazon Q : The most cost-effective way to configure the development environment would be to reduce the maximum number of EC2 instances in the development environment's Auto Scaling group (Option D). The most cost-effective way to configure the development environment would be to reduce the maximum number of EC2 instances in the development environment's Auto Scaling group (Option D). The most cost-effective way to configure the development environment would be to reduce the maximum number of EC2 instances in the development environment's Auto Scaling group (Option D).

upvoted 4 times

**meowruki** 2 years, 6 months ago

**Selected Answer: A**

Option D: Reducing the maximum number of EC2 instances in the development environment's Auto Scaling group could limit scalability but might not directly optimize costs. Min can still be the same number of EC2

upvoted 1 times

**saymolet** 2 years, 3 months ago

"might not directly optimize." No, it does, you're paying less for fewer machines. This is the most direct cost optimisation practice there could ever be. The correct answer is D

upvoted 1 times

**pentium75** 2 years, 5 months ago

No, you're confusing "target group" (of the ALB) with "Auto Scaling group". Answer A will direct ALB traffic only to one node, but it does not affect the number of nodes in any way (it will still be "at least two").

upvoted 4 times

**Chef\_couincouin** 2 years, 6 months ago

Answer is A but I'm not agree. We use only one instance with A and D. But with D, by default, instance is terminated whereas with A, instance still exist. Answer should be D

upvoted 3 times

**ravinperera** 2 years, 7 months ago

**Selected Answer: D**

This option is specific to the development environment and focuses on reducing the number of instances that can be spun up during scaling events. This means cost savings because fewer instances will be used even if the scaling policies are triggered.

Given the goal to configure the development environment in the most cost-effective way, without compromising the production environment, the best option is D

upvoted 3 times



A company runs a web application on Amazon EC2 instances in multiple Availability Zones. The EC2 instances are in private subnets. A solutions architect implements an internet-facing Application Load Balancer (ALB) and specifies the EC2 instances as the target group. However, the internet traffic is not reaching the EC2 instances.

How should the solutions architect reconfigure the architecture to resolve this issue?

- A. Replace the ALB with a Network Load Balancer. Configure a NAT gateway in a public subnet to allow internet traffic.
- B. Move the EC2 instances to public subnets. Add a rule to the EC2 instances' security groups to allow outbound traffic to 0.0.0.0/0.
- C. Update the route tables for the EC2 instances' subnets to send 0.0.0.0/0 traffic through the internet gateway route. Add a rule to the EC2 instances' security groups to allow outbound traffic to 0.0.0.0/0.
- D. Create public subnets in each Availability Zone. Associate the public subnets with the ALB. Update the route tables for the public subnets with a route to the private subnets. **Most Voted**

**Correct Answer: D**

Community vote distribution

D (86%)

8%

## Comments

**ktulu2602** **Highly Voted** 2 years, 9 months ago

I think either the question or the answers are not formulated correctly because of this document:  
<https://docs.aws.amazon.com/prescriptive-guidance/latest/load-balancer-stickiness/subnets-routing.html>

A - Might be possible but it's quite impractical

B - Not needed as the setup described should work as is provided the SGs of the EC2 instances accept traffic from the ALB

C - Update the route tables for the EC2 instances' subnets to send 0.0.0.0/0 traffic through the internet gateway route - not needed as the EC2 instances would receive the traffic from the ALB ENIs. Add a rule to the EC2 instances' security groups to allow outbound traffic to 0.0.0.0/0 - the default behaviour of the SG is to allow outbound traffic only.

D - Create public subnets in each Availability Zone. Associate the public subnets with the ALB - if it's a internet facing ALB these should already be in place. Update the route tables for the public subnets with a route to the private subnets - no need as the local prefix entry in the route tables would take care of this point

I'm 110% sure the question or answers or both are wrong. Prove me wrong! :)

upvoted 22 times

**UnluckyDucky** 2 years, 9 months ago

Completely agreed, I was looking for an option to allow HTTPS traffic on port 443 from the ALB to the EC2 instance's security group.

Either the question or the answers are wrong.

upvoted 8 times

**bdp123** **Highly Voted** 2 years, 9 months ago

**Selected Answer: D**

I change my answer to 'D' because of following link:

<https://aws.amazon.com/premiumsupport/knowledge-center/public-load-balancer-private-ec2/>

upvoted 15 times

**JackyCCK** **Most Recent** 1 year, 8 months ago

Can a EC2 in the private subnet sends traffic to the internet through ELB without using NAT gateway/instance?

if it's only about responses to requests coming through the ELB then no, you don't need NAT. If on the other hand you expect that your instances will need to initiate connections as oppose to just sending responses to the requests then yes, you will have to use NAT. The key point is that response to a request is not a new connection and will be sent to where it came from, i.e. to the ELB.

<https://serverfault.com/questions/986447/can-a-ec2-in-the-private-subnet-sends-traffic-to-the-internet-through-elb-withou>

upvoted 3 times

**sidharthwader** 1 year, 9 months ago

D looks the best but still it must have a internet gateway and once it has internet gateway we must add the route table for private subnet to talk to the public subnet so by using the it should be able to access. I don't think lb can act like internet gateway

upvoted 2 times

**bujuman** 1 year, 9 months ago

**Selected Answer: D**

Considering these statements:

-The EC2 instances are in private subnets.

- However, the internet traffic is not reaching the EC2 instances.

A reliable solution is D according to following link:

<https://repost.aws/knowledge-center/public-load-balancer-private-ec2>

Answer C could not satisfy the requirements because only outbound traffic rules are mentioned

upvoted 2 times

**pentium75** 1 year, 11 months ago

**Selected Answer: D**

A - "NAT gateway" is "to allow [outbound] internet traffic", but this is about inbound traffic

B - This is about outbound traffic while the problem is inbound

C - This is about outbound traffic while the problem is inbound

D - Sounds correct, though the "update the route tables" should not be required if both subnets are in same VPC

upvoted 7 times

**awsgeek75** 1 year, 10 months ago

D is the "least wrong" answer here. I was also confused by the route table part and thought I was missing something critical in the question.

upvoted 4 times

**David\_Ang** 2 years, 1 month ago

**Selected Answer: A**

this is a bad formulated question with gaps, but my reason tells me that if you want to connect something from a private subnet to internet you need a NAT (instance or gateway, bastion).

Creating public subnets in each Availability Zone and associating them with the Application Load Balancer (ALB) won't resolve the problem of allowing internet traffic to reach the private EC2 instances. Public subnets are typically used when you want your EC2 instances to have direct internet access, not when you want to keep them in private subnets with indirect access through a load balancer.

upvoted 3 times

**vijaykamal** 2 years, 2 months ago

**Selected Answer: D**

Option A (replace ALB with Network Load Balancer and add a NAT gateway) is not the most straightforward solution because it changes the load balancer type and introduces a NAT gateway, which might be unnecessary if the goal is to use an ALB for web traffic. ALBs are commonly used for internet-facing web applications.

Option B (move EC2 instances to public subnets and modify security group rules) involves placing instances in public subnets, which is generally not recommended for security reasons. Additionally, it suggests modifying security group rules for outbound traffic, which might not be the best practice to resolve the issue.

Option C (update route tables and security group rules) addresses the route table update, but it also suggests moving instances to public subnets, which is not ideal from a security perspective.

upvoted 2 times

**TariqKipkemei** 2 years, 2 months ago

**Selected Answer: D**

Create public subnets in each Availability Zone. Associate the public subnets with the ALB. Update the route tables for the public subnets with a route to the private subnets.

upvoted 3 times

**Its\_SaKar** 2 years, 2 months ago

**Selected Answer: D**

Option A is incorrect Internet traffic is http and https so it can't be configured to NLB

Option B and option C are incorrect because sending 0.0.0.0/0 is not best practices

Option D is correct because it's the only option left. and updating the route tables for the public subnets with a route to the private subnets ensures internet access to EC2 instances in private subnet.

upvoted 3 times

**Guru4Cloud** 2 years, 3 months ago

**Selected Answer: D**

D is the correct solution. By creating public subnets and associating them with the ALB, inbound internet traffic can reach the ALB. The route tables for the public subnets are updated to include a route to the private subnets, allowing traffic to reach the EC2 instances in the private subnets. This setup enables secure access to the application while allowing internet traffic to reach the EC2 instances through the ALB.

upvoted 4 times

**A1975** 2 years, 4 months ago

**Selected Answer: D**

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-example-private-subnets-nat.html>

upvoted 3 times

**cookieMr** 2 years, 5 months ago

**Selected Answer: D**

A. suggests using a different type of load balancer and configuring a NAT gateway, but it does not address the issue of internet traffic reaching the EC2 instances.

B. suggests exposing the EC2 instances to the public internet, which may pose security risks and does not address the issue of inbound internet traffic reaching the instances.

C. suggests configuring the EC2 instances to have outbound internet access, but it does not solve the problem of inbound internet traffic reaching the instances.

D. is the correct solution. By creating public subnets and associating them with the ALB, inbound internet traffic can reach the ALB. The route tables for the public subnets are updated to include a route to the private subnets, allowing traffic to reach the EC2 instances in the private subnets. This setup enables secure access to the application while allowing internet traffic to reach the EC2 instances through the ALB.

upvoted 6 times

**Vinhkewl** 2 years, 5 months ago

Should be C

It would normally make sense to segregate your ALBs into public or private zones by security group and target group, but this is configuration rather than architectural placement - there is nothing preventing you from adding a rule to route specific paths or ports to a public subnet from an ALB that has until then been serving private subnets only.

upvoted 1 times

**pentium75** 1 year, 11 months ago

C allows the EC2 instances to be accessed directly from the Internet, which we don't want. It's the ALB (not the Internet) that can't access them. We must make sure that the ALB can be reached from the Internet and that the EC2 instances can be reached from the ALB.

upvoted 2 times

**Abrar2022** 2 years, 6 months ago

**Selected Answer: D**

To attach Amazon EC2 instances that are located in a private subnet, first create public subnets

upvoted 5 times

**Bmarodi** 2 years, 6 months ago

**Selected Answer: D**

I vote with the option D.

upvoted 2 times

**antropaws** 2 years, 6 months ago

D is not quite accurate because subnets in a VPC have a local route by default, meaning that all subnets are able to communicate with each other: "Every route table contains a local route for communication within the VPC. This route is added by default to all route tables". This question is poorly formulated.

upvoted 4 times

A company has deployed a database in Amazon RDS for MySQL. Due to increased transactions, the database support team is reporting slow reads against the DB instance and recommends adding a read replica.

Which combination of actions should a solutions architect take before implementing this change? (Choose two.)

- A. Enable binlog replication on the RDS primary node.
- B. Choose a failover priority for the source DB instance.
- C. Allow long-running transactions to complete on the source DB instance. **Most Voted**
- D. Create a global table and specify the AWS Regions where the table will be available.
- E. Enable automatic backups on the source instance by setting the backup retention period to a value other than 0. **Most Voted**

**Correct Answer: CE**

Community vote distribution

CE (74%) AE (17%) 8%

## Comments

**fkier4** **Highly Voted** 3 years, 3 months ago

Who would know this stuff man...

upvoted 135 times

**presetacsing** 3 years ago

exactly

upvoted 3 times

**MNotABot** 2 years, 11 months ago

"Allow long-running transactions to complete on the source DB instance." --. Makes sense / Also a backup before changing anything again made a sense.

upvoted 3 times

**foha2012** 2 years, 4 months ago

Just take an intelligent guess. Eliminate 2 wrong answers and you will have a 50percent success chance.

upvoted 3 times

**KelvinEM** **Highly Voted** 3 years, 4 months ago

C,E

"An active, long-running transaction can slow the process of creating the read replica. We recommend that you wait for long-running transactions to complete before creating a read replica. If you create multiple read replicas in parallel from the same source DB instance, Amazon RDS takes only one snapshot at the start of the first create action.

When creating a read replica, there are a few things to consider. First, you must enable automatic backups on the source DB instance by setting the backup retention period to a value other than 0. This requirement also applies to a read replica that is the source DB instance for another read replica"

[https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER\\_ReadRepl.html](https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_ReadRepl.html)

upvoted 52 times

**607d657** **Most Recent** 10 months, 2 weeks ago

**Selected Answer: AC**

A: Enable binlog replication on RDS primary,this is essential for MySQL read replicas:

C: Allow long-running transactions to complete,this is a critical best practice

upvoted 1 times

**c12ab95** 1 year ago

**Selected Answer: CE**



C. Allow long-running transactions to complete on the source DB instance.

Active long-running transactions can delay read replica creation. AWS recommends resolving these first to avoid replication delays or failures .

E. Enable automatic backups on the source instance by setting the backup retention period to a value other than 0.

Automatic backups are mandatory for read replica creation. Enabling backups activates binary logging (binlog) on the source, which is required for replication

upvoted 1 times

**Yak\_Yeti** 1 year, 1 month ago

**Selected Answer: AE**

To create a read replica for Amazon RDS for MySQL, the following prerequisites must be met:

A. Enable binlog replication on the RDS primary node

Required. RDS read replicas rely on MySQL binary log (binlog) replication.

This setting is not enabled by default. You must set the DB parameter binlog\_format to ROW and enable binary logging by modifying the parameter group.

E. Enable automatic backups

Required. Enabling automatic backups by setting backup retention > 0 is necessary to support point-in-time recovery and to allow the creation of read replicas.

RDS uses the backup snapshot as a baseline to create the initial read replica.

Why not "C. Allow long-running transactions to complete"?

Not a prerequisite for creating a read replica.

Might improve performance temporarily, but it's unrelated to read replica setup.

upvoted 3 times

**Yak\_Yeti** 1 year, 1 month ago

Moderator - please delete

Enabling binary log replication on the RDS primary node is indeed automatically handled by Amazon RDS when you create a read replica for MySQL or MariaDB. You do not need to manually configure this beforehand.

upvoted 1 times

**Rcosmos** 1 year, 4 months ago

**Selected Answer: UE**

Opção A: Habilite a replicação de log binário no nó primário do RDS

A replicação de log binário é necessária para habilitar réplicas de leitura no RDS for MySQL. O banco de dados de origem precisa capturar e armazenar logs binários que serão usados para replicar alterações para as réplicas de leitura,

E: Habilite backups automáticos na instância de origem definindo o período de retenção de backup como um valor diferente de 0 Backups automáticos devem estar habilitados na instância de origem para criar réplicas de leitura. Isso garante que a réplica possa ser criada a partir de um ponto consistente do banco de dados de origem

upvoted 1 times

**derekxxxxxxx** 1 year, 6 months ago

**Selected Answer: CE**

Allow long-running transactions to complete on the source DB instance.

upvoted 1 times

**FlyingHawk** 1 year, 6 months ago

**Selected Answer: CE**

Based on the official documentation, the correct steps to prepare for creating a read replica are:

A. Enable binlog replication on the RDS primary node.

C. Allow long-running transactions to complete on the source DB instance.

E. Enable automatic backups on the source instance by setting the backup retention period to a value other than 0.

These ensure smooth and efficient setup while adhering to AWS best practices.

For more detailed information, you can refer to the AWS documentation:

[https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER\\_ReadRepl.Create.html](https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_ReadRepl.Create.html)

A (Enable binlog replication) is not a required manual step in Amazon RDS.

Amazon RDS automatically manages this requirement during the creation of the read replica for MySQL and MariaDB.

upvoted 3 times

**LeonSauveterre** 1 year, 6 months ago

**Selected Answer: CE**

Answer: CE.

A: In Amazon RDS, binary logging is automatically enabled when creating a read replica. You don't need to enable it manually.

B: Read replicas are used for improving read performance, not for failover purposes.

C: This ensures a clean state before replication begins.

D: Global tables are part of Amazon DynamoDB, not Amazon RDS.

E: RDS uses the automated backup feature to take a snapshot of the primary DB instance and initialize the replica.

upvoted 2 times

**tonybuivannghia** 1 year, 8 months ago

**Selected Answer: CE**

A isn't correct because binary log is just for external DB instance.

upvoted 3 times

**ccceb01** 1 year, 9 months ago

Answer is C and E

[https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER\\_ReadRepl.html](https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_ReadRepl.html)

upvoted 2 times

**1166ae3** 1 year, 11 months ago

**Selected Answer: AE**

A. Enable binlog replication on the RDS primary node:

Direct Impact: Enabling binlog replication is crucial for setting up read replicas, which will directly help in distributing the read load and improving read performance.

E. Enable automatic backups on the source instance by setting the backup retention period to a value other than 0:

Direct Impact: Automatic backups are necessary to ensure data integrity when creating read replicas. This setup is critical for maintaining consistent and reliable replicas.

upvoted 3 times

**awsgeek75** 2 years, 5 months ago

**Selected Answer: CE**

B and D don't have anything to do with the question.

E is a must have before doing major architecture changes

A is not something you need to do explicitly when creating read replicas as it is managed by RDS

C makes sense

\* I think the options are really badly worded which makes it confusing. I doubt this is a real question.

upvoted 4 times

**awsgeek75** 2 years, 5 months ago

Also, to add, binlog replication is needed if you are replicating to a non RDS instance. This is why I think the question is badly phrased as it does not specify the location of read replica.

<https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/MySQL.Procedural.Importing.External.Repl.html>

upvoted 1 times

**Ruffyt** 2 years, 6 months ago

An active, long-running transaction can slow the process of creating the read replica. We recommend that you wait for long-running transactions to complete before creating a read replica. If you create multiple read replicas in parallel from the same source DB instance, Amazon RDS takes only one snapshot at the start of the first create action.

When creating a read replica, there are a few things to consider. First, you must enable automatic backups on the source DB instance by setting the backup retention period to a value other than 0. This requirement also applies to a read replica that is the source DB instance for another read replica.

upvoted 2 times

**meowruki** 2 years, 6 months ago

**Selected Answer: AC**

To improve the read performance of a database in Amazon RDS for MySQL by adding a read replica, you should take the following actions:

Enable binlog replication on the RDS primary node: This allows the primary node to stream its binary logs to the read replica, enabling data replication.

A. Enable binlog replication on the RDS primary node.

Allow long-running transactions to complete on the source DB instance: Before creating a read replica, it's advisable to let any long-running transactions complete to ensure consistency between the source and the replica.

C. Allow long-running transactions to complete on the source DB instance.

The other options are not directly related to setting up a read replica:

upvoted 1 times

**meowruki** 2 years, 6 months ago

B. Choose a failover priority for the source DB instance: Failover priority is more relevant in a Multi-AZ setup where automatic failover might occur. It's not directly related to creating a read replica.

D. Create a global table and specify the AWS Regions where the table will be available: Global tables are used for cross-region replication, but they are not directly related to setting up a read replica for improved read performance.

E. Enable automatic backups on the source instance by setting the backup retention period to a value other than 0: While it's a good practice to have backups enabled, it is not a prerequisite for creating a read replica.

Therefore, the most appropriate actions are A and C.

upvoted 1 times

**xdkonorek2** 2 years, 6 months ago

**Selected Answer: AE**

A - it's essential for continuous replication

E - it's essential for setting up replication, initial data in replica is based on latest backup

other options:

B - we're not designing for HA, and it's related to mutli-AZ RDS deployments

C - is this needed for adding read replica?

D - it's not a dynamodb to create global table

upvoted 5 times

**vijaykamal** 2 years, 8 months ago

**Selected Answer: CE**

A. Enabling binlog replication is not something you need to do manually before creating a read replica. Amazon RDS for MySQL manages replication internally, and it's not necessary to enable binlog replication explicitly.

B. Choosing a failover priority is related to Multi-AZ configurations and automatic failover, but it is not specifically required when adding a read replica.

D. Creating a global table and specifying AWS Regions is related to Aurora Global Databases, which is not the same as creating a read replica for a standard RDS instance.

upvoted 2 times

A company runs analytics software on Amazon EC2 instances. The software accepts job requests from users to process data that has been uploaded to Amazon S3. Users report that some submitted data is not being processed Amazon CloudWatch reveals that the EC2 instances have a consistent CPU utilization at or near 100%. The company wants to improve system performance and scale the system based on user load.

What should a solutions architect do to meet these requirements?

- A. Create a copy of the instance. Place all instances behind an Application Load Balancer.
- B. Create an S3 VPC endpoint for Amazon S3. Update the software to reference the endpoint.
- C. Stop the EC2 instances. Modify the instance type to one with a more powerful CPU and more memory. Restart the instances.
- D. Route incoming requests to Amazon Simple Queue Service (Amazon SQS). Configure an EC2 Auto Scaling group based on queue size. Update the software to read from the queue. **Most Voted**

**Correct Answer: D**

Community vote distribution

D (97%) 3%

## Comments

**mhmt4438** **Highly Voted** 2 years, 11 months ago

**Selected Answer: D**

D. Route incoming requests to Amazon Simple Queue Service (Amazon SQS). Configure an EC2 Auto Scaling group based on queue size. Update the software to read from the queue.

By routing incoming requests to Amazon SQS, the company can decouple the job requests from the processing instances. This allows them to scale the number of instances based on the size of the queue, providing more resources when needed. Additionally, using an Auto Scaling group based on the queue size will automatically scale the number of instances up or down depending on the workload. Updating the software to read from the queue will allow it to process the job requests in a more efficient manner, improving the performance of the system.

upvoted 13 times

**b1e2cbe** **Most Recent** 1 year, 7 months ago

D is correct

upvoted 2 times

**Pangian** 1 year, 10 months ago

**Selected Answer: D**

The based on the queue size doesn't seem a perfect approach though

upvoted 2 times

**awsgeek75** 1 year, 11 months ago

**Selected Answer: D**

D: Because this whole exam seems to be selling more and more SQS solutions...

upvoted 3 times

**TariqKipkemei** 2 years, 2 months ago

**Selected Answer: D**

Route incoming requests to Amazon Simple Queue Service (Amazon SQS). Configure an EC2 Auto Scaling group based on queue size. Update the software to read from the queue

upvoted 2 times

**Guru4Cloud** 2 years, 3 months ago

**Selected Answer: D**

D. Route incoming requests to Amazon Simple Queue Service (Amazon SQS). Configure an EC2 Auto Scaling group based on queue size. Update the software to read from the queue.

upvoted 2 times

**Kill3rasp3r** 2 years, 3 months ago

**Selected Answer: D**

I would vote A if it was ALB targeting an EC2 auto scaling group.

I would vote D if the auto scaling group was based on CPU utilization rather than queue size.

So I think both answers are wrong but D is okay enough.

upvoted 4 times

**cookieMr** 2 years, 5 months ago

**Selected Answer: D**

A. Creating a copy of the instance and placing all instances behind an ALB does not address the high CPU utilization issue or provide scalability based on user load.

B. Creating an S3 VPC endpoint for S3 and updating the software to reference the endpoint improves network performance but does not address the high CPU utilization or provide scalability based on user load.

C. Stopping the EC2 instances and modifying the instance type to one with a more powerful CPU and more memory may improve performance, but it does not address scalability based on user load.

D. Routing incoming requests to SQS, configuring an EC2 ASG based on queue size, and updating the software to read from the queue improves system performance and provides scalability based on user load.

Therefore, option D is the correct choice as it addresses the high CPU utilization, improves system performance, and enables scalability based on user load.

upvoted 3 times

**WhericanIstart** 2 years, 9 months ago

**Selected Answer: D**

Autoscaling Group and SQS solves the problem.

SQS - Decouples the process

ASG - Autoscales the EC2 instances based on usage

upvoted 2 times

**ak1ak** 2 years, 9 months ago

**Selected Answer: A**

its definitely A

upvoted 1 times

**wRhIH** 2 years, 6 months ago

You don't "scale the system by load" by choosing A

upvoted 4 times

**AHUI** 2 years, 11 months ago

D is correct. Decouple the process. autoscale the EC2 based on query size. best choice

upvoted 4 times

**Aninina** 2 years, 11 months ago

I think it's A " A. Create a copy of the instance. Place all instances behind an Application Load Balancer.

upvoted 1 times

A company is implementing a shared storage solution for a media application that is hosted in the AWS Cloud. The company needs the ability to use SMB clients to access data. The solution must be fully managed.

Which AWS solution meets these requirements?

- A. Create an AWS Storage Gateway volume gateway. Create a file share that uses the required client protocol. Connect the application server to the file share.
- B. Create an AWS Storage Gateway tape gateway. Configure tapes to use Amazon S3. Connect the application server to the tape gateway.
- C. Create an Amazon EC2 Windows instance. Install and configure a Windows file share role on the instance. Connect the application server to the file share.
- D. Create an Amazon FSx for Windows File Server file system. Attach the file system to the origin server. Connect the application server to the file system. **Most Voted**

**Correct Answer:** D

Community vote distribution

D (97%)

3%

## Comments

**Morinator** **Highly Voted** 3 years, 4 months ago

**Selected Answer: D**

SMB + fully managed = fsx for windows imo  
upvoted 21 times

**devonwho** **Highly Voted** 3 years, 4 months ago

**Selected Answer: D**

Amazon FSx has native support for Windows file system features and for the industry-standard Server Message Block (SMB) protocol to access file storage over a network.

<https://docs.aws.amazon.com/fsx/latest/WindowsGuide/what-is.html>  
upvoted 7 times

**LeonSauveterre** 1 year, 6 months ago

As implied in the question, "The solution must be fully managed". Storage Gateway's SMB support is valid but not the best choice for this fully AWS hosted and fully managed requirement.

Storage Gateway requires the deployment of a gateway appliance (hardware or virtual). While AWS manages the back end (S3), setting up and maintaining the gateway appliance requires some operational effort, so it's not fully managed compared to FSx for Windows File Server.

Storage Gateway is ideal for hybrid environments but adds unnecessary complexity if the workload is fully cloud-based. Not to mention that Storage Gateway File Gateway is focused on storing objects in S3, not providing a shared file system for direct collaboration or use by applications.  
upvoted 2 times

**kg508** **Most Recent** 1 year, 3 months ago

**Selected Answer: D**

뜻밖이 windows가 나와 당황하고 왜 굳이 fsx for windows여야 했나 싶지만  
D는 문제의 요구사항에 모두 적합합니다  
upvoted 1 times

**0xE8D4A51000** 1 year, 11 months ago

**Selected Answer: A**

All the answers are wrong here by people. We don't know whether the user is using Windows based applications. AWS Storage Gateway also supports SMB protocol. This is the answer.

upvoted 2 times

**pentium75** 2 years, 5 months ago

**Selected Answer: D**

A: Volume Gateway provides virtual disks iSCSI, not SMB  
B: Tape Gateway provides virtual tapes via iSCSI, not SMB  
C: Not "fully managed"

upvoted 5 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: D**

SMB = Amazon FSx for Windows File Server

upvoted 5 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: D**

D. Create an Amazon FSx for Windows File Server file system. Attach the file system to the origin server. Connect the application server to the file system

upvoted 3 times

**Guru4Cloud** 2 years, 9 months ago

All who selected D. are correct - see more details from our community

upvoted 2 times

**animefan1** 2 years, 11 months ago

**Selected Answer: D**

Fsx is fully managed. Plus it supports SMB protocol

upvoted 2 times

**cookieMr** 2 years, 11 months ago

**Selected Answer: D**

A. involves using Storage Gateway, but it does not specifically mention support for SMB clients. It may not meet the requirement of using SMB client to access data.

B. involves using Storage Gateway with tape gateway configuration, which is primarily used for archiving data to S3. It does not provide native support for SMB clients to access data.

C. involves manually setting up and configuring a Windows file share on an EC2 Windows instance. While it allows SMB clients to access data, it is not a fully managed solution as it requires manual setup and maintenance.

D. involves creating an FSx for Windows File Server file system, which is a fully managed Windows file system that supports SMB clients. It provides an easy-to-use shared storage solution with native SMB support.

Based on the requirements of using SMB clients and needing a fully managed solution, option D is the most suitable choice.

upvoted 6 times

**LuckyAro** 3 years, 4 months ago

**Selected Answer: D**

Amazon FSx for Windows File Server file system

upvoted 2 times

**techhb** 3 years, 4 months ago

amazon fsx for smb connectivity.

upvoted 2 times

**Aninina** 3 years, 4 months ago

**Selected Answer: D**

FSX is the ans

upvoted 2 times

**mhmt4438** 3 years, 4 months ago

**Selected Answer: D**

<https://www.examttopics.com/discussions/amazon/view/81115-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 3 times

**bamishr** 3 years, 4 months ago

**Selected Answer: D**

D. Create an Amazon FSx for Windows File Server file system. Attach the file system to the origin server. Connect the application server to the file system.

upvoted 2 times



A company's security team requests that network traffic be captured in VPC Flow Logs. The logs will be frequently accessed for 90 days and then accessed intermittently.

What should a solutions architect do to meet these requirements when configuring the logs?

- A. Use Amazon CloudWatch as the target. Set the CloudWatch log group with an expiration of 90 days
- B. Use Amazon Kinesis as the target. Configure the Kinesis stream to always retain the logs for 90 days.
- C. Use AWS CloudTrail as the target. Configure CloudTrail to save to an Amazon S3 bucket, and enable S3 Intelligent-Tiering.
- D. Use Amazon S3 as the target. Enable an S3 Lifecycle policy to transition the logs to S3 Standard-Infrequent Access (S3 Standard-IA) after 90 days. **Most Voted**

**Correct Answer:** D

Community vote distribution

D (94%)

6%

## Comments

**cookieMr** **Highly Voted** 2 years, 11 months ago

**Selected Answer: D**

A. suggests using CloudWatch as the target for VPC Flow Logs. However, it does not provide a mechanism for managing the retention of the logs for 90 days and then accessing them intermittently.

B. suggests using Kinesis as the target for VPC Flow Logs. While it can retain the logs for 90 days, it does not address the requirement for intermittent access to the logs.

C. suggests using CloudTrail as the target for VPC Flow Logs. However, CloudTrail is designed for auditing and monitoring API activity, not for capturing network traffic logs. It does not meet the requirement of capturing VPC Flow Logs.

D. suggests using S3 as the target for VPC Flow Logs and leveraging S3 Lifecycle policies to transition the logs to a cost-effective storage class after 90 days. It meets the requirement of retaining the logs for 90 days and provides the flexibility for intermittent access while optimizing storage costs.

upvoted 10 times

**LuckyAro** **Highly Voted** 3 years, 4 months ago

**Selected Answer: D**

D is the correct answer.

upvoted 6 times

**manabpokhre17** **Most Recent** 1 year, 7 months ago

**Selected Answer: D**

D is the correct answer

upvoted 2 times

**RicardoD** 2 years, 3 months ago

**Selected Answer: A**

A is correct

You can change the log data retention setting for CloudWatch logs. By default, logs are kept indefinitely and never expire. You can adjust the retention policy for each log group, keeping the indefinite retention, or choosing a retention period between 10 years and one day.

<https://docs.aws.amazon.com/managedservices/latest/userguide/log-customize-retention.html>

upvoted 1 times

**jerry1** 1 year, 2 months ago

you need random access after 90 days but option A is removing the logs

upvoted 1 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: D**

Use Amazon S3 as the target. Enable an S3 Lifecycle policy to transition the logs to S3 Standard-Infrequent Access (S3 Standard-IA) after 90 days  
upvoted 2 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: D**

D. Use Amazon S3 as the target. Enable an S3 Lifecycle policy to transition the logs to S3 Standard-Infrequent Access (S3 Standard-IA) after 90 days.  
upvoted 2 times

**animefan1** 2 years, 11 months ago

**Selected Answer: D**

S3 will store logs. With life cycle, we can move it to different class. With Option A, log groups expiration will simply remove the logs and failing the 2nd request in question  
upvoted 3 times

**markw92** 2 years, 11 months ago

A doesn't solve "90 days and then accessed intermittently" this statement. It sets expire after 90. Not sure otherwise A seems to be right choice since you can create dashboards etc.  
upvoted 2 times

**Bmarodi** 3 years ago

**Selected Answer: A**

Option A meets these requirements.  
upvoted 1 times

**pentium75** 2 years, 5 months ago

"Expiration of 90 days", but you need to access the log AFTER 90 days, just "intermittently".  
upvoted 4 times

**ocbn3wby** 3 years, 4 months ago

**Selected Answer: D**

There's a table here that specifies that VPC Flow logs can go directly to S3. Does not need to go via CloudTrail and then to S3. Nor via CW.

<https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/AWS-logs-and-resource-policy.html#AWS-logs-infrastructure-S3>  
upvoted 4 times

**techhb** 3 years, 4 months ago

**Selected Answer: D**

we need to preserve logs hence D  
<https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/CloudWatchLogsConcepts.html>  
upvoted 3 times

**mp165** 3 years, 4 months ago

**Selected Answer: D**

D...agree that retention is the key word  
upvoted 3 times

**swolfgang** 3 years, 4 months ago

**Selected Answer: D**

a is not,retantion means delete after 90 days but questions say rarely access.  
upvoted 3 times

**mhmt4438** 3 years, 4 months ago

**Selected Answer: D**

D. Use Amazon S3 as the target. Enable an S3 Lifecycle policy to transition the logs to S3 Standard-Infrequent Access (S3 Standard-IA) after 90 days.

By using Amazon S3 as the target for the VPC Flow Logs, the logs can be easily stored and accessed by the security team. Enabling an S3 Lifecycle policy to transition the logs to S3 Standard-Infrequent Access (S3 Standard-IA) after 90 days will automatically move the logs to a storage class that is optimized for infrequent access, reducing the storage costs for the company. The security team will still be able to access the logs as needed, even after they have been transitioned to S3 Standard-IA, but the storage cost will be optimized.  
upvoted 5 times

**laicos** 3 years, 4 months ago

**Selected Answer: D**

I prefer D  
"accessed intermittently" need logs after 90 days

upvoted 2 times

**Parsons** 3 years, 4 months ago

**Selected Answer: D**

No, D should be is correct.

"The logs will be frequently accessed for 90 days and then accessed intermittently." => We still need to store instead of deleting as the answer A.

upvoted 3 times

**Aninina** 3 years, 4 months ago

**Selected Answer: D**

D looks correct. This will meet the requirements of frequently accessing the logs for the first 90 days and then intermittently accessing them after that. S3 standard-IA is a storage class that is less expensive than S3 standard for infrequently accessed data, so it would be a more cost-effective option for storing the logs after the first 90 days.

upvoted 2 times

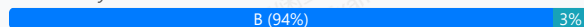
An Amazon EC2 instance is located in a private subnet in a new VPC. This subnet does not have outbound internet access, but the EC2 instance needs the ability to download monthly security updates from an outside vendor.

What should a solutions architect do to meet these requirements?

- A. Create an internet gateway, and attach it to the VPC. Configure the private subnet route table to use the internet gateway as the default route.
- B. Create a NAT gateway, and place it in a public subnet. Configure the private subnet route table to use the NAT gateway as the default route. **Most Voted**
- C. Create a NAT instance, and place it in the same subnet where the EC2 instance is located. Configure the private subnet route table to use the NAT instance as the default route.
- D. Create an internet gateway, and attach it to the VPC. Create a NAT instance, and place it in the same subnet where the EC2 instance is located. Configure the private subnet route table to use the internet gateway as the default route.

**Correct Answer: B**

Community vote distribution



## Comments

**mhmt4438** **Highly Voted** 2 years, 11 months ago

**Selected Answer: B**

B. Create a NAT gateway, and place it in a public subnet. Configure the private subnet route table to use the NAT gateway as the default route.

This approach will allow the EC2 instance to access the internet and download the monthly security updates while still being located in a private subnet. By creating a NAT gateway and placing it in a public subnet, it will allow the instances in the private subnet to access the internet through the NAT gateway. And then, configure the private subnet route table to use the NAT gateway as the default route. This will ensure that all outbound traffic is directed through the NAT gateway, allowing the EC2 instance to access the internet while still maintaining the security of the private subnet.  
upvoted 11 times

**Manjunathkb** 2 years, 8 months ago

NAT gateway does not allow internet on it's own. It needs internet gateway too. None of the answers make sense  
upvoted 11 times

**Manjunathkb** 2 years, 8 months ago

refer below link  
<https://aws.amazon.com/about-aws/whats-new/2021/06/aws-removes-nat-gateways-dependence-on-internet-gateway-for-private-communications/>  
upvoted 3 times

**TOR\_0511** 2 years ago

lol, thats for 'private connections'  
upvoted 1 times

**pentium75** 1 year, 11 months ago

B says "place it in a public subnet", a public subnet needs an Internet Gateway so that is included in the answer.  
upvoted 3 times

**pentium75** **Highly Voted** 1 year, 11 months ago

**Selected Answer: B**

A - if you "configure the private subnet route table to use the internet gateway" then it's no longer a private subnet  
B - Correct (you place NAT GW in a public subnet and add it to the private subnet's route table)  
C - NAT instance is deprecated, and it would still in a private subnet where it doesn't have Internet access  
D - NAT instance is deprecated, and in that answer it is created but not even used

upvoted 5 times

**Uzbekistan** **Most Recent** 1 year, 8 months ago

**Selected Answer: B**

B. Create a NAT gateway, and place it in a public subnet. Configure the private subnet route table to use the NAT gateway as the default route.

Explanation:

NAT Gateway: NAT (Network Address Translation) gateway is a managed service provided by AWS that allows EC2 instances in private subnets to access the internet while preventing inbound traffic from directly accessing them. You place the NAT gateway in a public subnet with an associated internet gateway, allowing it to send traffic to the internet.

Private Subnet Route Table: Configure the route table of the private subnet to route all outbound traffic (0.0.0.0/0) through the NAT gateway. This allows instances in the private subnet to access the internet through the NAT gateway while maintaining their private IP addresses and security.

upvoted 2 times

**EtherealBagel** 2 years ago

yes, the nat gateway on its own does not allow connection to the internet. But the question specifies that it has been placed in a public subnet. public subnets are public because they have access to the internet via an internet gateway.

upvoted 2 times

**xdkonorek2** 2 years ago

**Selected Answer: C**

<https://docs.aws.amazon.com/vpc/latest/userguide/configure-subnets.html>

Public subnet – The subnet has a direct route to an internet gateway. Resources in a public subnet can access the public internet.

Private subnet – The subnet does not have a direct route to an internet gateway. Resources in a private subnet require a NAT device to access the public internet.

Both B and C have caveats but are both viable:

C - NAT Instance is used as a NAT device instead of NAT gateway, but it's still viable option

B - Have 2 redundant components - IGW and public subnet, and NAT gateway still would route traffic to IGW, and if VPC is a custom VPC routing has to be set up

upvoted 1 times

**pentium75** 1 year, 11 months ago

"NAT instance in the same subnet where the EC2 instance is located", how would you "use the NAT instance as the default route" when it's in the same subnet?

upvoted 2 times

**oluolope** 2 years, 1 month ago

**Selected Answer: D**

A NAT Gateway should have one interface in each network it is connected to. I don't understand what it means when they say it is located either in the private or in the public network. It should be in both. Therefore, B and D do not really make sense.

I choose D over B because there is a requirement to access the internet and although it is possible for the NAT to exist without an internet gateway, the later is still needed when internet access is required which is the case in this scenario.

upvoted 1 times

**pentium75** 1 year, 11 months ago

NAT Gateway must be in a public subnet as it needs Internet access. It can be specified in a private subnet's route table as a destination.

D doesn't make sense because you created an (outdated) NAT instance but don't use it (you point the route table to the Internet Gateway).

upvoted 2 times

**TariqKipkemei** 2 years, 2 months ago

**Selected Answer: B**

Internet Gateway is required anyway to access the internet.

Option B makes more sense: Create a NAT gateway, and place it in a public subnet. Configure the private subnet route table to use the NAT gateway as the default route.

upvoted 2 times

**Guru4Cloud** 2 years, 3 months ago

**Selected Answer: B**

B. Create a NAT gateway, and place it in a public subnet. Configure the private subnet route table to use the NAT gateway as the default route.

upvoted 2 times

**cookieMr** 2 years, 5 months ago

A. provides direct internet access to the private subnet, which is not desired in this case as the goal is to restrict outbound internet access.

B. allows the EC2 in the private subnet to access the internet through the NAT gateway, which acts as a proxy. It provides controlled outbound internet access while maintaining the security of the private subnet.

C. is similar to using a NAT gateway, but it involves using a NAT instance. NAT instances require more manual configuration and management compared to NAT gateways, making them a less preferred option.

D. combines the use of an internet gateway and a NAT instance, which is not necessary. It introduces unnecessary complexity and adds a NAT instance that requires additional management.

Overall, option B is the most appropriate solution as it utilizes a NAT gateway placed in a public subnet to enable controlled outbound internet access for the EC2 instance in the private subnet.

NAT Gateways are preferred over NAT Instances by AWS and in general.

upvoted 4 times

**Bmarodi** 2 years, 6 months ago

**Selected Answer: B**

Option B meets the requirements, hence B is right choice.

upvoted 2 times

**Manjunathkb** 2 years, 8 months ago

D would have been the answer if NAT gateway is installed in public subnet and not where EC2 is located. None of the answers are correct.

upvoted 2 times

**AlessandraSAA** 2 years, 9 months ago

why not C?

upvoted 1 times

**UnluckyDucky** 2 years, 9 months ago

Because NAT Gateways are preferred over NAT Instances by AWS and in general.

I have yet to find a situation where a NAT Instance would be more applicable than NAT Gateway which is fully managed and is overall an easier solution to implement - both in AWS questions or the real world.

upvoted 3 times

**TungPham** 2 years, 10 months ago

**Selected Answer: B**

Require NAT gateway

upvoted 2 times

**techhb** 2 years, 10 months ago

**Selected Answer: B**

Answer explained here <https://medium.com/@tshemku/aws-internet-gateway-vs-nat-gateway-vs-nat-instance-30523096df22>

upvoted 2 times

**techhb** 2 years, 10 months ago

**Selected Answer: B**

NAT Gateway is right choice

upvoted 2 times

**bamishr** 2 years, 11 months ago

**Selected Answer: B**

<https://www.examttopics.com/discussions/amazon/view/59966-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 3 times

A solutions architect needs to design a system to store client case files. The files are core company assets and are important. The number of files will grow over time.

The files must be simultaneously accessible from multiple application servers that run on Amazon EC2 instances. The solution must have built-in redundancy.

Which solution meets these requirements?

A. Amazon Elastic File System (Amazon EFS) **Most Voted**

B. Amazon Elastic Block Store (Amazon EBS)

C. Amazon S3 Glacier Deep Archive

D. AWS Backup

**Correct Answer: A**

Community vote distribution

A (100%)

## Comments

**Aninina** **Highly Voted** 2 years, 4 months ago

**Selected Answer: A**

EFS Amazon Elastic File System (EFS) automatically grows and shrinks as you add and remove files with no need for management or provisioning.  
upvoted 8 times

**TungPham** **Highly Voted** 2 years, 4 months ago

**Selected Answer: A**

"accessible from multiple application servers that run on Amazon EC2 instances"  
upvoted 6 times

**Chiquitabandita** **Most Recent** 1 year, 8 months ago

**Selected Answer: A**

my choice is A but I think a better alternative would be S3 standard if offered wouldn't it be?  
upvoted 5 times

**sz3cur3** 10 months, 2 weeks ago

S3 can't be mounted like a shared file system as apps may require shared file system  
upvoted 1 times

**TariqKipkemei** 1 year, 8 months ago

**Selected Answer: A**

File system, scalable, multiple access = Amazon Elastic File System (Amazon EFS)  
upvoted 3 times

**Guru4Cloud** 1 year, 9 months ago

**Selected Answer: A**

Amazon Elastic File System (Amazon EFS)  
upvoted 2 times

**cookieMr** 1 year, 11 months ago

**Selected Answer: A**

EFS provides a scalable and fully managed file storage service that can be accessed concurrently from multiple EC2. It offers built-in redundancy by storing data across multiple AZs within a region. With EFS, the client case files can be accessed by multiple application servers simultaneously,

ensuring high availability and scalability as the number of files grows over time.

Option B, EBS, is a block-level storage service that is typically used for attaching to individual EC2 and does not provide concurrent access to multiple instances, making it unsuitable for this scenario.

Option C, S3 Glacier Deep Archive, is a long-term archival storage service and may not be suitable for active file access and simultaneous access from multiple application servers.

Option D, AWS Backup, is a centralized backup management service and does not provide the required simultaneous file access and redundancy features.

Therefore, the most suitable solution is Amazon EFS (option A).

upvoted 6 times

**Bmarodi** 2 years ago

**Selected Answer: A**

Option A meets the requirements, hence A is correct answer.

upvoted 2 times

**moiraqi** 2 years ago

What does "The solution must have built-in redundancy" mean

upvoted 1 times

**KZM** 2 years, 3 months ago

If the application servers are running on Linux or UNIX operating systems, EFS is a the most suitable solution for the given requirements.

upvoted 2 times

**mhmt4438** 2 years, 4 months ago

**Selected Answer: A**

Correct answer is A

upvoted 3 times

**bamishr** 2 years, 5 months ago

**Selected Answer: A**

<https://www.examttopics.com/discussions/amazon/view/68833-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 1 times



A solutions architect has created two IAM policies: Policy1 and Policy2. Both policies are attached to an IAM group.

#### Policy 1

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "iam:Get*",
        "iam:List*",
        "kms:List*",
        "ec2:*",
        "ds:*",
        "logs:Get*",
        "logs:Describe*"
      ],
      "Resource": "*"
    }
  ]
}
```

#### Policy 2

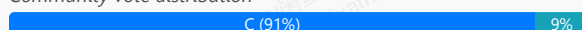
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Deny",
      "Action": "ds:Delete*",
      "Resource": "*"
    }
  ]
}
```

A cloud engineer is added as an IAM user to the IAM group. Which action will the cloud engineer be able to perform?

- A. Deleting IAM users
- B. Deleting directories
- C. Deleting Amazon EC2 instances **Most Voted**
- D. Deleting logs from Amazon CloudWatch Logs

**Correct Answer: C**

Community vote distribution



## Comments

**JayBee65** **Highly Voted** 3 years, 4 months ago  
ec2:\* Allows full control of EC2 instances, so C is correct

The policy only grants get and list permission on IAM users, so not A

ds>Delete deny denies delete-directory, so not B, see <https://awscli.amazonaws.com/v2/documentation/api/latest/reference/ds/index.html>  
The policy only grants get and describe permission on logs, so not D

upvoted 23 times

**vincent2023** 1 year, 4 months ago

ds is related to directory service, don't understand how is it related to deleting ec2 instances ??

upvoted 1 times

**mwwt2022** 2 years, 5 months ago

great explanation

upvoted 2 times

**Morinator** **Highly Voted** 3 years, 4 months ago

**Selected Answer: C**

Explicite deny on directories, only available action for deleting is EC2

upvoted 7 times

**surajkrishnamurthy** **Most Recent** 1 year, 4 months ago

**Selected Answer: C**

In Policy number 2 its explicit deny to deleting directories so the IAM user wont be able to delete any directories

However in policy number 1 the statement "ec2:\*", means the IAM user can perform all activities under the ec2

Therefore as per the given answer choice IAM user can delete an ec2 instance

so the choice is C: Deleting Amazon EC2 instances

upvoted 2 times

**1e22522** 1 year, 10 months ago

**Selected Answer: B**

This question and answers feel like a fever dream. what the hell is happening.

upvoted 2 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: C**

Deleting Amazon EC2 instances

upvoted 2 times

**Aninina** 3 years, 4 months ago

**Selected Answer: C**

C : Deleting Amazon EC2 instances

upvoted 2 times

**mhmt4438** 3 years, 4 months ago

**Selected Answer: C**

Answer is C

upvoted 3 times

**Aninina** 3 years, 4 months ago

C : Deleting Amazon EC2 instances

upvoted 2 times

**bamishr** 3 years, 4 months ago

**Selected Answer: C**

<https://www.examttopics.com/discussions/amazon/view/27873-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 4 times

A company is reviewing a recent migration of a three-tier application to a VPC. The security team discovers that the principle of least privilege is not being applied to Amazon EC2 security group ingress and egress rules between the application tiers.

What should a solutions architect do to correct this issue?

- A. Create security group rules using the instance ID as the source or destination.
- B. Create security group rules using the security group ID as the source or destination. **Most Voted**
- C. Create security group rules using the VPC CIDR blocks as the source or destination.
- D. Create security group rules using the subnet CIDR blocks as the source or destination.

**Correct Answer: B**

Community vote distribution

B (100%)

## Comments

Aninina **Highly Voted** 2 years, 11 months ago

**Selected Answer: B**

B. Create security group rules using the security group ID as the source or destination.

This way, the security team can ensure that the least privileged access is given to the application tiers by allowing only the necessary communication between the security groups. For example, the web tier security group should only allow incoming traffic from the load balancer security group and outgoing traffic to the application tier security group. This approach provides a more granular and secure way to control traffic between the different tiers of the application and also allows for easy modification of access if needed.

It's also worth noting that it's good practice to minimize the number of open ports and protocols, and use security groups as a first line of defense, in addition to network access control lists (ACLs) to control traffic between subnets.

upvoted 14 times

Wael216 **Highly Voted** 2 years, 9 months ago

**Selected Answer: B**

By using security group IDs, the ingress and egress rules can be restricted to only allow traffic from the necessary source or destination, and to deny all other traffic. This ensures that only the minimum required traffic is allowed between the application tiers.

Option A is not the best choice because using the instance ID as the source or destination would allow traffic from any instance with that ID, which may not be limited to the specific application tier.

Option C is also not the best choice because using VPC CIDR blocks would allow traffic from any IP address within the VPC, which may not be limited to the specific application tier.

Option D is not the best choice because using subnet CIDR blocks would allow traffic from any IP address within the subnet, which may not be limited to the specific application tier.

upvoted 9 times

Guru4Cloud **Most Recent** 2 years, 3 months ago

**Selected Answer: B**

Create security group rules using the security group ID as the source or destination.

This way, the security team can ensure that the least privileged access is given to the application tiers by allowing only the necessary communication between the security groups. For example, the web tier security group should only allow incoming traffic from the load balancer security group and outgoing traffic to the application tier security group. This approach provides a more granular and secure way to control traffic between the different tiers of the application and also allows for easy modification of access if needed.

It's also worth noting that it's good practice to minimize the number of open ports and protocols, and use security groups as a first line of defense, in addition to network access control lists (ACLs) to control traffic between subnets.

upvoted 3 times

cookieMr 2 years, 5 months ago

**Selected Answer: B**

A. would limit the traffic based on specific instances, which may not be the most suitable solution for applying the principle of least privilege between application tiers.

B. By using security group IDs in the rules, you can precisely control the traffic between application tiers, allowing only the necessary communication and adhering to the principle of least privilege.

C. would apply broad rules based on the entire VPC CIDR blocks, which may not provide the necessary level of granularity required for secure communication between specific application tiers.

D. would limit the traffic based on subnet CIDR blocks, which may not be sufficient for ensuring proper security between application tiers.

In summary, using security group IDs (Option B) is the recommended approach as it allows for precise control of traffic between application tiers, aligning with the principle of least privilege.

upvoted 8 times

**foha2012** 1 year, 10 months ago

with option A. How would you use instance ID in security group inbound rules ?

upvoted 2 times

**Bmarodi** 2 years, 6 months ago

**Selected Answer: B**

I vote for option B.

upvoted 2 times

**LuckyAro** 2 years, 10 months ago

**Selected Answer: B**

. Create security group rules using the security group ID as the source or destination

upvoted 2 times

**techhb** 2 years, 10 months ago

Security Group Rules apply to instances

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/security-group-rules.html>

upvoted 2 times

**mhmt4438** 2 years, 11 months ago

**Selected Answer: B**

Correct answer is B

upvoted 3 times

**bamishr** 2 years, 11 months ago

**Selected Answer: B**

<https://www.examttopics.com/discussions/amazon/view/46463-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 2 times

**Morinator** 2 years, 11 months ago

**Selected Answer: B**

B right

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/security-group-rules.html>

upvoted 2 times

A company has an ecommerce checkout workflow that writes an order to a database and calls a service to process the payment. Users are experiencing timeouts during the checkout process. When users resubmit the checkout form, multiple unique orders are created for the same desired transaction.

How should a solutions architect refactor this workflow to prevent the creation of multiple orders?

- A. Configure the web application to send an order message to Amazon Kinesis Data Firehose. Set the payment service to retrieve the message from Kinesis Data Firehose and process the order.
- B. Create a rule in AWS CloudTrail to invoke an AWS Lambda function based on the logged application path request. Use Lambda to query the database, call the payment service, and pass in the order information.
- C. Store the order in the database. Send a message that includes the order number to Amazon Simple Notification Service (Amazon SNS). Set the payment service to poll Amazon SNS, retrieve the message, and process the order.
- D. Store the order in the database. Send a message that includes the order number to an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Set the payment service to retrieve the message and process the order. Delete the message from the queue. **Most Voted**

**Correct Answer: D**

Community vote distribution

D (100%)

## Comments

**Aninina** **Highly Voted** 2 years, 11 months ago

**Selected Answer: D**

D. Store the order in the database. Send a message that includes the order number to an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Set the payment service to retrieve the message and process the order. Delete the message from the queue.

This approach ensures that the order creation and payment processing steps are separate and atomic. By sending the order information to an SQS FIFO queue, the payment service can process the order one at a time and in the order they were received. If the payment service is unable to process an order, it can be retried later, preventing the creation of multiple orders. The deletion of the message from the queue after it is processed will prevent the same message from being processed multiple times.

It's worth noting that FIFO queues guarantee that messages are processed in the order they are received, and prevent duplicates.

upvoted 11 times

**cookieMr** **Highly Voted** 2 years, 5 months ago

**Selected Answer: D**

A. is not a suitable solution for preventing the creation of multiple orders. This approach does not guarantee the sequential and reliable processing of orders.

B. is not an appropriate solution for preventing the creation of multiple orders. CloudTrail is primarily used for logging and auditing API activity, and invoking a Lambda based on the logged request does not ensure the correct order processing.

C. is not a suitable solution. SNS is a publish-subscribe messaging service, and polling it may result in delayed processing and potential order duplication.

D. is the correct solution. Using an SQS FIFO ensures that the orders are processed in a sequential and reliable manner, preventing the creation of multiple orders for the same transaction.

upvoted 7 times

**TariqKipkemei** **Most Recent** 2 years, 2 months ago

**Selected Answer: D**

if the backend can not keep up, queue the tasks.

upvoted 2 times

**Guru4Cloud** 2 years, 3 months ago

**Selected Answer: D**

D. Store the order in the database. Send a message that includes the order number to an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Set the payment service to retrieve the message and process the order. Delete the message from the queue.

upvoted 2 times

**animefan1** 2 years, 5 months ago

**Selected Answer: D**

The question is related in breaking down the flow. SQS is go-to choice to decouple & DB will be used to store

upvoted 2 times

**antropaws** 2 years, 5 months ago

Why not A?

upvoted 1 times

**pentium75** 1 year, 11 months ago

Because Kinesis Data Firehose is for ingestion of streaming data, not queuing items.

upvoted 4 times

**Wael216** 2 years, 9 months ago

**Selected Answer: D**

The use of a FIFO queue in Amazon SQS ensures that messages are processed in the order they are received.

upvoted 2 times

**mhmt4438** 2 years, 11 months ago

**Selected Answer: D**

<https://www.examttopics.com/discussions/amazon/view/95026-exam-aws-certified-solutions-architect-associate-saa-c03/>

upvoted 4 times

**bamishr** 2 years, 11 months ago

**Selected Answer: D**

asnwer is d

upvoted 3 times

A solutions architect is implementing a document review application using an Amazon S3 bucket for storage. The solution must prevent accidental deletion of the documents and ensure that all versions of the documents are available. Users must be able to download, modify, and upload documents.

Which combination of actions should be taken to meet these requirements? (Choose two.)

- A. Enable a read-only bucket ACL.
- B. Enable versioning on the bucket. **Most Voted**
- C. Attach an IAM policy to the bucket.
- D. Enable MFA Delete on the bucket. **Most Voted**
- E. Encrypt the bucket using AWS KMS.

**Correct Answer:** BD

Community vote distribution

BD (100%)

## Comments

**LoXoL** 1 year, 10 months ago

**Selected Answer: BD**

No Brainer: B & D

upvoted 3 times

**TariqKipkemei** 2 years, 2 months ago

**Selected Answer: BD**

Prevent accidental deletion of the documents = Enable MFA Delete on the bucket

Ensure that all versions of the documents are available = Enable versioning on the bucket

upvoted 4 times

**Guru4Cloud** 2 years, 3 months ago

**Selected Answer: BD**

Options B & D are the correct answers.

upvoted 2 times

**cookieMr** 2 years, 5 months ago

**Selected Answer: BD**

B. allows multiple versions of objects in the S3 bucket to be stored. This ensures that all versions of the documents are available, even if they are accidentally overwritten or deleted.

D. adds an extra layer of protection against accidental deletion of objects in the bucket. With MFA Delete enabled, a user would need to provide an additional authentication factor to successfully delete objects from the bucket. This helps prevent accidental or unauthorized deletions and provides an extra level of security for critical documents.

A. would restrict users from modifying or uploading documents. It would not meet the requirement of allowing users to download, modify, and upload documents.

C. can control access permissions to the bucket, it does not specifically address the requirement of preventing accidental deletion or ensuring availability of all versions of the documents.

E. Encryption focuses on data protection rather than versioning and deletion prevention.

upvoted 4 times

**Bmarodi** 2 years, 6 months ago

**Selected Answer: BD**

Options B & D are the correct answers.

upvoted 2 times

**Wael216** 2 years, 9 months ago

**Selected Answer: BD**

no doubts

upvoted 3 times

**akdavsan** 2 years, 10 months ago

**Selected Answer: BD**

b and d ofc

upvoted 2 times

**LuckyAro** 2 years, 10 months ago

**Selected Answer: BD**

B & D Definitely.

upvoted 2 times

**david76x** 2 years, 10 months ago

**Selected Answer: BD**

B & D is correct

upvoted 2 times

**Aninina** 2 years, 11 months ago

**Selected Answer: BD**

B and D for sure guys

upvoted 3 times

**mhmt4438** 2 years, 11 months ago

**Selected Answer: BD**

<https://www.examttopics.com/discussions/amazon/view/21969-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 3 times



A company is building a solution that will report Amazon EC2 Auto Scaling events across all the applications in an AWS account. The company needs to use a serverless solution to store the EC2 Auto Scaling status data in Amazon S3. The company then will use the data in Amazon S3 to provide near-real-time updates in a dashboard. The solution must not affect the speed of EC2 instance launches.

How should the company move the data to Amazon S3 to meet these requirements?

A. Use an Amazon CloudWatch metric stream to send the EC2 Auto Scaling status data to Amazon Kinesis Data Firehose. Store the data in Amazon S3. **Most Voted**

B. Launch an Amazon EMR cluster to collect the EC2 Auto Scaling status data and send the data to Amazon Kinesis Data Firehose. Store the data in Amazon S3.

C. Create an Amazon EventBridge rule to invoke an AWS Lambda function on a schedule. Configure the Lambda function to send the EC2 Auto Scaling status data directly to Amazon S3.

D. Use a bootstrap script during the launch of an EC2 instance to install Amazon Kinesis Agent. Configure Kinesis Agent to collect the EC2 Auto Scaling status data and send the data to Amazon Kinesis Data Firehose. Store the data in Amazon S3.

**Correct Answer:** A

Community vote distribution

A (89%)

11%

## Comments

**pentium75** **Highly Voted** 2 years, 5 months ago

**Selected Answer: A**

B - EMR cluster is for Big Data, has nothing to do with this

C - invokes the function "on a schedule", but you want to capture events

D - Could work, but would be overcomplex and would "affect the speed of EC2 instance launches" (which it should not)

upvoted 12 times

**LoXoL** 2 years, 4 months ago

Right.

upvoted 2 times

**Guru4Cloud** **Highly Voted** 2 years, 9 months ago

**Selected Answer: A**

This solution meets the requirements because it is serverless and does not affect the speed of EC2 instance launches. Amazon CloudWatch metric streams can continuously stream CloudWatch metrics to destinations such as Amazon S3. Amazon Kinesis Data Firehose can capture, transform, and deliver streaming data into data lakes, data stores, and analytics services. It can directly put the data into Amazon S3, which can then be used for near-real-time updates in a dashboard.

upvoted 9 times

**reviewmine** **Most Recent** 2 years, 3 months ago

**Selected Answer: A**

Answer A: Near real time --> Amazon Kinesis Data Firehose

upvoted 5 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: A**

You can use metric streams to continually stream CloudWatch metrics to a destination of your choice, with near-real-time delivery and low latency. Supported destinations include AWS destinations such as Amazon Simple Storage Service and several third-party service provider destinations. Main usage scenarios for CloudWatch metric streams: Data lake— Create a metric stream and direct it to an Amazon Kinesis Data Firehose delivery stream that delivers your CloudWatch metrics to a data lake such as Amazon S3.

<https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/CloudWatch-Metric-Streams.html#:~:text=CloudWatch%20metric%20streams>

upvoted 4 times

**Valder21** 2 years, 9 months ago

**Selected Answer: C**

Kinesis is for data streams not events. So, C

upvoted 1 times

**pentium75** 2 years, 5 months ago

C invokes the Lambda function "on a schedule". It would collect the scaling status during its runs. But you don't want the hourly status, you want to report "scaling events".

upvoted 2 times

**cookieMr** 2 years, 11 months ago

**Selected Answer: A**

B. introduces unnecessary complexity and overhead for collecting and sending the EC2 Auto Scaling status data to S3. It is not the most efficient serverless solution for this specific requirement.

C. would introduce delays in data updates, as it is not triggered in real-time. Additionally, it adds unnecessary overhead and complexity compared to using a direct data stream.

D. introduces additional dependencies and management overhead. It may also impact the speed of EC2 instance launches, which is a requirement that needs to be avoided.

Overall, option A provides a streamlined and serverless solution by leveraging CloudWatch metric streams and Kinesis Data Firehose to efficiently capture and store the EC2 Auto Scaling status data in S3 without affecting the speed of EC2 instance launches.

upvoted 6 times

**markw92** 2 years, 11 months ago

A: I was thinking D is the answer but the solution should not impact ec2 launches will make the difference and i fast read the question. A is a right choice.

upvoted 2 times

**Rahulbit34** 3 years, 1 month ago

A because of near real time scenario

upvoted 4 times

**UnluckyDucky** 3 years, 2 months ago

**Selected Answer: C**

Both A and C are applicable - no doubt there.

C is more straightforward and to the point of the question imho.

upvoted 3 times

**JohnYu** 1 year, 8 months ago

While EventBridge can capture events, scheduling Lambda functions to poll data is less efficient and may introduce latency, which would not meet the near-real-time requirement.

upvoted 2 times

**UnluckyDucky** 3 years, 2 months ago

Changing my answer to \*A\* as the dashboard will provide near-real updates.

Unless the lambda is configured to run every minute which is not common with schedules - it is not considered near real-time.

upvoted 4 times

**bdp123** 3 years, 3 months ago

**Selected Answer: A**

Serverless solution and near real time

upvoted 4 times

**Stanislav4907** 3 years, 3 months ago

**Selected Answer: A**

near real time -eliminates c

upvoted 2 times

**aakashkumar1999** 3 years, 4 months ago

**Selected Answer: A**

Answer is A

upvoted 2 times

**devonwho** 3 years, 4 months ago

**Selected Answer: A**

You can use metric streams to continually stream CloudWatch metrics to a destination of your choice, with near-real-time delivery and low latency. One of the use cases is Data Lake: create a metric stream and direct it to an Amazon Kinesis Data Firehose delivery stream that delivers your CloudWatch metrics to a data lake such as Amazon S3.

<https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/CloudWatch-Metric-Streams.html>

upvoted 3 times

**Stanislav4907** 3 years, 4 months ago

**Selected Answer: A**

Option C, using an Amazon EventBridge rule to invoke an AWS Lambda function on a schedule to send the EC2 Auto Scaling status data directly to Amazon S3, may not be the best choice because it may not provide real-time updates to the dashboard.

A schedule-based approach with an EventBridge rule and Lambda function may not be able to deliver the data in near real-time, as the EC2 Auto Scaling status data is generated dynamically and may not always align with the schedule set by the EventBridge rule.

Additionally, using a schedule-based approach with EventBridge and Lambda also has the potential to create latency, as there may be a delay between the time the data is generated and the time it is sent to S3.

In this scenario, using Amazon CloudWatch and Kinesis Data Firehose as described in Option A, provides a more reliable and near real-time solution.

upvoted 2 times

**MikelH93** 3 years, 4 months ago

**Selected Answer: A**

A seems to be the right answer. Don't think C could be correct as it says "near real-time" and C is on schedule

upvoted 2 times

**KAUS2** 3 years, 4 months ago

**Selected Answer: C**

C. Create an Amazon EventBridge rule to invoke an AWS Lambda function on a schedule. Configure the Lambda function to send the EC2 Auto Scaling status data directly to Amazon S3.

upvoted 2 times

**pentium75** 2 years, 5 months ago

"On a schedule" but you want to capture events, not a regular status report.

upvoted 2 times

**techhb** 3 years, 4 months ago

**Selected Answer: A**

A seems right choice but serverless keyword confuses, and cloud watch metric stream is serverless too.

upvoted 3 times

**pentium75** 2 years, 5 months ago

But A is serverless.

upvoted 2 times

A company has an application that places hundreds of .csv files into an Amazon S3 bucket every hour. The files are 1 GB in size. Each time a file is uploaded, the company needs to convert the file to Apache Parquet format and place the output file into an S3 bucket.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Create an AWS Lambda function to download the .csv files, convert the files to Parquet format, and place the output files in an S3 bucket. Invoke the Lambda function for each S3 PUT event.
- B. Create an Apache Spark job to read the .csv files, convert the files to Parquet format, and place the output files in an S3 bucket. Create an AWS Lambda function for each S3 PUT event to invoke the Spark job.
- C. Create an AWS Glue table and an AWS Glue crawler for the S3 bucket where the application places the .csv files. Schedule an AWS Lambda function to periodically use Amazon Athena to query the AWS Glue table, convert the query results into Parquet format, and place the output files into an S3 bucket.

D. Create an AWS Glue extract, transform, and load (ETL) job to convert the .csv files to Parquet format and place the output files into an S3 bucket. Create an AWS Lambda function for each S3 PUT event to invoke the ETL job. **Most Voted**

**Correct Answer:** D

Community vote distribution

D (92%)

8%

## Comments

**Parsons** **Highly Voted** 3 years, 4 months ago

**Selected Answer: D**

No, D should be correct.

"LEAST operational overhead" => Should you fully manage service like Glue instead of manually like the answer A.

upvoted 16 times

**awsgeek75** 2 years, 5 months ago

I also think it's D but remember that D requires writing ETL logic in AWS Glue (nothing in question says how complex it will be). AWS Lambda for CSV could be simple also (imagine NodeJS and millions of libraries support or Python's parsing) so both could be operationally on par to each other. Logically D makes more sense but in practice, AWS Glue rarely works with out of the box ETL and becomes a maintenance overhead in itself.

upvoted 1 times

**Guru4Cloud** **Highly Voted** 2 years, 9 months ago

**Selected Answer: D**

This solution meets the requirements with the least operational overhead because AWS Glue is a fully managed ETL service that makes it easy to move data between data stores. AWS Glue can read .csv files from an S3 bucket and write the data into Parquet format in another S3 bucket. The AWS Lambda function can be triggered by an S3 PUT event when a new .csv file is uploaded, and it can start the AWS Glue ETL job to convert the file to Parquet format. This solution does not require managing any servers or clusters, which reduces operational overhead.

upvoted 6 times

**DanRuggles** **Most Recent** 1 month, 1 week ago

**Selected Answer: D**

How It Works (Typical Architecture)A .csv file is uploaded to an S3 bucket.

An S3 Event triggers a Lambda function.

Lambda:Downloads the CSV from S3.

Converts it to Parquet using libraries like pandas + PyArrow.

Uploads the Parquet file to another S3 location (or the same bucket with a different prefix).

(Optional) Updates AWS Glue Data Catalog so the data is queryable via Athena.

upvoted 1 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: D**

AWS Glue can run your extract, transform, and load (ETL) jobs as new data arrives. For example, you can configure AWS Glue to initiate your ETL jobs to run as soon as new data becomes available in Amazon Simple Storage Service (S3). Clearly you don't need a lambda function to initiate the ETL job.

<https://aws.amazon.com/glue/#:~:text=to%20initiate%20your-,ETL,-jobs%20to%20run>

Option A requires writing code to perform the file conversion.

In the exam option D would be the best answer.

upvoted 4 times

**cookieMr** 2 years, 11 months ago

D is correct

upvoted 2 times

**cookieMr** 2 years, 11 months ago

A. introduces significant operational overhead. This approach requires managing the Lambda, handling concurrency, and ensuring proper error handling for large file sizes, which can be challenging.

B. adds unnecessary complexity and operational overhead. Managing the Spark job, handling scalability, and coordinating the Lambda invocations for each file upload can be cumbersome.

C. introduces additional complexity and may not be the most efficient solution. It involves managing Glue resources, scheduling Lambda, and querying data even when no new files are uploaded.

Option D leverages AWS Glue's ETL capabilities, allowing you to define and execute a data transformation job at scale. By invoking the ETL job using an Lambda function for each S3 PUT event, you can ensure that files are efficiently converted to Parquet format without the need for manual intervention. This approach minimizes operational overhead and provides a streamlined and scalable solution.

upvoted 5 times

**F629** 2 years, 11 months ago

**Selected Answer: A**

Both A and D can work, but A is more simple. It's more close to the "Least Operational effort".

upvoted 1 times

**pentium75** 2 years, 5 months ago

Creating, maintaining and supporting custom code that does the same as a ready-made serverless service is NEVER "least operational effort".

upvoted 2 times

**pentium75** 2 years, 5 months ago

Oh, and A can't handle 1 GB files.

upvoted 1 times

**jaswantr** 2 years, 4 months ago

Now Lambda support 1 GB to 10 GB.

upvoted 2 times

**JA2018** 1 year, 6 months ago

But what if the file conversion process exceeds more than 15 minutes for each file? How does Lambda fits into this picture?

upvoted 2 times

**shanwford** 3 years, 2 months ago

**Selected Answer: D**

The maximum size for a Lambda event payload is 256 KB - so (A) didn't work with 1GB Files. Glue is recommended for the Parquet Transformation of AWS.

upvoted 3 times

**jennyka76** 3 years, 4 months ago

ANS - d

<https://aws.amazon.com/blogs/database/how-to-extract-transform-and-load-data-for-analytic-processing-using-aws-glue-part-2/>

- READ ARTICLE -

upvoted 3 times

**aws4myself** 3 years, 4 months ago

Here A is the correct answer. The reason here is the least operational overhead.

A ==> S3 - Lambda - S3

D ==> S3 - Lambda - Glue - S3

Also, glue cannot convert on fly automatically, you need to write some code there. If you write the same code in lambda it will convert the same and push the file to S3

Lambda has max memory of 128 MB to 10 GB. So, it can handle it easily.

And we need to consider cost also, glue cost is more. Hope many from this forum realize these differences.  
upvoted 5 times

**LuckyAro** 3 years, 4 months ago

We also need to stay with the question, cost was not a consideration in the question.  
upvoted 2 times

**nder** 3 years, 3 months ago

Cost is not a factor. AWS Glue is a fully managed service therefore, it's the least operational overhead  
upvoted 5 times

**JA2018** 1 year, 6 months ago

But what if the file conversion process exceeds more than 15 minutes for each file? How does Lambda fits into this picture?  
upvoted 1 times

**JA2018** 1 year, 6 months ago

IIRC, many folks on this forum mentioned that, rule of thumbs for AWS exams, opt for AWS managed services...  
upvoted 1 times

**JayBee65** 3 years, 4 months ago

A is unlikely to work as Lambda may struggle with 1GB size: "< 64 MB, beyond which lambda is likely to hit memory caps", see <https://stackoverflow.com/questions/41504095/creating-a-parquet-file-on-aws-lambda-function>  
upvoted 3 times

**jainparag1** 3 years, 4 months ago

Should be D as Glue is self managed service and provides tel job for converting cab files to parquet off the shelf.  
upvoted 2 times

**Joxtat** 3 years, 4 months ago

**Selected Answer: D**

<https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/three-aws-glue-etl-job-types-for-converting-data-to-apache-parquet.html>  
upvoted 2 times

**techhb** 3 years, 4 months ago

AWS Glue is right solution here.  
upvoted 2 times

**mp165** 3 years, 4 months ago

**Selected Answer: D**

I am thinking D.

A says lambda will download the .csv...but to where? that seem manual based on that  
upvoted 2 times

**mhmt4438** 3 years, 4 months ago

**Selected Answer: A**

I think A  
upvoted 1 times

**bamishr** 3 years, 4 months ago

**Selected Answer: A**

<https://www.examttopics.com/discussions/amazon/view/83201-exam-aws-certified-solutions-architect-associate-saa-c02/>  
upvoted 1 times

A company is implementing new data retention policies for all databases that run on Amazon RDS DB instances. The company must retain daily backups for a minimum period of 2 years. The backups must be consistent and restorable.

Which solution should a solutions architect recommend to meet these requirements?

- A. Create a backup vault in AWS Backup to retain RDS backups. Create a new backup plan with a daily schedule and an expiration period of 2 years after creation. Assign the RDS DB instances to the backup plan. **Most Voted**
- B. Configure a backup window for the RDS DB instances for daily snapshots. Assign a snapshot retention policy of 2 years to each RDS DB instance. Use Amazon Data Lifecycle Manager (Amazon DLM) to schedule snapshot deletions.
- C. Configure database transaction logs to be automatically backed up to Amazon CloudWatch Logs with an expiration period of 2 years.
- D. Configure an AWS Database Migration Service (AWS DMS) replication task. Deploy a replication instance, and configure a change data capture (CDC) task to stream database changes to Amazon S3 as the target. Configure S3 Lifecycle policies to delete the snapshots after 2 years.

**Correct Answer: A**

Community vote distribution

A (93%)

7%

## Comments

**cookieMr** **Highly Voted** 2 years, 11 months ago

**Selected Answer: A**

A. suggests using AWS Backup, a centralized backup management service, to retain RDS backups. A backup vault is created, and a backup plan is defined with a daily schedule and a 2-year retention period for backups. RDS DB instances are assigned to this backup plan.

B. it does not address the requirement for consistent and restorable backups. Snapshots are point-in-time backups and may not provide the desired level of consistency.

C. it is not designed to provide the backup and restore functionality required for databases. It does not ensure the backups are consistent or provide an easy restore mechanism.

D. it does not address the requirement for daily backups and retention of consistent backups. It focuses more on replication and change data capture rather than backup and restore.

upvoted 11 times

**bamishr** **Highly Voted** 3 years, 4 months ago

**Selected Answer: A**

Create a backup vault in AWS Backup to retain RDS backups. Create a new backup plan with a daily schedule and an expiration period of 2 years after creation. Assign the RDS DB instances to the backup plan.

upvoted 5 times

**Ojonugwa** **Most Recent** 2 years, 2 months ago

The solution architect should recommend option B - Configure a backup window for the RDS DB instances for daily snapshots. Assign a snapshot retention policy of 2 years to each RDS DB instance. Use Amazon Data Lifecycle Manager (Amazon DLM) to schedule snapshot deletions.

This meets the requirements of:

Retaining daily backups for a minimum of 2 years. The daily snapshots captured within the backup window provide consistent, restorable backups on a daily basis.

Assigning a snapshot retention policy of 2 years ensures the snapshots are retained for the required period.

Using Amazon DLM allows automatically deleting snapshots older than 2 years to comply with the retention period in a cost-effective manner without manual administration.

The other option of using AWS Backup vault is not as suitable since it has limitations such as 35 day maximum retention for automated backups. Option B provides a native RDS solution capable of meeting the long term 2 year retention requirements.

upvoted 3 times

**kennylin689** 1 year, 9 months ago

If not specified, AWS Backup Vault Lock will not enforce a maximum retention period. If specified, backup and copy jobs to this vault with lifecycle retention periods longer than the maximum retention period will fail. Recovery points already saved in the vault prior to the vault lock's creation are not affected. The longest maximum retention period you can specify is 36500 days (approximately 100 years).

upvoted 2 times

**JA2018** 1 year, 6 months ago

I thought, for AWS exams, as a rule of thumbs, we should opt for AWS managed services?

upvoted 1 times

**vijaykamal** 2 years, 8 months ago

**Selected Answer: B**

Here's why Option B is the best choice:

Backup Window: Configuring a backup window for daily snapshots ensures that consistent backups are taken at the specified time each day. This helps maintain data integrity and consistency.

Snapshot Retention Policy: Assigning a snapshot retention policy of 2 years to each RDS DB instance ensures that the backups are retained for the required duration.

Amazon Data Lifecycle Manager (Amazon DLM): Amazon DLM can be used to automate the management of EBS snapshots, including RDS snapshots. You can configure Amazon DLM to schedule snapshot deletions, making it easier to manage the retention policy without manual intervention.

Option A (AWS Backup) is primarily used for managing backups of resources that may not have built-in backup capabilities, but for Amazon RDS, it's better to use the built-in snapshot capabilities and Amazon DLM for snapshot retention.

upvoted 3 times

**pentium75** 2 years, 5 months ago

"AWS Backup is primarily used for managing backups of resources that may not have built-in backup capabilities" says who?

upvoted 3 times

**TariqKipkemei** 2 years, 8 months ago

**Selected Answer: A**

Create a backup vault in AWS Backup to retain RDS backups. Create a new backup plan with a daily schedule and an expiration period of 2 years after creation. Assign the RDS DB instances to the backup plan.

upvoted 2 times

**Guru4Cloud** 2 years, 9 months ago

**Selected Answer: A**

A. Create a backup vault in AWS Backup to retain RDS backups. Create a new backup plan with a daily schedule and an expiration period of 2 years after creation. Assign the RDS DB instances to the backup plan

upvoted 3 times

**animefan1** 2 years, 11 months ago

**Selected Answer: A**

Backups work with EBS, FSX, RDS. Its managed & can has vault option for more better control over backup retention

upvoted 4 times

**markw92** 2 years, 11 months ago

Why not B?

upvoted 2 times

**\_deepsi\_dee29** 3 years ago

**Selected Answer: A**

A is correct

upvoted 1 times

**antropaws** 3 years ago

Why not D?

Creating tasks for ongoing replication using AWS DMS: You can create an AWS DMS task that captures ongoing changes from the source data store. You can do this capture while you are migrating your data. You can also create a task that captures ongoing changes after you complete your initial (full-load) migration to a supported target data store. This process is called ongoing replication or change data capture (CDC). AWS DMS uses this process when replicating ongoing changes from a source data store.